The Mass-Tracking Closure in Four-Dimensional Quasi-Topological Gravity: Double-Null Equations, an Exact Generalized-Mass Law and Evolution from Event-Horizon-Regular Data

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Abstract

The series' regular black hole is the Hayward metric, whose static sector is realized in four-dimensional quasi-topological gravity with $h(\psi) = \psi/(1-\ell^2\psi)$. We write the field equations with a massless scalar field in double-null form, derive them from the reduced action, and obtain an exact law for the generalized mass, $\partial_v m = 8\pi r^2 y \phi_v^2$, for every h. For data regular on the event horizon $y \geq \frac{1}{2}$, so m cannot decrease; the finite-time singularity of interior characteristic data belongs to the branch $y < 0$. A constrained evolution from such data agrees with an independent fourth-order code to 10^{-7} in r. For tail amplitudes 3×10^{-4}–3×10^{-2} the interior flux is the linear kernel of Paper G, the metric exponent is $-\kappa_-(m_f)$, and a front arrives behind which $\psi \to 1/\ell^2$, m grows linearly and the Kretschmann invariant exceeds $10^{25}\ell^{-4}$: the curvature bound of the static solutions does not extend to the dynamical interior. The accurate range of a scheme of order p is limited by an error growing as $h_v^p e^{\kappa_+ v}$, explained by a one-dimensional model with one fitted constant; evolving the difference from the exact horizon solution removes it (regular to $\kappa_+ v \simeq 69$). Behind the front the solution is slaved to the interior flux, and $K \propto v^{4k-2}$ for a tail v^{-k} is an extrapolation supported over the simulated range. A collapse of a compact scalar pulse generates the tail v^{-3}; for strong pulses its coefficient is $F \cdot 8 m_f |Q|$ with $F = 1 - Q_{\mathrm{abs}}/Q_0$ (Q_0 the Noether charge of the data, Q_{abs} the charge absorbed by the growing event horizon), to 0.05% on average in 22 cases. The identification of Q_{abs} with the horizon term of the time-inverted Newman–Penrose constant of Gajic and Kehrberger is conditional: the hypotheses of their theorem (a given metric settling to Schwarzschild) were not verified for the collapse data, and of that paper the statements of Sections 1–4, the derivation of the $\ell = 0$ constant (Proposition 7.1, Lemma 7.2), Sections 6, 8, 9 and the appendices were read, the energy estimates of Section 5 and the iteration of Sections 7.4–7.8 were not; for a Schwarzschild exterior the coefficient $8M|Q|$ is also obtained here independently, to first order in M and by a numerical evolution (1.000 to 10^{-3}). Not established: extendibility at $v = \infty$ and timelike tidal forces.

1 Introduction

The preceding papers of this series use the Hayward metric [Hayward, 2006]

$$f(r) = 1 - \frac{2mr^2}{r^3 + 2m\ell^2} \tag{1}$$

with a length ℓ fixed and a mass parameter m that tracks the mass of the hole (the closure $s = 0$ of Paper E [Ownby, 2026a]). Paper E asked whether the inner horizon of this metric survives perturbation. It answered with a thin-shell model that holds for any closure $M(m, r)$ and with a nonlinear double-null evolution that holds only for the *additive* closure $M = \mu + M_N(r)$, because the Einstein–nonlinear-electrodynamics system has no closure in which the mass parameter tracks the Misner–Sharp mass. Paper G [Ownby, 2026b] developed a late-time method for that additive system: data regular on the event horizon, the linear response of the static interior to the horizon flux, and exact integration of the mass along a ray. The open point is therefore the following: the series' metric is the static solution of a theory, if one exists, whose field equations couple m to the dynamics, and until that theory is specified one cannot evolve the interior of (1).

Such a theory now exists for the static sector. Bueno et al. [2025a] showed that quasi-topological gravities in dimensions $D \geq 5$ admit regular black holes whose metric function is fixed by a single function $h(\psi)$ of the curvature invariant $\psi = (1 - f)/r^2$; such polynomial theories do not exist in four dimensions, where non-polynomial quasi-topological gravities with the same structure were constructed by Bueno et al. [2026a], and Borissova and Carballo-Rubio [2026] identified the choice $h = \psi/(1 - \ell^2\psi)$ with the Hayward metric (their Eq. (156) solves their Eq. (152), $h = 2M/r^3$, with this h; this choice has a ψ^2 term and is not generated by the series of Bueno et al. 2026a). The formation of regular black holes by the collapse of thin dust shells in higher-dimensional ($D \geq 5$) quasi-topological gravity was studied by Bueno et al. [2025b], and the time-dependent spherically symmetric field equations with matter for the non-polynomial four-dimensional theories were obtained by Bueno et al. [2026a]. Frolov and Zelnikov [2026] (null-shell collisions in regular black holes of quasi-topological gravity: large amplification only for collisions extremely close to the inner horizon) and Di Filippo et al. [2026] (mass inflation and thin shells in quasi-topological gravity) studied shells and mass inflation in these theories; the latter argue that in standard distributional theory null thin shells are not available for the pure-gravity regular black hole solutions in the standard distributional theory (a conclusion drawn within that theory and, as far as we read, at the level of the abstract of that paper; Ref. Frolov and Zelnikov [2026] uses null shells with the Dray–'t Hooft–Barrabès–Israel junction conditions), so that the Ori thin-shell construction of Paper E need not carry over. This paper uses the first group of results to evolve the interior and compares with the second.

Related work. Mass inflation at the inner horizon of charged black holes was identified by Poisson and Israel [1989, 1990] and given an exact model by Ori [1991]; double-null evolutions of scalar-field collapse to charged black holes [Brady and Smith, 1995, Burko, 1997, Hod and Piran, 1998], the late-time analysis of the shock at the inner horizon [Marolf and Ori, 2012] and the mathematical theory [Dafermos, 2005] describe a weak null singularity. For regular black holes in general relativity with matter, Carballo-Rubio et al. [2022] construct regular black holes whose inner horizon has vanishing surface gravity (a degenerate inner horizon) and find no exponential mass inflation in the two perturbation models they study, Bonanno et al. [2026] find power-law growth of the Hayward mass function, map its basin of attraction, and find logarithmic growth in an asymptotically safe collapse model, Iofa [2022] computes the flux (in his abstract, the energy density) of scalar perturbations measured by a free-falling observer near the inner horizon of the Hayward black hole, which diverges exponentially there and which he states is not directly connected with mass inflation (the inner mass function itself stays finite in his calculation), and Shao et al. [2026] evolve a scalar field in the Hayward geometry sourced by non-linear electrodynamics and find that a strong field shrinks the inner horizon to zero. In quasi-topological gravity the static vacuum solutions satisfy a limiting-curvature property [Bueno et al., 2026b], minimally coupled matter breaks the mass-independent bound but sufficient conditions on the stress tensor keep the curvature bounded for static matter [Bueno et al., 2026b], null-shell collisions give large amplification only extremely close to the inner horizon [Frolov and Zelnikov, 2026], and thin shells are constrained differently from the Ori model [Di Filippo et al., 2026]. We are not aware of a double-null evolution, in the quasi-topological theory itself, of a continuous scalar flux from event-horizon-regular data through the inner-horizon region; the present paper is that evolution and reports what it shows about the generalized mass, the metric exponent and the curvature.

What is done. (1) We write the field equations of Ref. Bueno et al. [2026a] for a massless minimally coupled scalar field in double-null form (Section 3) and verify them by five independent checks that do not rely on the derivation (Section 4). (2) We derive an exact law for the generalized mass along an ingoing ray, $\partial_v m = 8\pi r^2 y \phi_v^2$ (Section 5), and prove that data regular on the event horizon have $y \geq \frac{1}{2}$, so that the generalized mass cannot decrease (Section 6). With the range of h the law classifies where ψ can go (Section 7). (3) We report what characteristic data on interior rays do in this system, a finite-advanced-time singularity of the data ray, and explain with the proposition that it belongs to the branch $y < 0$, which event-horizon-regular data exclude (Appendix A). (4) We construct event-horizon-regular data for an arbitrary h (Section 8) and a constrained evolution that is stable where the free evolution of Paper E is not (Section 9). (5) We report the evolution of the Hayward h from these data: the interior flux against the linear kernel, the metric exponent, the front of the inflating mass, the state behind it and the curvature (Section 12); and of a bounded h (Section 13). (6) We identify the numerical limit of the run, a growth of the truncation error as $h_v^p e^{\kappa_+ v}$ (p the order of the scheme; for the second-order code $\propto (B\, h_v)^2$), and remove it by refining the advanced-time step alone (Section 9); we show that the state behind the front is slaved to the interior flux and read the endpoint from that structure, with its asymptotics, for three tail exponents (Sections 12.5 and 12.6); (7) we repeat the evolution with a second, independent code of fourth order, which agrees with the first and allows the curvature invariant to be followed to $v \simeq 380$, and study that invariant and the tidal integrals along the ingoing null ray in that range (Sections 10 and 12.8); (8) we derive the mechanism of the limit, remove it by evolving the differences from the exact horizon solution, locate and remove a rounding-level seed of the remaining late-time growth (a step in the initial row), and follow the solution to $\kappa_+ v \simeq 50$–69 for five tail exponents, three amplitudes and three inner masses (Sections 11, 11.3 and 12.9); (9) we replace the prescribed horizon tail by the one generated by a collapse (a nonlinear double-null stage and a linear stage on the static exterior, with their validation), measure its exponent and amplitude in the Hayward and Einstein cases, drive code E with it, and repeat the construction with collapse data of nonzero Newman–Penrose constant, which gives the exponent $k = 2$ and a coefficient fixed by the data (Section 14); (10) we scan the strength of a pulse imposed on the horizon and then produce strong pulses by collapse into an existing hole (m_f/m_0 up to 5), which fixes the inner radius at $r_-(m_f)$ and shows that the v^{-3} coefficient is reduced by a factor $F = 0.37$–1 with respect to the linear identity $B = 8m|Q|$; (11) we explain F as one minus the fraction of the Noether charge of the data that the growing event horizon absorbs, verify this in 22 cases to 0.05% on average (0.09% at most; the reading of that fraction as the horizon term of a published theorem is conditional, Section 14), and give a priori estimates of that fraction; (12) we repeat the construction for data with a nonzero scalar charge at infinity (Section 14).

What is not done. The field equations are those of the reduced action of Refs. Bueno et al. [2026a], Borissova and Carballo-Rubio [2026]; we check that the covariant densities lift that action (Section 4(g)) and that the double-null equations are its Euler–Lagrange equations (Section 4(f)), but we take the four-dimensional theory itself, and the Hayward function as its resummation, from those references; for $\ell^2\psi \geq 1$ and for $\psi \to -\infty$ the time-dependent equations are an analytic continuation that we adopt and do not prove (the event-horizon-regular solutions of Sections 12–13 stay in $0 < \psi < 1/\ell^2$, inside the radius of convergence of the series). We do not include the Hawking flux, we do not simulate beyond $v \simeq 1400$ ($\kappa_+ v \simeq 69$ for $m/\ell = 5$; the endpoint is read from a structure verified to that range), we do not classify the singularity at $v = \infty$ (extendibility, tidal forces on timelike observers), and nothing here concerns the Ori shell (Ref. Di Filippo et al. [2026] argues that null thin shells are not available in standard distributional theory for these theories; Ref. Frolov and Zelnikov [2026] uses them with junction conditions). Section 15 lists what is and is not established.

2 The theory and the closure

In a four-dimensional quasi-topological gravity (Refs. Bueno et al. [2026a], Borissova and Carballo-Rubio [2026]; Ref. Bueno et al. [2025a] for the higher-dimensional polynomial theories with the same structure) the static spherically symmetric solutions are $ds^2 = -N^2 f\, dt^2 + dr^2/f + r^2 d\Omega^2$ with N constant and f determined by

$$h(\psi) = \frac{2m}{r^3}, \qquad \psi \equiv \frac{1-f}{r^2}, \tag{2}$$

where m is an integration constant (the *generalized mass*) and h is the characteristic function of the theory; Einstein gravity is $h = \psi$, for which $f = 1 - 2m/r$. The Hayward metric (1) is the solution of (2) for

$$h(\psi) = \frac{\psi}{1-\ell^2\psi}, \qquad h'(\psi) = \frac{1}{(1-\ell^2\psi)^2}, \tag{3}$$

because $1 - \ell^2\psi = r^3/(r^3 + 2m\ell^2)$ for (1). Two facts about this realization matter for what follows. First, the Misner–Sharp mass is $M = r^3\psi/2 = mr^3/(r^3 + 2m\ell^2)$, which is bounded by $r^3/2\ell^2$ for any m, whereas the generalized mass m is not; the mass-tracking closure of Paper E [Ownby, 2026a] is the statement that m is the quantity which is conserved in the vacuum and changed by the flux of matter, the Misner–Sharp mass being a derived function of (m, r). Second, in the quasi-topological theory the field equations determine how m changes (Eq. (16) below), so no closure has to be postulated; the additive closure of Paper E was needed because the Einstein–NED system does not supply one.

For the bounded model of Ref. Bueno et al. [2026a] (Eq. (51) there; "bounded" refers to the range of ψ, while h itself diverges at $|\psi| = 1/\alpha$)

$$h(\psi) = \frac{\psi}{\sqrt{1-\alpha^2\psi^2}}, \qquad h'(\psi) = \frac{1}{(1-\alpha^2\psi^2)^{3/2}}, \tag{4}$$

ψ is confined to $(-1/\alpha, 1/\alpha)$ in the static solutions and the generalized mass is unrestricted. We call this model 2 and the Hayward model 1; model 0 is Einstein gravity. The static interior of model 1 with $m = 5$, $\ell = 1$ has $r_- = 1.057475$, $r_+ = 9.897927$, $\kappa_- = 0.795649$ (the background of Paper E at $\mu = 0$), and that of model 2 with $m = 1$, $\alpha = 1$ has $\psi(r_-) = 0.8376$; the choice $m = 1$ keeps $\psi(r_-)$ away from the wall (interior-data runs, Appendix A). The event-horizon-regular runs of Section 13 use $m = 1.5$.

Static relations. From (2), $\psi'(r) = -3h/(rh')$ and so

$$f'(r) = -2r\psi - r^2\psi' = rQ(\psi), \qquad Q(\psi) \equiv \frac{3h}{h'} - 2\psi. \tag{5}$$

At a horizon $\psi = 1/r_h^2$ and $\kappa_h = |f'|/2 = r_h|Q(\psi_h)|/2$. For Hayward $Q = \psi(1-3\ell^2\psi)$, which gives $Q(1/r_-^2) = -1.5048$ and $\kappa_- = 0.7956$ at $r_- = 1.0575$, as it must; for $h = \psi$, $Q = \psi$. The sign of Q' at the inner horizon is opposite to that in Einstein gravity ($Q' = 1 - 6\ell^2\psi_- = -4.36$ for Hayward); we use Q only through (5) and (15) below.

3 Double-null form of the field equations

Take $ds^2 = -W\,du\,dv + r^2 d\Omega^2$, $W = e^\sigma$, a massless scalar $\phi(u, v)$ minimally coupled with the usual stress tensor $T_{\mu\nu} = \partial_\mu\phi\,\partial_\nu\phi - \frac{1}{2}g_{\mu\nu}(\partial\phi)^2$, and

$$\psi = \frac{1}{r^2}\Big(1 + \frac{4r_u r_v}{W}\Big), \qquad Q = \frac{3h}{h'} - 2\psi, \quad c_1 = \frac{1}{h'}, \quad c_2 = \frac{9\,h^2 h''}{4\,h'^3}, \quad c_3 = \frac{h''}{h'^3}. \tag{6}$$

In the gauge $\varphi = r$ the time-dependent equations of Ref. Bueno et al. [2026a] read

$$\partial_r\big[r^3 h(\psi)\big] = \frac{8\pi G}{N^2 f}\,r^2 T_{tt}, \qquad \partial_t f = -\frac{8\pi G}{h'(\psi)}\,r f\,T_{tr}, \qquad \partial_r N = \frac{4\pi G}{h'(\psi)}\,rN\Big(T_{rr} + \frac{T_{tt}}{N^2 f^2}\Big) \tag{7}$$

(Eqs. (36)–(38) there; T_{tt}, T_{tr}, T_{rr} are the components of the stress tensor of the scalar). Since $f = (\nabla r)^2$ and $\psi = (1-f)/r^2$ are scalars, the gauge $\varphi = r$ can be chosen wherever $\nabla r \neq 0$, and the first and third equations, rewritten in null coordinates with $T_{uu} = \phi_u^2$, $T_{vv} = \phi_v^2$ and $T_{uv} = 0$, give the two Raychaudhuri-type equations; the combination that defines the static solution, $f' = rQ(\psi)$, gives r_{uv}. The remaining equation, for σ_{uv}, is not independent: it is fixed by the compatibility of the others (Section 4(e)). With $G = 1$, then,

$$r_{uu} - \sigma_u r_u = -\frac{4\pi r}{h'}\,\phi_u^2, \qquad\qquad r_{vv} - \sigma_v r_v = -\frac{4\pi r}{h'}\,\phi_v^2, \tag{8}$$

$$r_{uv} = -\frac{Wr}{4}\,Q(\psi), \tag{9}$$

$$\sigma_{uv} = \frac{W\psi}{2} - Wc_2 - \frac{8\pi}{h'}\phi_u\phi_v + \frac{64\pi^2 c_3}{W}\phi_u^2\phi_v^2, \tag{10}$$

$$\phi_{uv} = -\frac{r_u\phi_v + r_v\phi_u}{r}. \tag{11}$$

For $h = \psi$ ($h' = 1$, $h'' = 0$, $c_2 = c_3 = 0$, $Q = \psi$) these are the Einstein–scalar equations in the form used in Paper E [Ownby, 2026a]: $r_{uv} = -\frac{1}{4}Wr\psi = -\frac{1}{4}(Wr^{-1} + 4r_u r_v/r)$, $\sigma_{uv} = \frac{1}{2}W\psi - 8\pi\phi_u\phi_v$. For a static solution $r_{uv} = ff'/4$ and $\sigma_{uv} = ff''/4$ (interior $f < 0$, $W = |f|$), which (9) and (10) must reproduce from (2) (check (b) below).

Status of the equations. Equations (8), (9) and (11) are our transcription of Ref. Bueno et al. [2026a] into null coordinates, and Eq. (10) follows from them by compatibility; we did not find them in this form in the literature. Check (f) below derives all of them, including the c_3 term, from the spherically reduced action of Refs. Bueno et al. [2026a], Borissova and Carballo-Rubio [2026], independently of the transcription. The checks of Section 4 test the transcription (not the field equations of the theory). The equations of Ref. Bueno et al. [2026a] are proved there for the class of h generated by the theory, which contains the bounded model (4); the Hayward h, which has a ψ^2 term, is realized by the action of Ref. Borissova and Carballo-Rubio [2026], whose characteristic function is arbitrary and whose spherical reduction has the same Horndeski form (check (f)).

4 Checks on the equations

The checks (a)–(e) below do not use the equations being tested to produce the reference; (f) derives them from the reduced action and (g) connects that action to the four-dimensional invariants.

(a) *Einstein limit.* The code with $h = \psi$ (model 0) and the code of Paper E [Ownby, 2026a] for the additive closure at $m_N = 0$ differ by 9.7×10^{-17} (maximum over the grid, field scale 3.4) on identical data (`check_gr_limit.py`).

(b) *Static interior.* For the Hayward h with $m = 5$ and for the bounded h with $m = 1$, equations (9)–(10) reproduce $ff'/4$ and $ff''/4$ computed by finite differences of the exact f to 1.3×10^{-7} and 1.1×10^{-6} (relative, nine radii between $1.05r_-$ and $0.97r_+$; limited by the finite difference) (`check_static_identity.py`).

(c) *Integrability.* The u- and v-Raychaudhuri equations (8) are compatible with the r_{uv} equation (9) only if, with $c = 4\pi/h'$ and $Q = 3h/h' - 2\psi$,

$$c\,Q' - c = c'\,(Q + 2\psi). \tag{12}$$

This is an identity in h; we verified it with numerical derivatives of h at six values of ψ for the Hayward h (maximum relative violation 1.8×10^{-11}) and five for the bounded h (2.1×10^{-9}). Using $c' = -4\pi h''/h'^2$, $Q' = 3 - 3hh''/h'^2 - 2$ and $Q + 2\psi = 3h/h'$, both sides equal $4\pi\big(1 - 3hh''/h'^2\big)/h'$ identically, so (12) is exact for every h.

(d) *Convergence of the constraints.* The evolution uses (9)–(11) only, not (8). The residuals $r_{uu} - \sigma_u r_u + 4\pi r\phi_u^2/h'$ and the v analog, computed afterwards, converge at second order in the mesh width for both models (rows: model and window in v; $A = 0.02$, $B = 0.002$; entries: relative L_2 residual in the u and v directions; `run_qtg_validate.py`, `validate_model1.log`, `validate_model2.log`). A wrong σ_{uv} or r_{uv} would leave an $O(1)$ residual.

model, window	$h = 0.08$ $(u,\ v)$	$h = 0.04$	$h = 0.02$	orders $(u;\ v)$
1, $v \in [0, 20]$	4.0×10^{-3}, 3.0×10^{-3}	9.9×10^{-4}, 7.5×10^{-4}	2.5×10^{-4}, 1.9×10^{-4}	2.0; 2.0
1, $v \in [20, 55.2]$	5.9×10^{-1}, 6.8×10^{-4}	1.4×10^{-1}, 1.7×10^{-4}	3.4×10^{-2}, 4.2×10^{-5}	2.0; 2.0
2, $v \in [0, 20]$	1.2×10^{-3}, 6.1×10^{-4}	2.9×10^{-4}, 1.6×10^{-4}	7.4×10^{-5}, 4.1×10^{-5}	2.0; 1.9
2, $v \in [20, 60]$	2.4×10^{-2}, 2.7×10^{-4}	5.9×10^{-3}, 6.8×10^{-5}	1.5×10^{-3}, 1.7×10^{-5}	2.0; 2.0

The static background itself converges at second order (maximum relative error in r: 9.3×10^{-4}, 2.3×10^{-4}, 5.8×10^{-5}, 1.5×10^{-5} for $h = 0.08, \ldots, 0.01$, model 1).

The u-residual of model 1 in the second window is large at the coarsest mesh (0.59) and falls at second order (0.14, 0.034); we attribute the size to the exponentially small r_u in that window (the residual is relative to $\|r_{uu}\|$) but have not separated that from truncation error in σ_u. The scheme, like that of Paper E, loses accuracy as v approaches the singular time of Appendix A.

(e) *Compatibility.* Given (8), (9) and (11), the mixed derivative of the Raychaudhuri equation must agree with the u-derivative of (9), $\partial_v(r_{uu}) = \partial_u(r_{uv})$. This is linear in σ_{uv} and determines it from the other equations at any point where $r_u \neq 0$ (the derivatives of all the fields that appear follow from the equations themselves). We solved it at 2000 random points (random $r, r_u, r_v, \sigma, \sigma_u, \sigma_v, \phi_u, \phi_v$ with ψ in the domain of h) for the Hayward h and the bounded h and compared with (10), including the coefficient 9/4 of c_2: the maximum relative difference is 3.4×10^{-12} and 3.3×10^{-10} (`check_compat.py`). So (10) is not an independent assumption: it is what makes the system consistent, for every h (the integrability identity (12) is the same statement for the coefficient of ϕ_u^2).

(f) *Derivation from the reduced action.* The tests (a)–(e) check the transcription and its consistency; they do not show that the equations are those of the theory. We therefore derived them from an action. Ref. Bueno et al. [2026a] (Eqs. (22)–(26) there) reduces the four-dimensional action on spherical symmetry, $ds^2 = \gamma_{ab}dx^a dx^b + \varphi^2 d\Omega^2$, to the two-dimensional Horndeski theory

$$S = \frac{1}{2G}\int d^2x\sqrt{-\gamma}\left[G_2 - \Box\varphi\, G_3 + G_4 R - 2G_{4,X}\left((\Box\varphi)^2 - \nabla_a\nabla_b\varphi\nabla^a\nabla^b\varphi\right)\right] - 2\pi\int d^2x\sqrt{-\gamma}\,\varphi^2(\nabla\phi)^2, \tag{13}$$

with $X = (\nabla\varphi)^2$, $\psi = (1-X)/\varphi^2$, $G_2 = \varphi^2(3h - 2\psi h')$, $G_3 = 2\varphi h'$ and $G_4 = -\frac{1}{2}\varphi^2\psi\int \mathrm{d}\psi\,\psi^{-2}h'$; for $h = \psi + \sum_{n\geq 3} a_n\psi^n$ this is the action of that paper, and Ref. Borissova and Carballo-Rubio [2026] (Eqs. (130) and (142) there, the former written with $H = 3h$; note that Eq. (133) of that paper, $H = 12M/r^3$, corresponds to $H = 6h$, i.e. to the Einstein normalization $H_2 = 2\psi$, $H_3 = 4$, so the two normalizations of H_i in that paper differ by a factor 2, which we absorb in the overall coefficient of the action) gives the same functions for an arbitrary h, in particular for terms ψ^2 as in the Hayward function, for which G_4 contains $-a_2\varphi^2\psi\ln\psi$. We took $\varphi = r$ and the double-null metric $-W\,du\,dv$ ($\sqrt{-\gamma} = W/2$, $X = -4r_u r_v/W$, $\Box r = -4r_{uv}/W$, $R = 4\sigma_{uv}/W$, $(\Box r)^2 - r_{ab}r^{ab} = 8(r_{uv}^2 - \mathcal{R}_u\mathcal{R}_v)/W^2$ with $\mathcal{R}_u = r_{uu} - \sigma_u r_u$) and computed the Euler–Lagrange expressions E_W and E_r by automatic differentiation of the Lagrangian with respect to $(r, r_u, r_v, r_{uu}, r_{uv}, r_{vv}, W, W_u, W_v, W_{uv})$, in exact rational arithmetic on bivariate Taylor jets (`action_check2.py`, `action_check3.py`; the logarithm of the ψ^2 term in float64). At random points, with h a random polynomial of degree four or five, the results are: (i) $E_W = 0$ is Eq. (9) (the difference is zero in every test); (ii) with the Raychaudhuri constraints (8) imposed at the point and (9) to second order in the jet, $E_r = 0$ is Eq. (10), including the term $64\pi^2 c_3\phi_u^2\phi_v^2/W$ (difference exactly zero in 12 tests with $a_2 = 0$ and at most 1.3×10^{-14} in 8 tests with $a_2 \neq 0$; without the c_3 term the difference is 10^{-5}–10^{-1} of order one of σ_{uv} at the same points, and for a Raychaudhuri coefficient 1.1 times larger it is 6×10^{-3} or more); E_r does not depend on r_{uuu}, as a second-order equation must not. (iii) A second action, built only from the static reduction $L = \frac{1}{2}N\,\mathrm{d}(r^3 h)/\mathrm{d}r$ (the Lagrangian-reduction property of these theories), a two-dimensional covariant completion $G_2(r,X) + G_3(r,X)\Box r + F(r)R$ that is analytic at $X = 0$, and unique with those requirements, has Euler–Lagrange expressions identical to those of (13) (difference zero), so the result does not depend on the form in which the reduced action is written (`action_check.py`, with the hand-derived partial derivatives). With it the coefficient of the Raychaudhuri equations is also derived rather than assumed: $\partial_v(r_{uu} - \sigma_u r_u + c\,r\phi_u^2)$ vanishes on the constraint surface, with σ_{uv}, r_{uv} and ϕ_{uv} from the field equations, exactly for $c = 4\pi/h'$ (30 tests, h of degree 4 and 6) and not for $1.1\,c$ or $c(1+\psi/5)$. The scripts and logs are in `code/eh_qtg/`. What this establishes is that (9)–(10) and the coefficient $4\pi/h'$ are the field equations of the reduced action of Refs. Bueno et al. [2026a], Borissova and Carballo-Rubio [2026]; check (g) below connects that action to the four-dimensional invariants.

(g) *The reduction from the four-dimensional invariants.* The action of Ref. Borissova and Carballo-Rubio [2026] (Eq. (66) there; Ref. Bueno et al. [2026a], Eqs. (3)–(5) and (20)–(21): the Lagrangians of the first curvature orders, the recursion and the action as a tower, for h a power series in ψ) is $S = \int \mathrm{d}^4x\sqrt{-g}\,[H_2(P) - H_3(P)\,H + H_4(P)\,K + 2H_4'(P)(H^2 - T)]$ with the four scalars P, H, T, K built from the Zakhary–McIntosh invariants of the Ricci and Weyl tensors (their Eqs. (55), (63)–(65)); on the warped product $g = q_{ab}\mathrm{d}y^a\mathrm{d}y^b + \varphi^2\mathrm{d}\Omega^2$ they are designed to equal $\psi = (1-(\nabla\varphi)^2)/\varphi^2$, $\eta = \Box\varphi/\varphi$, $\tau = \nabla_a\nabla_b\varphi\nabla^a\nabla^b\varphi/\varphi^2$ and the Ricci scalar $\mathcal{R}$ of q. We tested this, not assumed it: for five random spherically symmetric metrics (arbitrary $q_{ab}(y)$ and $\varphi(y)$ with no Killing vector, so including all time-dependent terms), the four-dimensional Riemann tensor was computed from the metric by fourth-order finite differences (no warped-product formula), the invariants I_R, $I_{\hat{R}^2}$, $I_{\hat{R}^3}$, I_{C^2}, I_{C^3}, I_{R^2C} were formed, and (i) their values agree with the reductions $R = \mathcal{R} - 4\eta + 2\psi$, $I_{C^2} = \Omega^2/3$, $I_{C^3} = \Omega^3/18$, $I_{\hat{R}^2} = \Sigma$, $I_{\hat{R}^3}$, I_{R^2C} of that reference ($\Omega = \mathcal{R} + 2\eta + 2\psi$) to $\leq 9\times 10^{-11}$; (ii) the lifts P, H, T, K reproduce $\psi, \eta, \tau, \mathcal{R}$ to $\leq 3\times 10^{-11}$ (`lift_check_4d.py`, `lift_check_4d.log`); (iii) the density with $H_2 = 2P$, $H_3 = 4$, $H_4 = 1$ equals the Ricci scalar ($\leq 3 \times 10^{-13}$). In Eq. (65) the third term of the numerator must be $4\, I_{\hat{R}^2} I_{C^3}^2 / I_{C^2}$ (the arXiv HTML source has the square; a text extraction of the PDF gave the first power): with I_{C^3} to the first power the term has dimension L^{-6} against L^{-12} for the others, and the test fails ($K \sim 10^4$ instead of $\mathcal{R} \sim 10^{-1}$ in the script log). Since $\sqrt{-g} = \varphi^2 \sqrt{-q} \sin\theta$ and $\int \mathrm{d}\Omega = 4\pi$, the density reduces to $4\pi\varphi^2[H_2 - H_3\eta + H_4\mathcal{R} + 2H_4'(\eta^2 - \tau)]$, which is (13) with $G_2 = \varphi^2 H_2$, $G_3 = \varphi H_3$, $G_4 = \varphi^2 H_4$ and $G_{4,X} = -H_4'$ (using $\partial\psi/\partial X = -1/\varphi^2$) when the H_i are in the normalization of Eq. (142) of Ref. Borissova and Carballo-Rubio [2026] ($H_2 = 3h - 2\psi h'$, $H_3 = 2h'$, $H_4 = -\frac{1}{2}\psi \int \mathrm{d}\psi\, h'/\psi^2$, which for $h = \psi$ is one half of the Einstein-normalized $H_2 = 2\psi$, $H_3 = 4$, $H_4 = 1$ used in (iii)) and the overall factor is $1/(16\pi G)$ (so that $4\pi\varphi^2 \cdot 2H_i/(16\pi G) = \varphi^2 H_i/(2G)$). By the principle of symmetric criticality for the compact group $SO(3)$ the field equations of the reduced action are the spherically symmetric sector of those of the four-dimensional theory. Limits: the lifts are non-polynomial (ratios of invariants; I_{C^2} and $I_{C^2} I_{R^2 C} + 2 I_{\hat{R}^2} I_{C^3}$ must not vanish, which fails for flat or conformally flat regions) and are not unique (Ref. Borissova and Carballo-Rubio [2026] notes that other eliminations give inequivalent four-dimensional theories with the same spherical sector), so the check shows that a four-dimensional theory with this reduction exists, which is all the time-dependent spherical dynamics uses; it says nothing about perturbations that break spherical symmetry (hyperbolicity of the full theory is not addressed).

5 An exact law for the generalized mass along an ingoing ray

Let $\rho = -r_v$ and $y = -r_u/W$, so that $\psi = (1 + 4y\rho)/r^2$ and the Misner–Sharp mass is $M = \frac{r}{2}(1 + 4y\rho) = r^3\psi/2$ (the variables of the nonlinear test of Paper E [Ownby, 2026a]). Equations (8) and (9) give, along a ray $u = \mathrm{const}$,

$$r' = -\rho, \qquad \rho' = \sigma_v \rho + \frac{4\pi r}{h'} \phi_v^2, \qquad y' = \frac{rQ}{4} - \sigma_v y, \tag{14}$$

and differentiating $r^2\psi = 1 + 4y\rho$,

$$\psi_v = \frac{\rho}{r}\big(Q + 2\psi\big) + \frac{16\pi}{r\, h'(\psi)}\, y\, \phi_v^2, \qquad Q + 2\psi = \frac{3h}{h'}. \tag{15}$$

The metric function σ_v cancels. Multiplying by h' and using $r_v = -\rho$,

$$\boxed{\frac{\partial m}{\partial v} = 8\pi\, r^2\, y\, \phi_v^2, \qquad m \equiv \tfrac{1}{2} r^3 h(\psi)} \tag{16}$$

for every h: $(r^3 h)_v = 3r^2 r_v h + r^3 h' \psi_v = 16\pi r^2 y \phi_v^2$. This law is not written in Ref. Bueno et al. [2026a]; it is the combination of its Eqs. (36) and (37) along an ingoing null ray. In the gauge $\varphi = r$ they give $\mathrm{d}(r^3 h) = 8\pi r^2 [T_{tt}\, \mathrm{d}r/(N^2 f) + f\, T_{tr}\, \mathrm{d}t]$; on $\mathrm{d}r = -Nf\, \mathrm{d}t$, for the massless scalar $T_{tt} - NfT_{tr} = \frac{1}{2}(\phi_t - Nf\phi_r)^2$, and with $\partial_v = \lambda(\partial_t - Nf\partial_r)$, $y = -1/(4\lambda N)$, this is Eq. (16). It also follows from Eqs. (8)–(9). For $h = \psi$ it is the mass equation of the Einstein–scalar system ($M_v = 8\pi r^2 y \phi_v^2$; for $M \gg r$ it gives $d\ln M/dv = 4\pi r \phi_v^2/\rho$, the ray equation of Paper E). Its content in the present context is that the *sign of y is the sign of the change of the generalized mass*, independently of the theory: m increases on rays with $y > 0$ and decreases on rays with $y < 0$. We verified (16) on the converged solution of Appendix A (`check_mass_law.py`): $|\partial_v m - 8\pi r^2 y \phi_v^2| / |8\pi r^2 y \phi_v^2| = 2 \times 10^{-4}$ at $v = 5$, 3×10^{-4} at $v = 45$, 1.5×10^{-4} at 55, 3×10^{-4} at 60 and 2×10^{-2} at 64 on $u = 0$ (central differences of the stored samples at $\Delta v = 0.05$, where m changes by a factor 2 per 4 units of v; the larger value is the finite-difference error of $\partial_v m$ at the steepest point). On the event-horizon-regular runs the same check is made in Section 12.

6 A sign theorem for data regular on the event horizon

The sign of $y = -r_u/W$ decides the sign of the change of the generalized mass in (16). Equations (8) give a monotonicity law for y and for its ingoing counterpart $q = -r_v/W = \rho/W$. Since $r_{uu} - \sigma_u r_u = W\,\partial_u(r_u/W)$ and likewise in v,

$$\partial_u y = \frac{4\pi r}{h'(\psi)}\,\frac{\phi_u^2}{W}, \qquad \partial_v q = \frac{4\pi r}{h'(\psi)}\,\frac{\phi_v^2}{W}, \qquad \psi = \frac{1+4yqW}{r^2}. \tag{17}$$

Both are exact consequences of (8). They are the content of the sign theorem below, and they are the equations that the evolution of Section 9 integrates in place of evolving r_u and r_v.

Proposition 1. *Let the matter be the real massless scalar field of* (11) *(so that its stress tensor satisfies the null energy condition,* $T_{uu} \propto \phi_u^2 \geq 0$ *and likewise in* v*), let* $h' > 0$ *on the range of* ψ *attained, let* $r > 0$*, and let* (r, σ, ϕ) *solve* (8)–(11) *on* $\{U \geq 0,\ v \geq 0\}$ *in coordinates* (U, v) *regular across the event horizon* $\mathcal{H}^+ = \{U = 0\}$*, with* $y = -r_U/W = \frac{1}{2}$ *on* $\mathcal{H}^+$*. Then*

(i) $y \geq \frac{1}{2}$ *everywhere;*

(ii) q *is non-decreasing in* v *and* $\partial_v m \geq 4\pi r^2 \phi_v^2 \geq 0$ *along every ingoing ray: the generalized mass is non-decreasing;*

(iii) if $\rho \geq 0$ *on* $v = 0$*, then* $\rho \geq 0$ *and* $\psi \geq 1/r^2 > 0$ *everywhere.*

Proof. (i) By (17), $\partial_u y \geq 0$ at fixed v, and $y = \frac{1}{2}$ at $U = 0$. (ii) The second of (17) gives $\partial_v q \geq 0$; then (16) and (i) give $\partial_v m = 8\pi r^2 y \phi_v^2 \geq 4\pi r^2 \phi_v^2$. (iii) $q = \rho/W$ is non-decreasing from a non-negative value, and $\psi = (1 + 4yqW)/r^2 \geq 1/r^2$. □

Remarks. (1) The value $\frac{1}{2}$ is a normalization of v: under $v = b(\tilde{v})$ one has $\tilde{W} = b'W$ and hence $\tilde{y} = y/b'$, while $u \to a(u)$ leaves y unchanged, so on a non-degenerate horizon ($r_U < 0$ on $\mathcal{H}^+$) the normalization $y = \frac{1}{2}$ can always be arranged. The statement that matters, $y > 0$ everywhere, does not depend on it. (2) For $h = \psi$ ($h' = 1$) the first law of (17) is the standard monotonicity of the Einstein–scalar system in double-null form (cf. the analogous identity in Ref. Dafermos [2005]); the point here is that it holds for every h with $h' > 0$, for the Hayward h ($h' = (1 - \ell^2\psi)^{-2}$) and the bounded h ($h' = (1 - \alpha^2\psi^2)^{-3/2}$) alike. (3) The proposition does not say that a solution exists globally; it says that on the domain where one exists the negative-mass branch of Section 7 is not available. Data given on a pair of interior rays have no such constraint: y is set by the data on $u = u_0$ and can be negative (Appendix A). (4) The proposition is a statement about the sign of the source: it holds on the evolved solution, not only to linear order, and does not need the tail to be small. (5) Part (i) uses only the null energy condition: the sign of $\partial_u y$ in (17) is that of ϕ_u^2; part (ii) through (16) also uses $T_{uv} = 0$ (massless, no potential), since otherwise $\partial_v m$ has an extra term. For matter that violates it (a ghost scalar, $\phi_u^2 \to -\phi_u^2$) y decreases along u and $y \geq \frac{1}{2}$ is lost; the horizon value $y = \frac{1}{2}$ is also an unstable one under forward integration along $\mathcal{H}^+$ (perturbations of r_v there grow as $e^{\kappa_+ v}$, Eq. (19) and Section 9), which is why the data are built by integrating backwards in v (Section 8).

7 Where ψ can go: classification by h

Equation (16) and the definition $h(\psi) = 2m/r^3$ determine ψ along the ray from $m(v)$ and $r(v)$ by inverting h. The image of h therefore decides what can happen.

Hayward. $h = \psi/(1-\ell^2\psi)$ maps $\psi \in (-\infty, 1/\ell^2)$ one-to-one onto $h \in (-1/\ell^2, \infty)$, with $h \to \infty$ at the pole $\psi \to 1/\ell^2$ and $h \to -1/\ell^2$ as $\psi \to -\infty$. Hence, with $2m/r^3 = h$:

- *(i)* $y > 0$*:* m increases monotonically (mass inflation of the generalized mass; without bound if the flux persists, as in the runs below); as $2m/r^3 \to \infty$, $\psi \to 1/\ell^2$ from below. The pole is never reached at finite v: in (15) with $z = 1 - \ell^2\psi$ one has $\psi_v = 3\rho\psi z/r + 16\pi y\phi_v^2 z^2/r$, whose right-hand side is smooth and vanishes at $z = 0$, so $\psi = 1/\ell^2$ is an invariant manifold. The Misner–Sharp mass $M = r^3\psi/2$ stays below $r^3/2\ell^2$ however large m becomes. The curvature invariants of the Hayward metric are bounded by ℓ for $m \to \infty$ at fixed r (Hayward's regularity), so nothing singular occurs by this route.

- *(ii)* $y < 0$*:* m decreases, becomes negative, and $2m/r^3$ reaches $-1/\ell^2$ when $r^3 = 2|m|\ell^2$. There h leaves its range and $\psi \to -\infty$ (equivalently $r_u r_v/W \to -\infty$ in $\psi = (1 + 4r_u r_v/W)/r^2$). This is the radius at which the Hayward metric with a negative mass parameter, $f = 1 - 2mr^2/(r^3 + 2m\ell^2)$, has its pole, and at which the source $1/h' = z^2$ in the Raychaudhuri equations (8) diverges together with σ_v. It is reached at a finite advanced time provided r decreases to $(2|m|\ell^2)^{1/3}$, which the dynamics decides: $|m|$ is increasing along the ray and, for $\rho > 0$, r is decreasing, so the two meet unless r stops at a value above $(2|m|\ell^2)^{1/3}$ first. Near the inner horizon $r \approx r_- = 1.06$, so the singularity forms when $|m| \simeq r_-^3/2\ell^2 = 0.59$, i.e. when m has changed by about 5.6 from its static value 5. This branch needs $y < 0$, which data regular on the event horizon exclude (Proposition 1).

Both statements are consequences of (16) and the range of h; they hold on any ray. The singular case is not a numerical failure: it is the end of the range of h.

Bounded h**.** $h = \psi/\sqrt{1-\alpha^2\psi^2}$ maps $(-1/\alpha, 1/\alpha)$ onto all of $\mathbb{R}$. So for every $m(v) \in \mathbb{R}$ and $r > 0$ there is exactly one $\psi \in (-1/\alpha, 1/\alpha)$: $\psi \to \pm 1/\alpha$ as $m/r^3 \to \pm\infty$, never reached at finite m. The walls are invariant ($\psi_v = 3\rho\psi s^2/r + 16\pi y\phi_v^2 s^3/r$ with $s = \sqrt{1-\alpha^2\psi^2}$ vanishes there) and nothing diverges: mass inflation of $|m|$ of either sign ends at a wall, with $M = r^3\psi/2$ bounded by $r^3/2\alpha$ and σ_v finite.

What this says. Two characteristic functions that agree with Einstein gravity at small ψ ($\psi \ll 1/\ell^2$) differ qualitatively in what the inflating generalized mass does to the geometry when the mass can decrease: the Hayward h ends the negative-mass side in a curvature singularity at finite radius, the bounded h does not. The positive-mass side, which is the one that data regular on the event horizon select (Proposition 1), behaves in both as an inflation of m with ψ approaching the pole or the wall without reaching it. This is a statement about the ray equations; it uses neither the Ori shell nor the additive closure of Paper E [Ownby, 2026a]. In Paper E's additive closure the Misner–Sharp mass inflates without bound on both sides at finite radius (the sign-mechanism and curvature sections there).

8 Event-horizon-regular data in the theory

Paper G [Ownby, 2026b] builds data regular on the event horizon for the additive closure from three ingredients: the exact static solution in Kruskal-type coordinates, the teleological data on the horizon, and the integration of the transverse equations along $v = 0$. Each carries over to an arbitrary h, because in this theory the static metric is a function f with simple zeros $r_- < r_+$ and the dynamical equations enter only through $Q(\psi)$ and $c_1 = 1/h'$. The domain is $U \geq 0$, the interior of the event horizon, with $v \geq 0$.

(a) Static solution. With $\kappa_\pm = |f'(r_\pm)|/2$ and $x = U e^{\kappa_+ v}$, the static metric $-f\,du_s\,dv$ with $U = e^{-\kappa_+ u_s}$ has

$$x\frac{dr}{dx} = \frac{f(r)}{2\kappa_+}, \quad \widetilde{W} = -\frac{f}{\kappa_+ U}, \quad r_U = -\tfrac{1}{2}\widetilde{W}, \quad r_v = \tfrac{1}{2}f, \quad \tilde{\sigma}_v = \tfrac{1}{2}f', \quad \tilde{\sigma}_U = \frac{1}{U}\Big(\frac{f'}{2\kappa_+} - 1\Big), \tag{18}$$

with $r \to r_+$ as $x \to 0$ (so $y = -r_U/\widetilde{W} = \frac{1}{2}$ on $\mathcal{H}^+$) and $r \to r_-$ as $x \to \infty$. For $x < 1$ we write $r = r_+ + z$, $z = -x\,w(x)$ with w regular, and for $x \geq 1$ $r = r_- + e^L$ with L integrated in $\ln x$; the divided differences $f(r_+ + z)/z$, $[f'(r_+ + z) - f'(r_+)]/z$ and $f(r_- + \rho)/\rho$ that appear are evaluated from Taylor series of order 14 near the zeros (truncated-Taylor arithmetic in double precision; no symbolic or multiprecision library is needed), so that every field keeps full relative precision down to $x \sim 10^{-12}$ and $\rho = r - r_- \sim 10^{-76}$. The implementation (`static_gen.py`) accepts any f and is checked against the Hayward polynomial implementation of Paper E [Ownby, 2026a]: the maximum relative difference in $(r, \sigma, r_U, r_v, \sigma_U, \sigma_v)$ over six points up to $v = 98$ is 5.8×10^{-13} (`static_gen.log`).

(b) Teleological horizon data. On $\mathcal{H}^+$ we impose $y = \frac{1}{2}$, i.e. $r_U = -W/2$, so that $\psi_H = (1 - 2r_v)/r^2$, and the ray equations (8)–(9) reduce to

$$\sigma_v = \tfrac{1}{2}\,r\,Q(\psi_H), \qquad r_{vv} = \sigma_v r_v - \frac{4\pi r}{h'(\psi_H)}\phi_v^2, \tag{19}$$

which for $h = \psi$ is the horizon equation of Paper E. The tail $\phi(0, v) = B(1 + v)^{-k}$ is prescribed on $\mathcal{H}^+$. The event horizon is the one on which $r_v \to 0$ at late times: we integrate (19) backwards from $v = v_1 + 300$ with $r_v = 0$ and a radius r_f fixed by the condition $r(0) = r_+$ (shooting, tolerance 10^{-16}). The final radius is $r_f = r_H(v_1) > r_+$ for $B \neq 0$, and the generalized mass accumulated on the horizon is

$$m_f = \tfrac{1}{2}\,r_f^3\,h(r_f^{-2}), \tag{20}$$

which is $r_f/[2(1 - \ell^2/r_f^2)]$ for the Hayward h and $\frac{1}{2}r_f^3\psi_f/\sqrt{1 - \alpha^2\psi_f^2}$, $\psi_f = r_f^{-2}$, for the bounded h. For the Hayward h, $m = 5$ and $\ell = 1$ this gives $m_f = 5.000001$, 5.000099 and 5.00996 for $B = 3 \times 10^{-4}$, 3×10^{-3} and 3×10^{-2} (`kap_check_m1_B*.log`).

(c) Initial row. On $v = 0$ the fields r, σ, r_U, σ_U are the exact static ones (18); r_v, σ_v and ϕ_v follow from the transverse equations in U (the u-derivatives of (8)–(11)) with $\phi_U = 0$ on $v = 0$ (no ingoing perturbation) and the horizon values as the starting point, integrated to 10^{-12} (`dnU_qtg.py`). The data on $\mathcal{H}^+$ and on $v = 0$ then agree at the corner.

(d) Grid. The evolution is performed in (s, v) with $U = c\,(e^{\kappa_+ s} - 1)/\kappa_+$, $c = 0.2\,e^{-\kappa_+ v_1}$, uniform in s and v ($h_s = h_v = h$), which resolves the layer $x = U e^{\kappa_+ v}$ at all v with a fixed number of points; the fields are transformed ($\sigma_s = \tilde{\sigma} + \ln g'$, $r_s = r_U g'$, $\phi_s = \phi_U g'$, $g' = c\,e^{\kappa_+ s}$) so that the equations keep their form, and $y, q, \psi, g_\phi = \phi_u\phi_v/W$ are invariant. We store four rays at $U \simeq 0.3$, 0.6, 1.0, 1.2.

9 A constrained evolution

The free evolution fails. The scheme of Paper E [Ownby, 2026a] evolves r, σ, ϕ and the first derivatives $r_u, r_v, \sigma_u, \sigma_v, \phi_u, \phi_v$ by a predictor–corrector on the null diamond, using (9)–(11) only; the Raychaudhuri equations (8) are constraints. On the exact static solution of Section 8(a) (so

$B = 0$ and the data are exact) on $U \le 3$, $v \le 100$, with four meshes $h = 0.1$, 0.05, 0.025, 0.0125, this free evolution is second-order convergent at early times (extrapolated error in ψ at $v = 40$: 6×10^{-14} on $U \simeq 0.3$, 8×10^{-11} on $U \simeq 0.6$, but 2×10^{-8} and 1.3×10^{-6} on $U \simeq 1.0$ and 1.2), and then loses all accuracy, earlier on the rays with larger U (on $U \gtrsim 0.6$ by $v = 60$, on $U \simeq 0.3$ by $v = 80$; Table 1): the error in ψ on $U \simeq 0.6$ at $v = 60$ is 1.3×10^{2}, 3.5, 0.21, 1.3×10^{-2} on the four meshes, the solution is non-finite on $U \simeq 0.6$ at $v = 80$, and at $v = 98$ on every ray and mesh (Table 1, `t_static_free_U3.log`). Because the data are exact, this is a failure of the discretization, not of the data; we have not analyzed its mechanism. The cure below removes the evolved first derivatives.

The constrained scheme. We keep $\sigma, \sigma_u, \sigma_v, \phi, \phi_u, \phi_v$ on the diamond scheme and replace the evolution of r, r_u, r_v by the integrals of (17):

1. y along u at fixed v, starting from $y = \frac{1}{2}$ on $\mathcal{H}^+$: $y_{i+1} = y_i + \frac{1}{2} h_s(\gamma_i + \gamma_{i+1})$, $\gamma = 4\pi r c_1 \phi_s^2 / W$ (trapezoid rule);
2. q along v at fixed u, starting from the initial row: $q^{j+1} = q^j + \frac{1}{2} h_v(\gamma_v^j + \gamma_v^{j+1})$, $\gamma_v = 4\pi r c_1 \phi_v^2 / W$;
3. r along v from $r_v = -qW$: $r^{j+1} = r^j + \frac{1}{2} h_v(r_v^j + r_v^{j+1})$, so that r is not evolved through r_{uv};
4. $\psi = (1 + 4yqW)/r^2$, which enters c_1 and Q; the implicit coupling is resolved by four fixed-point sweeps inside each cell, and the diamond corrector for $(\sigma, \sigma_u, \sigma_v, \phi_u, \phi_v)$ is iterated three times.

The scheme is second order and streams (memory $O(n_u)$; `qtg_dn.c`, `evolve_stream_qtg_cy`, mode 3). Two intermediate variants, mode 1 (project r_u through y only) and mode 2 (project r_u and r_v), are also in the code; on static data they give extrapolated errors of at most 1.1×10^{-9} and 1.1×10^{-8}, against 4.7×10^{-11} for mode 3, which we use throughout. By construction y and q are non-decreasing, as Proposition 1 requires: the discrete solution cannot violate the sign theorem.

Validation. (i) *Static data.* Table 1 gives the error in ψ against the exact solution on $U \simeq 0.3$ and $U \simeq 1.0$: second order in h, saturated rather than growing in v, and $\le 3 \times 10^{-12}$ after Richardson–Romberg extrapolation in h^2 over four meshes; over all four rays, four times and the three variables (r, σ_v, ψ) the largest extrapolated error is 4.7×10^{-11} (`t_static_cy_mode3.log`). (ii) *Free and constrained evolutions on tail data* ($B = 3 \times 10^{-3}$, meshes $h = 0.05 \ldots 0.00625$): the two differ by $O(h^2)$ at all times at which the free evolution is still usable, e.g. on $U \simeq 1$ at $v = 30$ the difference in σ is 4.8×10^{-4}, 1.2×10^{-4}, 3.0×10^{-5}, 7.5×10^{-6}, and at $v = 40$ 1.0×10^{-2}, 2.5×10^{-3}, 6.3×10^{-4}, 1.6×10^{-4} (`free_vs_con.log`). (iii) *An equation that is not used.* In mode 3 the r_{uv} equation (9) does not update r. Its finite-difference residual on stored rows $v = \text{const}$ is 1.1×10^{-3}, 2.7×10^{-4}, 6.7×10^{-5} (relative L^2, meshes $h = 0.05$, 0.025, 0.0125) at $v = 40$ and 3.3×10^{-3}, 8.9×10^{-4}, 2.6×10^{-4} at $v = 100$: second order, through the front layer of Section 12 (`check_ruv_B0.003.log`). This tests σ_{uv}, including its $\phi_u^2 \phi_v^2$ term, together with the Raychaudhuri integrals, where the flux terms are large (behind the front $r_{uv} \sim 10^{-15}$ is below the roundoff of the finite difference and the test is uninformative there). (iv) *Bounded h.* The same static test with $m = 1.5$, $\alpha = 1$ gives an extrapolated error $\le 4.1 \times 10^{-11}$ (`t_static_b_a1_M1.5.log`). It fails for $m = 5$, $\alpha = 1$, where $\psi(r_-)$ is within 5×10^{-3} of the wall (extrapolated error in σ_v of 6.5×10^{-3} at $v = 98$), and that case is not used. (v) *Two discretizations of the same equations.* Mode 2 advances r with the r_{uv} equation (9), mode 3 does not use it; both integrate the Raychaudhuri equations for r_u and r_v. On the tail data ($B = 3 \times 10^{-3}$, $v_1 = 110$, four isotropic meshes $h = 0.05, \ldots, 0.00625$, Romberg) the two extrapolated solutions on $U \simeq 1$ differ by $\leq 7.3\times10^{-9}$ in σ_v at $v = 30, 40, \ldots, 105$ (3.3×10^{-9} at $v = 105$), by 1.6×10^{-6} (relative) in $z = 1 - \ell^2\psi$ at $v = 105$ ($z = 4.8 \times 10^{-3}$) and by 9×10^{-10} in r (`modecmp.log`), through the front ($v \simeq 70$) and 35 units of v behind it, where Σ is dominated by the c_3 term. This tests the Raychaudhuri-integrated solution against the r_{uv} equation where the flux terms are large, including a region where the finite-difference residual of item (iii) is below round-off. (vi) *Sensitivity to* c_3*.* The agreement of (v) is not sensitive to the coefficient of the c_3 term and therefore does not test it. Replacing c_3 by 0.9 and 1.1 times itself changes σ_v on $U \simeq 1$ at $v = 70$ from -0.437 to -0.463 and -0.413 (unextrapolated, $h = 0.025$) and the modes differ by the same mesh error for all values ($|\delta z/z| = 0.8$–2.6×10^{-2} at that mesh), and with $c_3 = 0$ there is no front ($\sigma_v = -0.795$ at $v = 90$, `c3_sensitivity.log`): the rise of σ_v is driven by the c_3 term, and what fixes its coefficient is the reduced action (Section 4(f)), equivalently the compatibility condition of Section 4(e), repeated in 80-digit decimal arithmetic for states like those behind the front ($1 - \ell^2\psi$ from 10^{-6} to 10^{-1}, W down to 10^{-30}, g_ϕ up to 10^{12}, 2000 states per decade): the relative defect of σ_{uv} is $\leq 7.1 \times 10^{-50}$ (`check_compat_pole.log`).

Table 1: Error in ψ against the exact static solution at the finest mesh $h = 0.0125$ ($B = 0$, $\ell = 1$, $m = 5$), for the free evolution of Paper E and the constrained evolution, and the constrained evolution after Richardson–Romberg extrapolation over four meshes (`t_static_free_U3.log`, `t_static_cy_mode3.log`). nf: non-finite.

ray	evolution	$v = 40$	$v = 60$	$v = 80$	$v = 98$
$U \simeq 0.3$	free	2.5×10^{-9}	2.1×10^{-5}	2.0	nf
	constrained	1.3×10^{-9}	5.7×10^{-5}	5.8×10^{-5}	5.8×10^{-5}
	constrained, extrapolated	4×10^{-14}	2.5×10^{-12}	2.9×10^{-12}	2.9×10^{-12}
$U \simeq 1.0$	free	2.9×10^{-5}	5.6×10^{1}	nf	nf
	constrained	5.4×10^{-5}	5.4×10^{-5}	5.4×10^{-5}	5.4×10^{-5}
	constrained, extrapolated	1.9×10^{-12}	1.9×10^{-12}	1.9×10^{-12}	1.9×10^{-12}

Extrapolation. The production runs use Richardson–Romberg extrapolation in the step (the T-table of h^2-extrapolation; the top level is quoted, and its difference from the next-lower level of the table is the error estimate). Its accuracy is limited in v; the next two paragraphs say by what.

What limits the window. With four isotropic meshes $h_s = h_v = h = 0.05, \ldots, 0.00625$ the extrapolated solution is accurate only up to a window in v. For $B = 3 \times 10^{-3}$ the inter-level difference of σ_v on $U \simeq 0.3$ is 3×10^{-10} at $v = 80$, 1.4×10^{-6} at 160, 3.4×10^{-5} at 180 and 2.2×10^{-4} at 190 (`endpoint_m1_B0.003_v200.log`); for $B = 3 \times 10^{-2}$ the inter-level difference of Σ on $U \simeq 1$ is 1.0×10^{-2} (relative) at $v = 90$ and 0.19 at $v = 100$, and the run is non-finite at $v = 120$; for $B = 3 \times 10^{-4}$ the run ends at $v = 110$ inside the window. We have measured how the limit depends on the discretization. Every run becomes non-finite at an advanced time v_b. For $B = 3 \times 10^{-3}$, v_b depends on h_v alone and shifts by the same amount at each halving: the first non-finite v is 205.25, 233.58, 261.9, 290.2 for $h_v = 0.05$, 0.025, 0.0125, 0.00625 (`blow1.log`–`blow3.log`) and, at integer sampling, 319, 347 for $h_v = 0.003125$, 0.0015625 (Table 2); the shift is 28.3 per halving, and $\ln 4/\kappa_+ = 28.31$. It does not depend on h_s ($h_s = 0.0125$, $h_v = 0.05$: 205.3; $h_s = 0.05$, $h_v = 0.0125$: 261.85; `vb1.log`) nor on the end v_1 of the grid ($v_1 = 250$ and 400: 205.25 in both). It depends on the amplitude: for $B = 3 \times 10^{-2}$ the first non-finite v is 169, 197, 225, 254 at $h_v = 0.0125, \ldots, 0.0015625$, earlier than for $B = 3 \times 10^{-3}$ by 93, and $93 \simeq (2/\kappa_+) \ln 10 = 94.0$; for $B = 3 \times 10^{-4}$ no level fails before the end of the grid ($v = 298$), as the same law implies ($v_b \geq 356$

at $h_v = 0.0125$). The data of Table 2 are described to 1.5 units of v (the sampling is integer) by

$$v_b = 205.25 + \frac{2}{\kappa_+} \ln \frac{1.5 \times 10^{-4}}{B\, h_v}, \tag{21}$$

that is, the truncation error grows as $\epsilon \sim C\,(B\,h_v)^2\, e^{\kappa_+ v}$ with C independent of h_s, and the run fails when ϵ reaches a fixed size. The factor B^2 is the square of the flux, as for the source of the Raychaudhuri equation. We found (21) by measurement. The rate κ_+ is the rate at which a perturbation of r_v on the horizon grows under forward integration of the horizon equation (19) ($\delta r_{vv} = \sigma_v \delta r_v$ with $\sigma_v = \kappa_+$ there; equivalently the separation $x = U e^{\kappa_+ v}$ of neighboring rays from the horizon), and the dependence on the order of the scheme predicted by this reading, a shift of $p \ln 2/\kappa_+$ per halving of h_v for a scheme of order p, is confirmed by an independent fourth-order code (Section 10); the prefactor and the dependence on B are derived, for the fourth-order code, in Section 11. The relative error of $\rho = -r_v$ behind the front, between $h = 0.05$ and 0.025, is 6.6×10^{-3}, 1.4×10^{-2} and 0.12 at $v = 100, 150, 180$ on $U = 1.5$ (`errprof.log`); behind the front σ_v is the logarithmic derivative of ρ (Section 12.5), so a growing error in ρ ends the window.

Table 2: First non-finite sample v of the run on $U \simeq 1$ ($h_s = 0.05$; sampled at integer v) and Eq. (21) (`vb_law.log`; the first three values for $B = 3 \times 10^{-3}$ at $h_v \geq 0.0125$ are 205.25, 233.58, 261.9 at the sampling of the step, `blow1.log`–`blow3.log`). For $B = 3 \times 10^{-4}$ no level fails before the end of the grid.

	h_v					
B	0.05	0.025	0.0125	0.00625	0.003125	0.0015625
3×10^{-3} measured	206	234	262	291	319	347
Eq. (21)	205.3	233.6	261.9	290.2	318.5	346.8
3×10^{-2} measured	–	–	169	197	225	254
Eq. (21)	111.2	139.5	167.8	196.1	224.4	252.8
3×10^{-4} measured	–	–	> 298	> 298	> 298	> 298
Eq. (21)	299.3	327.6	355.9	384.2	412.5	440.8

Refining the advanced-time step alone. Because v_b is fixed by $B\,h_v$, only h_v needs to be refined. We keep $h_s = 0.05$ and halve h_v from 0.05 to 0.0015625 (six levels, grid to $v_1 = 400$, `gen_q2.py`; the finest level has 2.6×10^5 steps and took 1370 s), and apply the Romberg extrapolation in h_v^2 to the finest $K \leq 4$ levels that are finite at least 30 units of v beyond the point considered (a level is used only where its error has not amplified by more than $e^{30\kappa_+} = 4.3$ toward its own blow-up). The error estimate is the difference between the top of the table and the extrapolation from the finest $K - 1$ levels. For $B = 3 \times 10^{-3}$ this gives four levels to $v = 230$ and two to $v \simeq 280$ with the levels above (more levels in Section 12.6). The discretization error in h_s then remains as a smooth systematic. We measured it with a second family at $h_s = 0.025$ (finest level $h_v = 0.003125$): σ_v on $U \simeq 1$ of the $h_s = 0.05$ family is more negative by 1.2×10^{-4} than that of the $h_s = 0.025$ family for $150 \leq v \leq 220$ (constant to 2%), so by 1.6×10^{-4} than the $h_s \to 0$ limit (assuming $O(h_s^2)$), and the $h_s = 0.025$ family by 4×10^{-5} (`hs_extrap.log`). The finite-difference derivative $\partial_v \sigma$ of the stored σ is less sensitive: at $h_s = 0.05$ it differs from the $h_s \to 0$ value of σ_v by 3×10^{-6} (-0.043838 against -0.043835 at $v = 200$), while the stored σ_v differs from it by 1.6×10^{-4} (`rho_check.log`). For $|\sigma_v| \gtrsim 10^{-2}$ the h_s-systematic is below 2% of σ_v and we quote the $h_s = 0.05$ values unless stated; near $\sigma_v \to 0$ it would set the floor of the accuracy, 1.6×10^{-4}, which is reached at $v \approx 3 \times 10^4$ if $\sigma_v \simeq -5/v$ (Section 12.6). It does not close the window.

10 A second implementation

The constrained scheme of Section 9 (code A) is one discretization. Its validation (Section 9) tests the solution against exact static data and against the r_{uv} equation, but a coding error that also appears in the data and the evolution could survive it. We therefore wrote a second code (code B; `code/eh_qtg/codeb/qtg_b.c`) that shares the equations and the data construction (`dnU_qtg.py`) but nothing else: the primary variables are different, the discretization is different, and the order is four.

Code B. The method of lines in v. A row $v = \text{const}$ carries $(r, \sigma, \phi_s, y, q)$ on the s grid of Section 8(d); $r_v = -qW$, $q_v = 4\pi r c_1 \phi_v^2/W$ and $\phi_{s,v}$, $y_v = \frac{1}{4}rQ - \sigma_v y$ (from (14)) are the evolution equations, and σ_v, ϕ_v are not evolved but obtained on every row by integrating in s from the horizon column, $\partial_s\sigma_v = \frac{1}{2}W\psi - Wc_2 - 8\pi c_1\phi_s\phi_v + 64\pi^2 c_3\phi_s^2\phi_v^2/W$ and $\partial_s\phi_v = (yW\phi_v + qW\phi_s)/r$ (the u-derivatives of σ_v and ϕ_v from (10) and (11)). Both the s-integration and the time stepping are Runge–Kutta of order four (classical RK4 in v with the horizon column evaluated at the half-step; RK4 along s with cubic interpolation of the row to the mid-points), so the scheme is not a diamond scheme and does not use r_{uv} or σ_u. In the constrained version (the one used below) y is recomputed on each Runge–Kutta stage from $\partial_s y = 4\pi r c_1\phi_s^2/W$, $y = \frac{1}{2}$ on $\mathcal{H}^+$ (Eq. (17)); the unconstrained version evolves y by its ODE.

Tests of code B. (i) *Exact static data* ($B = 0$, same test as Table 1, $h_s = h_v = h$): without extrapolation the maximum error over the four rays at $v = 98$ is 1.1×10^{-6}, 6.9×10^{-8}, 4.3×10^{-9} in r, 1.5×10^{-5}, 9.4×10^{-7}, 5.9×10^{-8} in σ_v and 1.9×10^{-6}, 1.2×10^{-7}, 7.3×10^{-9} in ψ for $h = 0.1$, 0.05, 0.025: observed order 4.00 (from the last pair of meshes) in every quantity at every time (`t_static_b4.log`); the unconstrained version has the same order (`t_static_b4_proj0.log`). These data have no flux, so the test does not exercise the $\phi_u\phi_v$ terms. (ii) *Mesh dependence on tail data* ($B = 3 \times 10^{-3}$, $k = \frac{3}{2}$, ray $U \simeq 1$, $h_v = 0.05$): the (relative) difference between $h_s = 0.05$ and $h_s = 0.0125$ is 1.6×10^{-8} in r (and 1.5×10^{-8} between 0.05 and 0.025) and 5.9×10^{-8} in σ_v at $v = 60$; the errors of $h_s = 0.05$ and $h_s = 0.025$ against $h_s = 0.0125$ are in the ratio 17.0 (the fourth-order value) in r, ϕ_v and σ_v for $v \lesssim 200$, and the clean ratio is lost beyond that (4.5 at $v = 250$, 3.2 at 300 in σ_v) where the truncation error in h_v of the $h_v = 0.05$ runs takes over (`cmp_hs.log`). The h_s-systematic of code A (-1.6×10^{-4} in σ_v, Section 9) is therefore absent in B at $h_s = 0.05$ (`cmp_hs.py`). (iii) *The error law* (Eq. (21)). The first non-finite v of B at $h_s = 0.05$ is 252, 307, 364, 420, 477 for $h_v = 0.2$, 0.1, 0.05, 0.025, 0.0125, a shift of 55, 57, 56, 57 per halving, against $p \ln 2/\kappa_+ = 56.6$ for $p = 4$ (28.3 for code A, $p = 2$); the blow-up does not depend on the end of the grid ($v_1 = 360$ and 400: 307 at $h_v = 0.1$). The usable range of a single level ends about 80–100 units of v before its blow-up: at $h_v = 0.0125$ the change of σ_v from the next coarser level is 2×10^{-7} at $v = 200$ and 2.2×10^{-5} at 300, and the change of K is 2.4×10^{-3} (relative) at 300, 1.8×10^{-2} at 340 and 0.15 at 380, so that, taking the error of the finer level as 1/15 of the change, the window of B at $h_v = 0.0125$ is $v \lesssim 380$ for K to 1% (one run, 627 s on one core, `gen_b_v600.log`, against 1370 s for the finest level alone of code A, whose Romberg window ends at $v \simeq 300$). (iv) *The unconstrained version* (with y evolved by its ODE rather than recomputed from the constraint) is accurate on static data (i) but fails on the tail data: it is non-finite at $v = 70$, 95, 112 for $h_v = 0.1$, 0.05, 0.025, against 307, 364, 420 for the constrained version at the same h_v (`cmp_table.log`). A related instability of the unconstrained evolution therefore appears in both codes, although the pattern differs (code A's free evolution fails already on exact static data by $v \simeq 60$–80, code B's unconstrained version is accurate on static data to $v = 98$ and fails on tail data); the mechanism remains unanalyzed.

Comparison with code A. Code A: $h_s = 0.05$ and 0.025 families, Romberg in h_v^2 over the finest levels that are finite 30 units of v beyond the point, then the $h_s \to 0$ extrapolation of Section 9; code B: $h_s = 0.05$, $h_v = 0.05$ (the same ray, $U = 0.99928$). The relative differences of the two codes for the $v_1 = 400$ run are 1.0×10^{-7} in r at $40 \leq v \leq 250$ and, in σ_v, 4.6×10^{-7}, 6.3×10^{-7}, 2.5×10^{-7}, 6.2×10^{-6} and 8.1×10^{-5} at $v = 40, 60, 100, 150, 200$; in ϕ_v 3.8×10^{-7} to 5.9×10^{-7} for $v \leq 100$ and 1.6×10^{-5} at 200. The differences at $v \geq 150$ are the size of the error of B at $h_v = 0.05$ itself (the change from $h_v = 0.1$, divided by 15, is 6×10^{-6} and 7×10^{-5} in σ_v at $v = 150, 200$). Table 3 compares the finest levels of the two codes up to $v = 300$: with the h_s correction of Section 9, $\sigma_v^{\mathrm{A}} + 1.6 \times 10^{-4}$, the two agree to 4×10^{-6} (absolute) for $120 \leq v \leq 240$ and 1.3×10^{-5} at 300 (a constant shift is the weakest form of the correction: at $v = 100$ the difference is 1.0×10^{-5}, whereas the $h_s \to 0$ extrapolation of the two code A families agrees with B to 6×10^{-7} relative) (the rays differ by 1.2×10^{-8} in U; σ_v is independent of U there to 10^{-4}, Section 12.5), which confirms the extrapolated values of Table 10 and the size of the h_s-systematic, measured in Section 9 by a second mesh family. The error estimate $\delta\sigma_v$ of Table 10 (difference of the top two Romberg levels) is 5×10^{-9} and 4×10^{-8} at $v = 260$ and 280, three orders below the gap to code B there (5×10^{-6}, 7×10^{-6}): near the end of the window it is not a bound. Code B was run only for $B = 3 \times 10^{-3}$, $k = \frac{3}{2}$ and the ray $U \simeq 1$.

Table 3: σ_v on $U \simeq 1$ ($B = 3 \times 10^{-3}$, $k = \frac{3}{2}$) from code A (Table 10, $h_s = 0.05$), the same plus the $h_s \to 0$ correction $+1.6 \times 10^{-4}$, and from code B ($h_s = 0.05$, $h_v = 0.0125$; `cmp_AB_final.log`, `codeb/data/`). Δ_{AB}: B minus corrected A; Δ_h: $\sigma_v^B(h_v = 0.0125) - \sigma_v^B(h_v = 0.025)$ (the error of the finer level is about $\Delta_h/15$).

v	A	A $+1.6\times10^{-4}$	B	Δ_{AB}	Δ_h
100	-0.094251	-0.094091	-0.094081	$+1.0\times10^{-5}$	1×10^{-9}
140	-0.079755	-0.079595	-0.079593	$+2.5\times10^{-6}$	-2×10^{-8}
180	-0.056255	-0.056095	-0.056092	$+3.4\times10^{-6}$	-9×10^{-8}
220	-0.034253	-0.034093	-0.034090	$+3.5\times10^{-6}$	-5×10^{-7}
260	-0.023090	-0.022930	-0.022925	$+5.4\times10^{-6}$	-3×10^{-6}
300	-0.018223	-0.018063	-0.018050	$+1.3\times10^{-5}$	-2×10^{-5}

What this does and does not show. The two codes share the equations, the static data and the data construction, so their agreement does not test the equations (that is the task of Section 4) or the data; it tests the evolution scheme, the h_s-systematic, the extrapolation procedure of code A and the error law, whose dependence on the order of the scheme (shift per halving of h_v: 28.3 at second order, 56.6 at fourth) is predicted correctly. The remaining unchecked step is a code in a different gauge or with different variables for the equations themselves (Section 15).

11 Why the window closes, and how to remove it

Sections 9 and 10 established the error law (21) by measurement. This section derives it, tests the derivation against runs that were not used to build it, and uses it to remove the limit.

11.1 The mechanism

Every row of code B (and, in the diamond form, of code A) advances r_v in v by the Raychaudhuri equation $r_{vv} = \sigma_v r_v - S$, $S = 4\pi r c_1 \phi_v^2$, whatever its distance from the horizon. On the horizon column this is Eq. (19) with $\sigma_v = \kappa_+$ at late times. Its homogeneous solution grows as $e^{\kappa_+ v}$; the solution that is regular at $v \to \infty$ (the event horizon) is the particular solution selected by the final condition and is, for the forward integration, an unstable one. This is the teleological character of the event horizon, which is why event-horizon finders integrate backward in time [Libson et al., 1996, Diener, 2003], and why Section 8 builds the horizon column by a backward shot. The column supplies $r_v^H(v)$ exactly, but a row near the horizon is advanced forward from $v = 0$: the truncation error committed at early v in each step is carried by the homogeneous solution and amplified as $e^{\kappa_+(v-v')}$. The error is $\epsilon_p(v) \simeq C_p B^2 h_v^p \, e^{\kappa_+ v}$ for a scheme of order p: h_v^p is the truncation error per unit v, and B^2 is the size of the source $S \propto \phi_v^2$ that is being integrated.

A one-dimensional model. To test this reading, without any of the machinery of the two-dimensional code, we integrate forward the horizon equation (19) for $y \equiv r_v$,

$$y' = \sigma_v^H(v)\, y - 4\pi\, r_H(v)\, c_1(\psi)\, \phi_v^H(v)^2, \qquad \psi = \frac{1-2y}{r_H(v)^2}, \tag{22}$$

with the exact column $(r_H, \sigma_v^H, \phi_v^H)$ of the data (at the grid and half-step values that code B uses), by classical RK4 ($p = 4$) or by the midpoint rule ($p = 2$) with the step h_v, starting from the exact $y(0) = r_v^H(0)$, and compare with $r_v^H(v)$ (`mech_vb.py`, `codee/`). Table 4 gives the error. It scales as h_v^p (ratio per halving 15.1–16.0 for $p = 4$ and 3.8–4.0 for $p = 2$), and between $v = 100$ and $v = 250$ it grows by 1548, against $e^{150\kappa_+} = 1549$. With $B = 3 \times 10^{-3}$, $\epsilon_4 = 20.4\, B^2 h_v^4 e^{\kappa_+ v}$ and $\epsilon_2 = 54\, B^2 h_v^2 e^{\kappa_+ v}$ (fine steps, $v = 100$); each halving of h_v lowers the error at fixed v by 2^p, which delays the blow-up by $p \ln 2/\kappa_+$: 28.3 for $p = 2$ and 56.6 for $p = 4$, as measured in Sections 9 and 10.

Table 4: One-dimensional model (22) ($B = 3 \times 10^{-3}$, $k = \frac{3}{2}$, $m = 5$): error $|y - r_v^H|$ of the forward integration of the horizon equation by RK4 ($p = 4$) and the midpoint rule ($p = 2$) at $v = 100$ and 250, and the ratio to the previous step (`mech_vb_B3e-3.log`). The error grows between the two times by 1548 ($e^{150\kappa_+} = 1549$).

	$p = 4$			$p = 2$		
h_v	$v = 100$	$v = 250$	ratio	$v = 100$	$v = 250$	ratio
0.2	3.63×10^{-5}	5.62×10^{-2}	–	2.46×10^{-3}	3.80	–
0.1	2.41×10^{-6}	3.73×10^{-3}	15.1	6.42×10^{-4}	0.994	3.8
0.05	1.53×10^{-7}	2.37×10^{-4}	15.7	1.63×10^{-4}	0.252	3.9
0.025	9.59×10^{-9}	1.49×10^{-5}	15.9	4.08×10^{-5}	6.33×10^{-2}	4.0
0.0125	6.00×10^{-10}	9.30×10^{-7}	16.0	1.02×10^{-5}	1.58×10^{-2}	4.0

Does the model describe the code? Code B fails at $v_b = 252, 307, 364, 420, 477$ for $h_v = 0.2, \ldots, 0.0125$ (Section 10). The model error at these five advanced times is 0.0620, 0.0608, 0.0629, 0.0613, 0.0625: the code becomes non-finite when the error of the model reaches $\delta^* = 0.062 \pm 0.001$, the same value for all five step sizes. This is the only constant fitted. The law

$$v_b = \frac{1}{\kappa_+} \ln \frac{\delta^*}{C_p\, B^2 h_v^p} \tag{23}$$

then predicts, without further input, the dependence on the amplitude. We ran code B (`gen_b.py`, $h_s = 0.05$, $k = \frac{3}{2}$) at two amplitudes not used to calibrate: the first non-finite advanced time is 158, 214, 270 for $h_v = 0.2, 0.1, 0.05$ at $B = 3 \times 10^{-2}$ and 346 for $h_v = 0.2$ at $B = 3 \times 10^{-4}$, against the model's 158.4, 213.9, 270.3 and 346.2 (Table 5; `vb_prediction.log`). The shift with the amplitude, $v_b(3\times10^{-2})-v_b(3\times10^{-3}) = -94,\ -93,\ -94$ for the three steps against $-(2/\kappa_+)\ln 10 = -94.0$, is that of the square of the flux, as the source of the equation requires. For code A (second order, a diamond scheme that is not the midpoint rule) the same form holds with $p=2$ (Eq. (21): δ^*/C_2 inferred from the fit is 5.2×10^{-4}, against 1.1×10^{-3} for the midpoint model; the two schemes are not the same, and only the form is predicted for code A). The prefactor and the B-dependence that Section 9 left open are therefore derived (for the fourth-order code, where the model is exact in its integrator) and the origin of the whole limit is the forward integration of the horizon Raychaudhuri equation. The threshold $\delta^* = 0.062$ itself is empirical: it is the size of δr_v at which the nonlinear couplings of the rows make the scheme non-finite, and we have not derived it.

Table 5: First non-finite advanced time of code B (measured, $h_s = 0.05$) and the prediction (23) of the one-dimensional model with $\delta^* = 0.0619$, fitted to the row $B = 3\times10^{-3}$ only. -: not run.

			h_v		
	0.2	0.1	0.05	0.025	0.0125
$B = 3\times10^{-3}$ measured	252	307	364	420	477
model (calibration)	252.0	307.4	363.7	420.2	476.8
$B = 3\times10^{-2}$ measured	158	214	270	–	–
model (prediction)	158.4	213.9	270.3	326.9	383.6
$B = 3\times10^{-4}$ measured	346	–	–	–	–
model (prediction)	346.2	401.5	457.8	514.2	570.8

11.2 Variables measured from the horizon (code E)

The remedy follows from the mechanism. The exact solution of (19) is known at every v from the backward shot; the amplification acts on the difference between a row and that solution. Code E (`codee/qtg_e.c`; the method of lines of code B, the same equations, the same data) therefore evolves the differences

$$\Delta = r - r_H(v), \qquad D = r_v - r_v^H(v), \tag{24}$$

with $\Delta_v = D$ and $D_v = \sigma_v D + (\sigma_v - \sigma_v^H)\, r_v^H - (S - S_H)$, and builds on every row, by integrating along s from the horizon column, the increments $\mathrm{d}y = y - \frac{1}{2}$, $\mathrm{d}\sigma = \sigma_v - \sigma_v^H$ and $\mathrm{d}\phi = \phi_v - \phi_v^H$ (RK4 along s, as in code B), never as differences of large numbers. The source difference is formed in difference form,

$$\begin{aligned} S - S_H &= 4\pi\Big[\Delta\, c_1\phi_v^2 + r_H\,(c_1 - c_{1,H})\,\phi_v^2 + r_H c_{1,H}\,\mathrm{d}\phi\,(\phi_v + \phi_v^H)\Big], \\ \psi - \psi_H &= (1 - 2r_v^H)\Big(\frac{1}{r^2} - \frac{1}{r_H^2}\Big) - \frac{2D + 4\,\mathrm{d}y\, r_v}{r^2}, \end{aligned} \tag{25}$$

and $c_1 - c_{1,H} = (z + z_H)(z - z_H)$ with $z = 1 - \ell^2\psi$ for the Hayward h (for other h the difference is formed directly). The initial row uses $\Delta_0 = r - r_H(0) = z(U)$, the static $r - r_+$ without cancellation, which vanishes at $s = 0$ (it must be continuous with the column, Section 11.3), and D_0 from integrating the D equation along s (the form of a Raychaudhuri integral in the difference variable) to 10^{-13} relative. The scheme remains fourth order, and y is recomputed from its constraint on every stage (the constrained version of code B). No difference of two nearly equal numbers is formed near the horizon, and the truncation error of a step is an error of the difference ($\propto h_v^4 D$, not $\propto h_v^4 r_v^H$): there is nothing for the homogeneous mode to amplify, since the exact column that carries it is not integrated. A first attempt with a hierarchy of variables (code C) was ill conditioned and abandoned; passing the horizon values through the row (a projection of r_v on the column, tested in a variant of code B) made the error larger, and the version in which the variables are subtracted but the sources are still evaluated as differences of absolute fields (code D) is limited by roundoff growth to $v \simeq 500$ (a limit not re-examined after the correction of Section 11.3; code E is regular beyond 1400).

Tests of code E. (i) *Exact static data* (same test as for code B, $h_s = h_v = h$): the maximum error over four rays at $v = 98$ is 1.2×10^{-5}, 8.0×10^{-7}, 5.0×10^{-8} in r, 2.3×10^{-4}, 1.4×10^{-5}, 9.0×10^{-7} in σ_v and 2.1×10^{-5}, 1.4×10^{-6}, 8.5×10^{-8} in ψ for $h = 0.2$, 0.1, 0.05, observed order 3.98–3.99 at every time (`codee/logs/t_static_e.log`), the size of code B's errors at the same h. (ii) *Convergence on tail data* ($B = 3 \times 10^{-3}$, $k = \frac{3}{2}$, ray $U = 0.9968$, reference $h_s = 0.2$, $h_v = 0.1$): the change of σ_v on halving h_v is 5×10^{-7} and constant in v for $100 \le v \le 1100$ (code B's changes grow like $e^{\kappa_+ v}$), the change on halving h_s is 1.6×10^{-7} at $v = 100$ and decreases to $\le 5 \times 10^{-9}$ for $300 \le v \le 1000$, and the relative change of r is 3.3×10^{-7} (h_v) and 3.9×10^{-6} (h_s), constant in v (`conv_e2.log`). (iii) *Agreement with code B.* Code E ($h_s = 0.1$, $h_v = 0.05$) against code B ($h_s = 0.05$, $h_v = 0.0125$, ray $U = 0.9993$): the relative differences of σ_v are 8.7×10^{-6}, 3.6×10^{-6}, 3.8×10^{-6}, 1.2×10^{-5} at $v = 140$, 180, 220, 260 (the rays differ by 2.4×10^{-3} in U; from the dependence of σ_v on U among the rays of code E this accounts for 7×10^{-6}, 2.5×10^{-6}, 1.6×10^{-6}, 1.1×10^{-6} of them, and for 7×10^{-5} of the 1.0×10^{-4} at $v = 100$), of r 2.6×10^{-7} and of ϕ_v $(5\text{–}7) \times 10^{-7}$ for $v \le 220$; the differences at $v \ge 300$ (8×10^{-5} at 300, 5.6×10^{-3} at 380) are the growth of code B's own error toward its window (Table 3: $\Delta_h = -2 \times 10^{-5}$ at 300). The Kretschmann invariant of code E on its ray is $1.1195 \times 10^{25}\ell^{-4}$ at $v = 100$ against 1.09×10^{25} of code B; K depends on the ray ($\mathrm{d}\ln K/\mathrm{d}U = -9.5$ at $v = 100$ and -5.4 at $v = 300$ between the rays $U = 0.997$ and 1.20 of code E), which accounts for the difference (-2.3% predicted for $\Delta U = 2.4 \times 10^{-3}$ against -2.6% observed, K of code B being quoted to three digits; -1.3% against -1.4% at $v = 300$); all values of K quoted below are on the ray $U = 0.9968$. (iv) *Cost.* The run to $v_1 = 800$ with $h_s = 0.2$ and $h_v = 0.1$ on $U \le 1.5$ takes about one minute on one core (code B at $h_v = 0.0125$ to $v = 380$: 627 s).

11.3 The seed of the late-time departure, and the range of code E

Code E does not fail at v_b of the law (23). In its first version it departed from the smooth continuation beyond $\kappa_+ v \simeq 33$ ($v \simeq 650$–700 for $m/\ell = 5$) and ran into a non-finite state at $v \simeq 800$, whatever h_s or h_v, and for double and long-double arithmetic alike when both used the same horizon data. The cause is not in the scheme or in the arithmetic; it is a defect of the initial row, found as follows.

Search. (i) Double and long-double runs that share the horizon column agree to 10^{-11} in $v\sigma_v$ at $v \le 650$ (an earlier comparison of a double run with $v_1 = 800$ against a long-double run with $v_1 = 1100$ had compared different columns, and its apparent $e^{\kappa_+ v}$ growth was the dependence on v_1). (ii) The departure is independent of h_s, h_v and of $v_1 \ge 950$. (iii) Changing the integrator tolerances that produce the horizon column and the initial row (relative tolerances between 10^{-10} and 2×10^{-14}, against 10^{-12} before; the column then changes by a few ulps only) moves the departure by tens of units of v: $v\sigma_v$ at $v = 740$ is -4.47, -4.94, -5.23. (iv) Replacing, one channel at a time, the column fields r_H, σ_H, r^H_U, r^H_v, σ^H_U, σ^H_v, ϕ^H_U by those of the most accurate column, one at a time, changes nothing visible (five digits of $v\sigma_v$ at $v \le 780$), whereas replacing the initial row does the whole of the change. (v) The initial row was built as $\Delta_0 = \big(r(U,0) - r_+\big) + \big(r_+ - r_H(0)\big)$. The last bracket is the rounding-level mismatch between the exact static radius $r_+ = 9.8981276645\ldots$ and the shooting solution, $r_+ - r_H(0) = 3.55 \times 10^{-15}$ (two ulps), and it changes with the tolerances. It is added at every $s > 0$ but not at $s = 0$, where $\Delta = 0$ by definition: a *step* of 3.6×10^{-15} in Δ_0 between the horizon and every other ray, whereas $\Delta_0 \propto U$ is as small as 10^{-22} on the first rays of the grid. The earlier input-sensitivity tests (relative biases and noise of 10^{-12} in the fields of the row and column, `noise_tests.log`) did not see it because they were multiplicative: they vanish with the field at $s = 0$ and cannot create a step.

Mechanism and test. A step ϵ in Δ_0 is a perturbation of the initial radius of every ray off the horizon that does not vanish with U; it is carried by the homogeneous mode of the r_{vv} equation, $e^{\kappa_+ v}$, that Section 11 identified. Adding to the corrected initial row ($\Delta_0 = -x\,w(x)$, which is continuous at $s = 0$; see below) an offset ϵ at every $s > 0$ gives Table 6: the advanced time at which $v\sigma_v$ departs from the unperturbed run by 0.05 is shifted by -19.8 per e-fold of ϵ (fit over $10^{-15} \le \epsilon \le 10^{-12}$, rms residual 0.2), against $-1/\kappa_+ = -20.4$ expected for an amplification $\propto \epsilon\, e^{\kappa_+ v}$, and the run with $\epsilon = 3.55 \times 10^{-15}$ reproduces the former behavior. Every number in the table is a separate run (`codee/logs/seed_amplitude.log`).

Table 6: Advanced time at which $|v\sigma_v - (v\sigma_v)_{\text{ref}}|$ first exceeds 0.05 and 0.5 on the ray $U = 0.9968$ ($B = 3 \times 10^{-3}$, $k = \frac{3}{2}$, $m = 5$, $h_s = 0.2$, $h_v = 0.1$, $v_1 = 1000$) when a constant offset ϵ is added to Δ_0 at every $s > 0$; the reference is the run with $\epsilon = 0$ ($v_1 = 1200$). $\epsilon = 3.55 \times 10^{-15}$ is the offset $r_+ - r_H(0)$ of the former initial row ($v_1 = 950$).

ϵ	0	10^{-15}	3.55×10^{-15}	10^{-14}	10^{-13}	10^{-12}
v (> 0.05)	none (≤ 880)	747	722	701	656	610
v (> 0.5)	none (≤ 880)	788	763	742	696	650

Correction. The initial row now uses $\Delta_0 = r(U,0) - r_H(0)$ with $r(U,0) = r_+ + z(U)$ and $z = -x\,w(x)$, that is $\Delta_0 = z(U)$, which vanishes at $s = 0$ and needs no knowledge of $r_+ - r_H(0)$: the initial data are then the static data shifted by the (irrelevant) 10^{-15} in r, and they are continuous with the column. The same continuity is automatic for D_0 (integrated from $D = 0$ at $s = 0$) and for the other fields, whose errors are multiplicative. The condition is the discrete form of the statement that data regular on the event horizon must join the horizon column with all their derivatives; the horizon solution is the separatrix of the r equation, and a step of any size is a component along the unstable direction.

Range of code E after the correction. No departure and no non-finite value is found to $v = 1400$ ($\kappa_+ v = 69$, $m/\ell = 5$, $B = 3 \times 10^{-3}$, $k = \frac{3}{2}$; $h_s = 0.2$, $h_v = 0.1$, $v_1 = 1500$), where $v\sigma_v = -5.038$. Tests, all at constant level in v (no growth like $e^{\kappa_+ v}$; `codee/logs/conv_e2.log`): (a) changing v_1 from 1200 to 1500 changes K by $\le 6 \times 10^{-9}$ ($v \le 650$), 4×10^{-7} ($v = 1000$) and 3×10^{-6} ($v = 1100$, within 10^2 of v_1, where the terminal data begin to act); (b) halving h_v changes K by 2×10^{-5}, r by 3×10^{-7} and σ_v by 5×10^{-7} (absolute); halving h_s changes K by 3×10^{-4} (5×10^{-4} at $v = 100$) and r by 4×10^{-6}; (c) long double against double with the same data: K differs by $\le 5 \times 10^{-7}$ and σ_v by $\le 6 \times 10^{-13}$ (absolute) at $v \le 1100$ (the K values are computed from the shipped 12-digit data, which limits this comparison). The window law of Section 11 (code B) is therefore not a limit of code E: the range of code E is set by the discretization error only, which is flat in v, and by the choice of v_1.

12 Results for the Hayward characteristic function

Data: $m = 5$, $\ell = 1$ ($r_- = 1.057475$, $r_+ = 9.897927$, $\kappa_- = 0.795649$, $\kappa_+ = 0.0489687$), tail $\phi(0,v) = B(1+v)^{-3/2}$ on $\mathcal{H}^+$, $B = 3 \times 10^{-4}$, 3×10^{-3}, 3×10^{-2}, runs to $v_1 = 110$ (and $v_1 = 200$ for $B = 3 \times 10^{-3}$, 3×10^{-2}). Figure 1 shows the results.

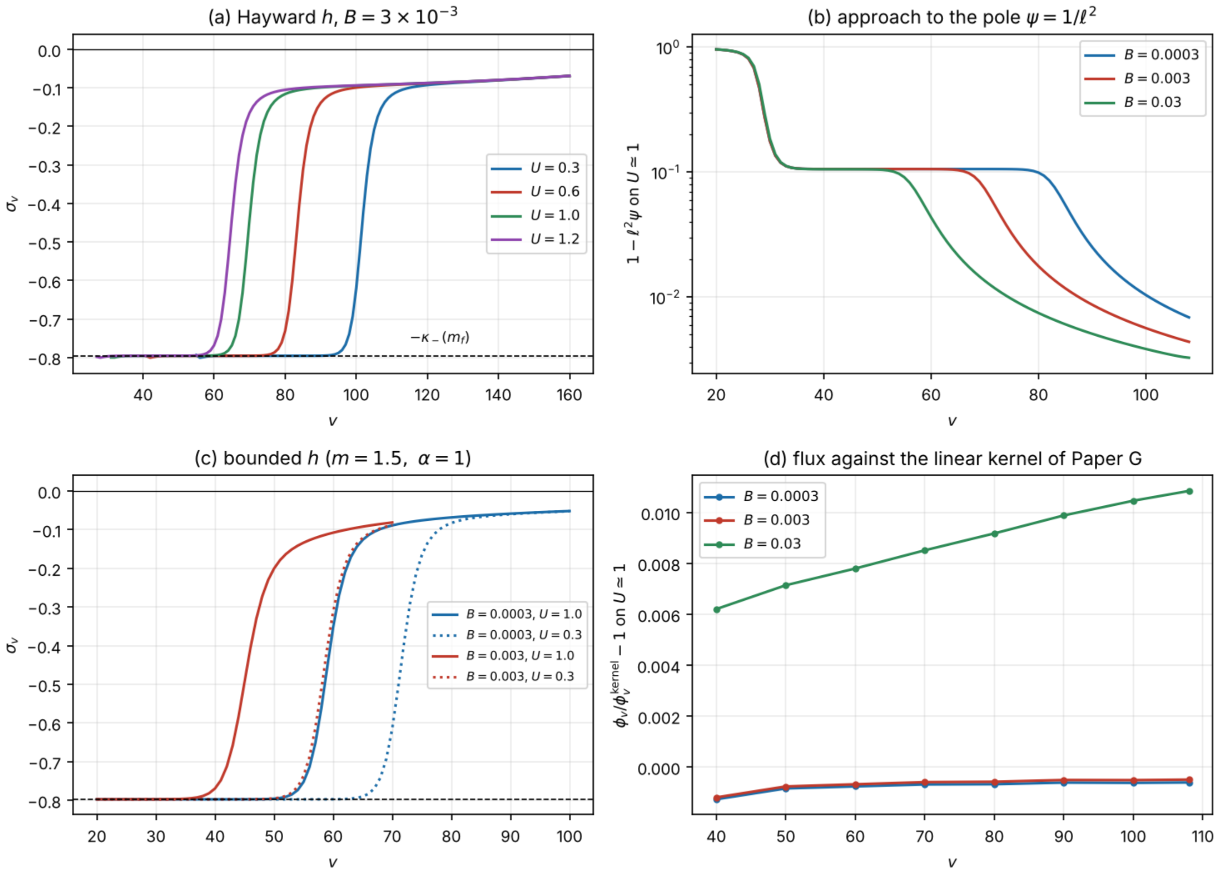

Figure 1: Evolution from event-horizon-regular data. (a) σ_v on four rays for the Hayward h, $B = 3\times10^{-3}$ (Richardson–Romberg, converged window $v \le 160$; the rays are plotted from the end of the static transition at $x \approx 1$); dashed: $-\kappa_-(m_f)$. (b) $1 - \ell^2\psi$ on $U \simeq 1$ for three amplitudes; the drop near $v = 20$–35 is the static transition from $\psi \approx 0$ near the horizon to $\psi = 1/r_-^2$. (c) Bounded h, $m = 1.5$, $\alpha = 1$: σ_v on $U \simeq 1$ (full) and $U \simeq 0.3$ (dotted). (d) Ratio of the flux ϕ_v to the linear-response kernel of Paper G [Ownby, 2026b], minus one, on $U \simeq 1$.

12.1 The flux through the interior against the linear kernel

The scalar equation (11) contains h only through the metric. At linear order in B the field propagates on the static background, which for the Hayward h is the Hayward metric whatever the closure; the theory enters the flux only through the back-reaction, at $O(B^2)$. The linear response of the static interior to the horizon flux and its kernel are therefore those of Paper G. Table 7 gives the ratio of the evolved flux ϕ_v on $U \simeq 1$ to the kernel prediction. For $B = 3\times10^{-4}$ and 3×10^{-3} the ratio is 0.9992–0.9995 for $v \ge 60$ and the offsets from unity at the two amplitudes differ by at most 1.1×10^{-4}; for $B = 3\times10^{-2}$ it is 1.0078–1.0108, a nonlinear departure of 0.8–1.1%. The offset of 5–8×10^{-4} at small B is not a back-reaction effect (it does not scale with B) and is shared with the additive pipeline: the difference between the two evolutions is -3.2×10^{-4} at $B = 3\times10^{-3}$ on the ray $U \simeq 1$, so the additive flux departs from the kernel by about 3×10^{-4}, consistent with the 4×10^{-4} of Papers E and G [Ownby, 2026a,b]. The difference of the two theories is quadratic in B: relative to the additive closure (same data, same four meshes; `cmp_add_B*.log`) the QTG flux differs by -2.4×10^{-4} to -3.7×10^{-4} at $B = 3\times10^{-3}$ and by exactly 10^{-2} of that at $B = 3\times10^{-4}$

(for $v = 60$–80 on the four rays), i.e. by about $-(27\text{–}41)B^2$. This is the first quantity at which the theory is visible, and it is small. It stays small only until the front of Section 12.3 reaches the ray: at $v = 80$ on $U \simeq 1$ the metric exponents of the two evolutions differ by 0.68.

Table 7: Flux ϕ_v of the evolved solution divided by the linear-response kernel prediction of Paper G, ray $U \simeq 1$ (`kernel_ratio_B*.log`).

B	$v = 60$	$v = 80$	$v = 100$	$v = 108$
3×10^{-4}	0.999244	0.999331	0.999383	0.999399
3×10^{-3}	0.999328	0.999428	0.999492	0.999512
3×10^{-2}	1.007802	1.009181	1.010465	1.010849

12.2 The metric exponent

Before the front, the interior is the static interior of the Hayward hole with the mass parameter reached on the horizon. On the rays where this regime lasts longest ($U \simeq 0.3$) σ_v is constant at a value that agrees with $-\kappa_-(m_f)$, with m_f of (20) and $\kappa_-(m)$ the surface gravity of the static Hayward hole with mass parameter m, and not with the static value $-\kappa_-(5) = -0.7956493$ (Table 8; the plateau value is the median of the extrapolated σ_v over the points $v \geq 45$ with $\sigma_v < -0.79$). At $B = 3 \times 10^{-2}$ the shift from the static value is 4.2×10^{-4} and is reproduced to 1.3×10^{-6}. This is the property found in Paper E [Ownby, 2026a] for the additive closure, here in the theory whose static sector is the Hayward metric (`plateau_m1_B*.log`).

Table 8: Plateau of σ_v on $U \simeq 0.3$ against $-\kappa_-(m_f)$, Hayward h.

B	m_f	$-\kappa_-(m_f)$	plateau of σ_v	difference
3×10^{-4}	5.000001	−0.7956493	−0.7956493	-1.5×10^{-10}
3×10^{-3}	5.000099	−0.7956534	−0.7956534	-4.1×10^{-9}
3×10^{-2}	5.00996	−0.7960656	−0.7960643	$+1.3 \times 10^{-6}$

12.3 The front

At a time $v_f(U)$ the solution leaves the static regime: σ_v rises from $-\kappa_-(m_f)$ to about -0.1, over 7.2–7.6 units of v between its 10% and 90% points on every ray and amplitude (Fig. 1a; `width_m1.log`), $1 - \ell^2\psi$ falls from $1 - r_-^{-2} = 0.1057$ (Fig. 1b), the generalized mass m starts to grow, and y and $g_\phi = \phi_u\phi_v/W$ grow from $\frac{1}{2}$ and $< 10^{-2}$ to 10^7–10^8 (`late_m1_B*.log`). Defining v_f by $\sigma_v = -0.4$, at $B = 3 \times 10^{-3}$ it is 102.3, 84.0, 70.4, 65.4 on $U \simeq 0.3, 0.6, 1.0, 1.2$, and on $U \simeq 1$ it is 83.5, 70.4, 57.3 for $B = 3 \times 10^{-4}$, 3×10^{-3}, 3×10^{-2}: a delay of 13.1 per decade of $1/B$, i.e. 5.7 per e-fold (`front_m1_B*.log`). The front is not at a fixed Kruskal value $x = Ue^{\kappa_+ v}$: $x_f = 45, 37, 31, 30$ on the four rays at $B = 3 \times 10^{-3}$, and 60, 31, 16.5 on $U \simeq 1$ for the three amplitudes. We have not derived the delay; its logarithmic dependence on B is that expected for a quantity that grows exponentially from an initial value $\propto B$ or B^2 (the finite-advanced-time singularity of the interior-data ray, Appendix A, has 12.3 per decade).

12.4 Behind the front

Table 9 follows $U \simeq 1$ for $B = 3 \times 10^{-3}$ from the run to $v_1 = 200$, with the inter-level differences that define the window. Behind the front: (1) r stays within 3×10^{-6} of $r_- = 1.057475$; (2)

$\psi \to 1/\ell^2$ from below, algebraically: $1 - \ell^2\psi = \psi r^3/(2m)$ exactly (from $h = 2m/r^3$), so with $r \simeq r_-$ one has $(1-\ell^2\psi)\, m \to \frac{1}{2}r_-^3/\ell^2 = 0.59$ (Table 9), and since m grows linearly in the window, $\mathrm{d}m/\mathrm{d}v = 3.7 \to 4.1$, the pole is approached as $1/v$ and never reached; (3) $\mathrm{d}m/\mathrm{d}v = 8\pi r^2 y\phi_v^2$ with $y\phi_v^2 \simeq 0.13$–0.145 nearly constant (y grows as ϕ_v^2 decays); (4) σ_v rises monotonically from -0.094 at $v = 100$ to -0.056 at $v = 180$, with an increment of 6.3–6.5×10^{-4} per unit v over $160 \le v \le 190$ (the same on all four rays); it has not reached zero (and does not in the extended runs of Section 12.6).

Table 9: Hayward h, $B = 3 \times 10^{-3}$, ray $U \simeq 1$ (run to $v_1 = 200$, extrapolated over four meshes). $\delta\sigma_v$ and δz: difference between the top two levels of the Romberg table (absolute for σ_v, relative for $z = 1 - \ell^2\psi$); $\mathrm{d}m/\mathrm{d}v$ by central differences over ± 5; $y\phi_v^2$ as defined in the text (`endpoint_m1_B0.003_v200.log`).

v	σ_v	$\delta\sigma_v$	z	$\delta z/z$	m	$\mathrm{d}m/\mathrm{d}v$	zm	$y\phi_v^2$
80	-0.1159	1.9×10^{-9}	1.77×10^{-2}	5.5×10^{-8}	32.8	3.21	0.581	0.116
100	-0.0941	2.3×10^{-10}	5.65×10^{-3}	2.6×10^{-7}	104.1	3.72	0.588	0.133
120	-0.0876	4.0×10^{-9}	3.28×10^{-3}	1.3×10^{-7}	179.9	3.84	0.589	0.137
140	-0.0796	6.9×10^{-8}	2.29×10^{-3}	4.1×10^{-6}	257.5	3.92	0.590	0.139
160	-0.0688	1.4×10^{-6}	1.75×10^{-3}	2.2×10^{-4}	336.7	4.00	0.590	0.142
180	-0.0561	3.4×10^{-5}	1.41×10^{-3}	8.6×10^{-3}	417.4	4.07	0.590	0.145

The generalized mass of the ray therefore inflates without bound (linearly in v in the window), ψ approaches the pole, and the Misner–Sharp mass $M = \frac{1}{2}r^3\psi$ stays at $\frac{1}{2}r_-^3/\ell^2 \cdot \psi \approx 0.59$. This is the inflation of the generalized mass of Section 7, with a front and a rate that the characteristic data of Appendix A could not show. The mass law is verified on the extrapolated rays: over $[80, 90]$, $[90, 100]$ and $[100, 108]$ on $U \simeq 1$ the mass increment equals the trapezoid integral of $8\pi r^2 y\phi_v^2$ (samples every unit of v) to 1.1×10^{-4}, 2.1×10^{-5} and 7.5×10^{-6} ($B = 3 \times 10^{-3}$; `masslaw_m1_B*.log`), with the larger values, 1.3×10^{-2} over $[60, 70]$ and 1.1×10^{-3} over $[70, 80]$, in the windows where the front passes and the trapezoid rule at $\Delta v = 1$ is the limit; at $B = 3 \times 10^{-2}$ it holds to 6.4×10^{-4} and 5.9×10^{-5} in $[60, 70]$ and $[70, 80]$ and degrades beyond $v = 100$, outside the converged window.

12.5 Behind the front the solution is slaved to the interior flux

The following are properties of the evolved solution for $B = 3 \times 10^{-3}$, from snapshots on rows $v = \text{const}$ at $v = 100$, 150, 190 or 200 ($h_s = 0.05$, $h_v = 0.003125$ or 0.0125, not extrapolated; `slaving.log`, `slaving_prof.log`, `rho_check.log`); they are observations, and the consequences below are derived from them. With $x = Ue^{\kappa_+ v}$ and $\rho = -r_v$:

(a) *The flux is independent of u.* ϕ_v varies by 3×10^{-6} (max−min over mean) across $10 \le x \le 2 \times 10^4$ at the three times. We call it $\Phi(v)$. It is the flux through the interior of the linear response of Paper G: $\Phi_{\mathrm{PDE}}/\Phi_{\mathrm{kernel}} = 1.0001$ at $v = 100$ and 1.0006 for $150 \le v \le 250$ (`gpred_final.log`), the $O(B^2)$ difference of Table 7.

(b) *ρ is independent of u behind the edge of the plateau.* Between $x = 40$ and $x \gtrsim 200$ it varies by 1×10^{-4}, 2×10^{-3} and 2×10^{-2} (relative) at $v = 100$, 150, 200 (the variation grows with v; a difference of 10^{-4} in ρ_v/ρ between two values of x, which is the size of the h_s-systematic of Section 9 and of the spread in (c), accumulates to 1% over 100 units of v).

(c) *The Raychaudhuri equation* (8) reads $\rho_v = \sigma_v\rho + S$, $S = 4\pi r c_1\phi_v^2$, so that $\rho_v/\rho = \sigma_v + S/\rho$ at every point. This sum is independent of x: $-0.0922 \pm 1 \times 10^{-4}$ for $30 \le x \le 200$ at $v = 100$,

$-0.07438 \pm 2 \times 10^{-5}$ for $40 \le x \le 2300$ at $v = 150$, $-0.0432 \pm 1 \times 10^{-4}$ for $60 \le x \le 10^4$ at $v = 200$ (`slaving_prof.log`). Behind the front ($x \ge 2x_f$) $c_1 = z^2 \lesssim 10^{-4}$, so $S/\rho \lesssim 5 \times 10^{-3}$ and $\sigma_v = \rho_v/\rho$ up to that: the metric exponent is the logarithmic derivative of ρ. In the plateau ($30 \lesssim x \lesssim x_f$) $\sigma_v = -\kappa_-(m_f)$ and $S/\rho = 0.70$–0.73.

(d) *An exact identity.* By the definition of ψ, $y\rho = \beta \equiv (\psi r^2 - 1)/4$ (to 10^{-6}, `slaving.log`). With $W = -r_u/y$ this gives $\sigma_v = \rho_v/\rho + r_{uv}/r_u - \beta_v/\beta$ identically; the last two terms are -9×10^{-5} at $v = 200$ ($\sigma_v = -0.0438$) and decrease faster than $1/v$ (`rho_check.log`).

Since ρ_v/ρ takes the same value at every x in the region covered by (b), it can be evaluated where the equation is simplest, in the plateau, where $\sigma_v = -\kappa_-$ and $c_1 = c_1^{\rm II} = (1 - 1/r_-^2)^2$ are constants and $r = r_-$:

$$\rho_v = -\kappa_-\rho + S^{\rm II}(v), \qquad S^{\rm II} = 4\pi r_- c_1^{\rm II}\,\Phi(v)^2, \qquad \sigma_v \;\to\; G(v) \equiv \frac{S^{\rm II}}{\rho} - \kappa_-. \tag{26}$$

The amplitude of Φ cancels in G: it depends on the shape of $\Phi(v)$ alone, and so does not depend on B. Two further consequences follow from the exact mass law (16) and (d): $\mathrm{d}m/\mathrm{d}v = 8\pi r^2 y \phi_v^2 = 2r\beta\,(S^{\rm II}/\rho)/c_1^{\rm II}$, that is

$$\frac{\mathrm{d}m}{\mathrm{d}v} = \frac{2r\beta}{c_1^{\rm II}}\left(\kappa_- + G\right) \;\longrightarrow\; \frac{2r_-\beta_\infty\kappa_-}{c_1^{\rm II}} = 4.45, \qquad \beta_\infty = \frac{r_-^2 - 1}{4}, \tag{27}$$

and $zm = \psi r^3/2$ exactly. *Tests.* With Φ the kernel flux of Paper G, the solution of (26) (ρ is the convolution of $S^{\rm II}$ with $e^{-\kappa_- v}$ from $v = 20$; `gpred_q.py`) predicts $G(v)$ with no further input. Table 10 gives it beside the extrapolated σ_v, and Table 11 for other amplitudes and tail exponents. For $B = 3 \times 10^{-3}$ the difference $\sigma_v - G$ is -2.1×10^{-3} at $v = 100$ (the transient after the front has not died out: the front is only ~ 50 units of v earlier on this ray), -3.2×10^{-4} at 150 and -2.2×10^{-4} at 200, -2.0×10^{-4} at 250, where 1.6×10^{-4} is the h_s-systematic of Section 9; after the $h_s \to 0$ extrapolation the residual is -7.5×10^{-5}, -5.6×10^{-5} and -4.3×10^{-5} at $v = 180, 200, 220$ (0.13% of G; `hs_extrap.log`), the size of the 0.06% difference of the PDE flux from the kernel in (a). The closed form (27), with the local β and the extrapolated σ_v, gives $\mathrm{d}m/\mathrm{d}v$ to 0.8% (3.693 against 3.730 at $v = 100$, 4.135 against 4.162 at $v = 200$; Table 10). In the bounded model the relation $\sigma_v = \rho_v/\rho$ holds with $\sigma_v - \rho_v/\rho = -3.6 \times 10^{-4}$, -1.8×10^{-4}, -1.0×10^{-4} at $v = 95, 110, 125$ ($B = 3 \times 10^{-4}$, $m = 1.5$; $\sigma_v = -0.0552, -0.0470, -0.0411$), while $2\,\mathrm{d}\ln|\Phi|/\mathrm{d}v$ is within 2–3% of ρ_v/ρ (`bounded_aniso.log`); we did not build the bounded-h kernel.

12.6 The endpoint

What the extended runs show. With the remedy of Section 9 the run for $B = 3 \times 10^{-3}$ is accurate to $v = 300$ (inter-level difference of σ_v at most 9×10^{-6}) and usable to $v \simeq 320$ (2×10^{-4}); eight levels $h_v = 0.05, \ldots, 3.9 \times 10^{-4}$ from two runs of the same problem (`genq2_hs0.05_m1_B0.003_k1.5_v400.json` and `_v360.json`, whose rays coincide to 10^{-7} in U; the convergence ratio of successive levels is 4.0 across the two files, `merge_levels.py`) are used (Table 10). σ_v increases monotonically from -0.094 at $v = 100$ and does not reach zero: it is -0.0182 at $v = 300$, whereas the linear continuation from the end of the earlier window (6.4×10^{-4} per unit v at $v = 180$) reached zero near $v \simeq 270$. The increment decreases instead: $\mathrm{d}\sigma_v/\mathrm{d}v = 6.4 \times 10^{-4}$ at $v = 180$, 3.4×10^{-4} at 230 and 1.2×10^{-4} at 280. The product $v\sigma_v$ has a minimum -11.2 near $v = 150$ and rises to -5.47 at $v = 300$ (and to -5.58 at $v = 290$ for $B = 3 \times 10^{-4}$, `endpoint_B0.0003.log`), toward the value -5 derived below. The generalized mass has $\mathrm{d}m/\mathrm{d}v = 4.30$ at $v = 280$, the closed form (27) giving 4.29 (it tends to 4.45), and $zm = \frac{1}{2}\psi r^3$ stays at 0.59. The curvature invariant increases monotonically after the front (Table 10; Section 12.7).

Table 10: Hayward h, $B = 3\times10^{-3}$, $k = 3/2$, $m = 5$, ray $U \simeq 1$ ($h_s = 0.05$; Romberg in h_v^2 over the finest levels that are finite 30 units of v beyond the point, Section 9; `endpoint_B0.003_merged.log`). $\delta\sigma_v$: difference between the top of the table and the extrapolation from the finest $K-1$ levels (it does not include the h_s-systematic of 1.6×10^{-4}). G: prediction of Eq. (26) from the kernel flux. $\mathrm{d}m/\mathrm{d}v$ by central differences over ±5; "closed": Eq. (27) with the local β and the measured σ_v. K: Kretschmann invariant in units ℓ^{-4}.

v	σ_v	$\delta\sigma_v$	G	$v\sigma_v$	z	$\mathrm{d}m/\mathrm{d}v$	closed	K
100	-0.094251	3×10^{-11}	-0.092186	-9.425	5.64×10^{-3}	3.730	3.693	1.1×10^{25}
120	-0.087733	2×10^{-11}	-0.087031	-10.528	3.27×10^{-3}	3.848	3.816	1.7×10^{25}
140	-0.079755	3×10^{-11}	-0.079363	-11.166	2.29×10^{-3}	3.926	3.896	6.0×10^{25}
160	-0.068968	6×10^{-11}	-0.068687	-11.035	1.75×10^{-3}	4.004	3.976	2.5×10^{26}
180	-0.056255	2×10^{-10}	-0.056017	-10.126	1.41×10^{-3}	4.086	4.058	8.6×10^{26}
200	-0.043998	4×10^{-10}	-0.043779	-8.800	1.18×10^{-3}	4.162	4.135	2.2×10^{27}
220	-0.034253	1×10^{-9}	-0.034044	-7.536	1.01×10^{-3}	4.222	4.195	4.3×10^{27}
240	-0.027494	3×10^{-9}	-0.027293	-6.599	8.82×10^{-4}	4.264	4.238	6.5×10^{27}
260	-0.023090	5×10^{-9}	-0.022894	-6.003	7.82×10^{-4}	4.292	4.266	8.7×10^{27}
280	-0.020206	4×10^{-8}	-0.020014	-5.658	7.02×10^{-4}	4.310	4.285	1.1×10^{28}
300	-0.018223	9×10^{-6}	-0.018030	-5.467	6.37×10^{-4}	4.372	4.299	1.3×10^{28}

Other amplitudes and tail exponents. Table 11 compares σ_v with the kernel prediction G for $k = \frac{1}{2}, \frac{3}{2}, 3$ and for $B = 3\times10^{-4}, 3\times10^{-3}, 3\times10^{-2}$ (runs `genq2_hs0.05_m1_B*_k*_v*.json`, four levels each, Romberg as above; `ktable.log`). After the transient, $\sigma_v - G = -(1.9\text{–}3.0)\times10^{-4}$ for all five cases at $v \geq 200$ ($-(2.7\text{–}4.4)\times10^{-4}$ at $v = 150$) (it is the h_s-systematic -1.6×10^{-4} and the $O(B^2)$ flux difference of Section 12.5), although σ_v itself differs by a factor 5.8 between $k = \frac{1}{2}$ and $k = 3$ at $v = 200$. For $k = 3$ the tail is steep, Φ is dominated by the exponential of the kernel for $v \lesssim (k+1)/\kappa_+ = 82$ and well beyond, and σ_v sits on a plateau -0.098 to -0.099 at $120 \leq v \leq 190$, which is $-2\kappa_+ = -0.0979$ (the kernel prediction is still -0.0979 at $v = 200$ and reaches the power law, -0.0419 at $v = 400$, only at $v \gtrsim 300$). For $k = \frac{1}{2}$, $v\sigma_v = -4.49, -3.75, -3.36, -3.22$ at $v = 100, 150, 200, 290$ and tends to $-3 = -(2k+2)$ (the kernel gives $-3.31, -3.09, -3.03, -3.02$ at $200, 400, 10^3, 2\times10^3$). The windows: $k = \frac{1}{2}$ is accurate to $v \simeq 290$; $k = 3$ to $v \simeq 190$ (the first non-finite sample is 196, 225, 253, 281 at $h_v = 0.025, \ldots, 0.003125$, shifted by -37.6 with respect to Eq. (21), whose constant depends on k); $B = 3\times10^{-2}$ to $v \simeq 160$.

Asymptotics of the slaved description. Eq. (26) fixes the late behavior once $\Phi(v)$ is known. (i) *Sign.* If $S^{\mathrm{II}} \propto \Phi^2$ decreases, then $\rho(v) = \int_{v_0}^{v} S^{\mathrm{II}}(v')e^{-\kappa_-(v-v')}\mathrm{d}v' \geq S^{\mathrm{II}}(v)/\kappa_-$ and $G = S^{\mathrm{II}}/\rho - \kappa_- \leq 0$: no zero crossing. The kernel flux $|\Phi|$ is strictly decreasing on the unit grid $20 \leq v \leq 1200$ for $k = \frac{1}{2}, 1, \frac{3}{2}, 2, 3$, with $|\Phi(v+1)/\Phi(v)| < 0.9988$ (`phi_monotone.py`, `logs/phi_monotone.log`), and the exact identity (d) adds the term $-\beta_v/\beta < 0$. (ii) *Rate.* The kernel of Paper G tends to $K(t) \to R_1 e^{-\kappa_+ t}$, so $\Phi(v) = \int K(v-v')j(v')\mathrm{d}v' \to (R_1/\kappa_+)\, j(v)\,[1 + O((k+1)/\kappa_+ v)]$, with $j \propto v^{-(k+1)}$ the horizon flux of the tail $\phi \propto v^{-k}$. Then $\rho \to S^{\mathrm{II}}/\kappa_-$ and

$$\sigma_v \;\to\; \frac{\mathrm{d}\ln S^{\mathrm{II}}}{\mathrm{d}v} = 2\frac{\mathrm{d}\ln|\Phi|}{\mathrm{d}v} \;\to\; -\frac{2(k+1)}{v} = -\frac{5}{v} \quad (k = \tfrac{3}{2}), \tag{28}$$

with corrections of relative size $(k+1)/(\kappa_+ v)$, which is 0.26 at $v = 200$: the approach is slow. The kernel prediction for vG ($k = \frac{3}{2}$) is -8.76 at $v = 200$, -5.16 at 400, -5.05 at 10^3 and -5.03 at

Table 11: σ_v on $U \simeq 1$ ($h_s = 0.05$) and its difference from the kernel prediction G of Eq. (26) (in units of 10^{-4}), for the tail exponents $k = \frac{1}{2}, \frac{3}{2}, 3$ and amplitudes B (`ktable.log`); –: beyond the accurate range. The limiting values $-(2k+2)/v$ are -0.010, -0.017, -0.028 at $v = 290$ for $k = \frac{1}{2}, \frac{3}{2}, 3$. In all cases the h_s-systematic is -1.6×10^{-4}.

	$k = \frac{1}{2}$		$k = \frac{3}{2}$		$k = 3$
v	$B = 3\times10^{-3}$	3×10^{-4}	3×10^{-3}	3×10^{-2}	3×10^{-3}
100	-0.04488 (-16)	-0.09908 (-69)	-0.09425 (-21)	-0.09306 (-8.7)	-0.10104 (-22)
150	-0.02497 (-3.7)	-0.07482 (-4.4)	-0.07471 (-3.2)	-0.07469 (-3.1)	-0.09839 (-2.7)
200	-0.01681 (-2.4)	-0.04402 (-2.4)	-0.04400 (-2.2)	–	-0.09812 (-1.9)
250	-0.01295 (-2.2)	-0.02507 (-2.2)	-0.02505 (-2.0)	–	–
290	-0.01111 (-2.8)	-0.01923 (-3.0)	-0.01913 (-1.9)	–	–

2×10^3 (`gpred_final.log`), against -8.80 at 200 and -5.47 at 300 in Table 10. (iii) *Consequences* (all in the slaved description, as $v \to \infty$): $W = -r_u\rho/\beta \propto \rho \propto v^{-(2k+2)}$, with r_u independent of v ($\partial_v \ln|r_u| = -1.4 \times 10^{-7}$ at $v = 200$, `rho_check.log`), so the affine parameter $\lambda = \int W\,\mathrm{d}v$ of the ingoing null geodesic at fixed u (along which $W\,\mathrm{d}v/\mathrm{d}\lambda$ is constant, since $g_{uv}k^v$ is conserved) has a finite range to $v = \infty$, with remaining length $\Delta\lambda \propto v^{-(2k+1)}$; $m \simeq 4.45\,(v - v_0)$ in the limit ($\mathrm{d}m/\mathrm{d}v = 4.3$ in the window) and $z = \psi r^3/2m \propto 1/(v - v_0)$ with $v_0 \approx 85$ (from $m(v)$, Section 12.8); $\psi \to 1/\ell^2$, $r \to r_-$ and $\sigma_v \to 0^-$; and the invariant is $K \simeq 16\Sigma^2 \propto z^6\Phi^{-4}$, which in this description tends to v^{4k-2} ($= v^4 \propto \Delta\lambda^{-1}$ for $k = 3/2$; we regard this as a conjecture about the endpoint, supported by the structure but not reached by the simulation; the structural relation $Kz^{-6}\Phi^4 = \text{const}$ holds to a factor 1.7–2.2 over $100 \leq v \leq 280$ while K changes by 10^3, `curv_hv.log`, `curv_hv_B0.0003.log`). The offset v_0 delays the asymptotic exponents: the local exponent of z is -1.43 at $v = 280$ and -1.29 at 380 and that of K is $-6v/(v - v_0) + 2|vG|$ (Eq. (30)), which is 2.5 at $v = 300$ and 3.4 at $v = 10^3$; it was measured to be 2.5 on $300 \leq v \leq 380$ with code B (Section 12.8; 2.1 at $v = 280$ for $B = 3 \times 10^{-4}$ and 4.2 at $v = 240$ for $B = 3 \times 10^{-3}$ with code A) and returns to 4 only for $v \gtrsim 10^3$. For the exponent $k = \frac{1}{2}$ the same relation gives $K \propto v^0$: in the run K decreases after the front (2.1×10^{21} at $v = 100$, 3.0×10^{19} at $v = 290$, `endpoint_k0.5.log`), the sign that the structure predicts at these v and an illustration that the growth of the curvature depends on the tail; for $k = 3$ it rises steeply (4.9×10^{25} at $v = 100$, 3.8×10^{29} at 190).

What is and is not shown. Shown: the structure (a)–(d) of Section 12.5 between $v = 100$ and $v \simeq 300$ for $B = 3 \times 10^{-3}$, with a second code to $v \simeq 380$ (Sections 10 and 12.8); that the prediction (26) reproduces σ_v to the level of the numerical systematic (2×10^{-4}, 0.13% after $h_s \to 0$ for $180 \leq v \leq 230$) for three tail exponents and three amplitudes; that σ_v approaches zero from below with a decreasing rate; and the monotonic growth of K for $k \geq \frac{3}{2}$. Not shown: anything about $v \gtrsim 1400$ (and, with codes A and B, $v \gtrsim 400$), which rests on the continuation of the slaved structure and of the kernel (Eq. (28)) and is not a simulation. In particular a finite affine range to $v = \infty$ (the simulated $W \propto v^{-5.1}$ is consistent with it) and $K \propto v^{4k-2}$ (not reached: local slope 2.5 at $v = 300$–380 and 3.64 at 1400, Section 12.9) are predictions of the effective description; we do not claim that the Cauchy horizon of the hole is a singularity of a given strength, only that within this description the invariant diverges as a power of v for $k > \frac{1}{2}$. The description itself is classical and ceases to apply when $K \gg \ell^{-4}$ (Section 15), which is the case already at 10^{25}.

12.7 Curvature

The Kretschmann invariant of the metric $-W\,du\,dv + r^2 d\Omega^2$ is, for any h and in terms of the solved fields (Appendix B),

$$K = (4\Sigma)^2 + 8\,\frac{H_{ab}H^{ab}}{r^2} + 4\psi^2, \tag{29}$$

$$\Sigma = \frac{\sigma_{uv}}{W} = \frac{\psi}{2} - c_2 - 8\pi c_1 g_\phi + 64\pi^2 c_3 g_\phi^2, \qquad \frac{H_{ab}H^{ab}}{r^2} = 128\pi^2 c_1^2 g_\phi^2 + \frac{Q^2}{2},$$

with $g_\phi = \phi_u\phi_v/W$ and $H_{ab} = \nabla_a\nabla_b r$. For $g_\phi = 0$ and the Hayward h at the pole ($\psi = 1/\ell^2$, $c_2 = 0$, $Q = -2/\ell^2$) this gives $K = 24\,\ell^{-4}$, the de Sitter value, and for $h = \psi$ it gives $48M^2/r^6$; on the static Hayward and bounded interiors it agrees with $f''^2 + 4f'^2/r^2 + 4(1-f)^2/r^4$ to 3.4×10^{-7} and 7.2×10^{-8} (finite differences; `check_curv_static.log`).

Behind the front the static part stays at $O(\ell^{-4})$ and the invariant is carried by the flux terms. At $B = 3\times10^{-3}$ the maximum of K over the sampled v is 2.9×10^{32}, 4.7×10^{29}, 2.8×10^{27} (ℓ^{-4}) on $U \simeq 0.3, 0.6, 1.0$, at the passage of the front ($v = 103, 85, 72$). On $U \simeq 1$, $K = 1.4\times10^{26}$ at $v = 80$, falls to 1.1×10^{25} at $v = 100$, and then rises again: 1.7×10^{25}, 5.8×10^{25}, 2.4×10^{26} at $v = 120, 140, 160$ (`curv_m1_B*.log`, `late_m1_B0.003_v200.log`). At $v = 100$ the terms of Σ on $U \simeq 1$ are $\psi/2 = 0.50$, $-c_2 = -0.025$, $-8\pi c_1 g_\phi = -4.8\times10^4$ and $64\pi^2 c_3 g_\phi^2 = 8.3\times10^{11}$ (`curv_terms_m1_B0.003.log`): the invariant is $\simeq 16\Sigma^2$ with Σ dominated by the c_3 term, which is specific to this theory (it is absent for $h = \psi$) and is fixed by the compatibility condition (Section 4(e)). The bound on the curvature of the static Hayward metric does not extend to the dynamical solution with flux. A curvature of $10^{25}\ell^{-4}$ means a curvature radius $10^{-6}\ell$; for any ℓ above the Planck length the classical description by this theory is not reliable there, and the solution should be read as the classical evolution of the theory, not as a prediction for the physical hole. We note also that the size of K rests on the c_3 term of (10): it is the field equation of the reduced action (Section 4(f), exact in rational arithmetic) and agrees with the compatibility condition (3×10^{-12}); the second-order convergence of the Raychaudhuri residuals and the r_{uv} residual through the front (Section 9(iii)) test the scheme, and there is no numerical test of the term in the region behind the front. *Later times.* With the extended runs of Section 9, K on $U \simeq 1$ ($B = 3\times10^{-3}$) keeps rising after $v = 160$: 1.4×10^{27}, 4.3×10^{27}, 6.5×10^{27} at $v = 190, 220, 240$ and 1.3×10^{28} at 300 (Table 10); its local logarithmic slope $\mathrm{d}\ln K/\mathrm{d}\ln v$ is 10.9 at $v = 160$ and 4.2 at $v = 240$. For $B = 3\times10^{-4}$ (front later) $K = 3.6\times10^{34}$ at $v = 100$, falls to 1.1×10^{34} at 120 and rises to 1.7×10^{36} at $v = 290$, with slope 9.1 at $v = 170$ and 2.1 at $v = 280$ (`curv_hv_B0.0003.log`). Behind the front the slaved structure of Section 12.5 gives $K \simeq 16\Sigma^2 \propto z^6\Phi^{-4}$ (the combination $Kz^{-6}\Phi^4$ changes by a factor 1.7–2.2 over $100 \le v \le 280$ while K changes by 10^3), so the growth rate depends on the tail exponent k through $\Phi \propto v^{-(k+1)}$ (Section 12.6).

12.8 The curvature invariant and the tidal integrals to $v = 380$

Code B (Section 10, $h_s = 0.05$, $h_v = 0.0125$, $B = 3\times10^{-3}$, $k = \frac{3}{2}$, ray $U \simeq 1$) allows K to be followed from $v \simeq 300$ to $v \simeq 380$ (change from the next coarser level 2.4×10^{-3} at $v = 300$, 1.8×10^{-2} at 340, 0.15 at 380, i.e. an error of about 1% at 380; `tidal_v600.log`). The scripts are `kstudy.py` and `tidal.py`; all numbers below are from them.

The invariant. $K = 1.09\times10^{25}$, 2.17×10^{27}, 1.26×10^{28}, 1.98×10^{28} and $2.27\times10^{28}\ell^{-4}$ at $v = 100, 200, 300, 360$ and 380. Code A at $h_s = 0.05$ gives K higher by 3% (1.12×10^{25}, 2.22×10^{27}, 1.29×10^{28} at 100, 200, 300): this is its h_s-systematic, and the K column of Table 10, and the values quoted in Sections 12.6 and 12.7, carry it (it does not affect logarithmic slopes at the level used here). At $v = 340$ the terms of Σ are $\psi/2 = 0.4997$, $-c_2 = -2.4\times10^{-3}$, $-8\pi c_1 g_\phi = -9.4\times10^4$ and $64\pi^2 c_3 g_\phi^2 = 3.3\times10^{13}$: the last is $1+3\times10^{-9}$ times Σ, and the other two terms of (29) are of relative size 10^{-17} and smaller. For the Hayward h, $c_3 = h''/h'^3 = 2z^3$ ($\ell = 1$), so $K \simeq (512\pi^2)^2 z^6 g_\phi^4$ to better than 10^{-8}: the curvature of the dynamical solution is carried entirely by the $\phi_u^2\phi_v^2$ term, which the minimally coupled scalar sources through the quartic part of the theory. (Behind the front $\Omega = \mathcal{R} + 2\eta + 2\psi \simeq 4\Sigma \geq 8\times10^{11}$, so $I_{C^2} = \Omega^2/3$ and the denominators of the lifts of Section 4(g) do not vanish.) The local slope $\mathrm{d}\ln K/\mathrm{d}\ln v$ is 10.9 at $v = 160$, 7.9 at 200, 4.2 at 240, 3.2 at 260, 2.7 at 280, 2.5 at 300, and then 2.47, 2.48, 2.50, 2.45 at $v = 320, 340, 360, 380$ (centered differences over ±10): in the simulated range K does *not* follow $v^{4k-2} = v^4$, and its slope has a maximum at $v \simeq 160$ followed by a plateau near 2.5. The slaved structure accounts for this. With $K \propto z^6\Phi^{-4}$ (Section 12.6), $z = \psi r^3/2m \propto 1/(v - v_0)$ and $\sigma_v \simeq 2\,\mathrm{d}\ln|\Phi|/\mathrm{d}v$ one has $\mathrm{d}\ln K/\mathrm{d}\ln v = -6v/(v-v_0) + 2|v\sigma_v|$, that is

$$\frac{\mathrm{d}\ln K}{\mathrm{d}\ln v} \simeq 4 - \frac{6\,v_0}{v - v_0} + 2\big(|v\sigma_v| - 5\big), \qquad v_0 \simeq 86, \tag{30}$$

which gives 2.42, 2.45, 2.56 at $v = 300, 340, 380$ with the measured $|v\sigma_v| = 5.42, 5.24, 5.16$, against the measured 2.52, 2.48, 2.45 (within 0.15; the structural relation also contains the factor $\phi_u^4(1 + G/\kappa_-)^4$ whose variation is not included). The offset v_0 is obtained from the exact generalized mass $m = r^3\psi/2z$: $m = 926.3, 1099.5, 1273$ and $\mathrm{d}m/\mathrm{d}v = 4.32, 4.34, 4.34$ at $v = 300, 340, 380$, so m extrapolates to zero at $v_0 = 85.7, 86.6, 86.5$ (72 and 80 at $v = 100$ and 200: v_0 is still drifting and $\mathrm{d}m/\mathrm{d}v$ rises from 3.7 to 4.3); the local exponent of z is -1.40, -1.34, -1.29, which is $v/(v - v_0)$ with $v_0 = 86$ (`kstudy_v600.log`). A value $v_0 \simeq 100$ would put Eq. (30) at 1.8, 2.0 and 2.2, off by 0.3–0.7. If v_0 and $|v\sigma_v| \to 5$ stay at these values, Eq. (30) gives 3.4 at $v = 10^3$ and 3.95 at 10^4: the exponent $4k - 2$ would be reached to 10% only for $v \gtrsim 1.4\times10^3$, a factor four beyond the simulated range (an indicative extrapolation, since v_0 drifts). The simulation thus gives neither evidence for nor against the asymptotic law; it confirms the slaved structure that predicts the approach, and the plateau at 2.5 is a transient that Eq. (30) says will rise. The product $v\sigma_v$ is -5.42, -5.24, -5.16 at $v = 300, 340, 380$ and tends to $-(2k + 2) = -5$ from below, as the kernel prediction requires. Section 12.9 extends the range to $v = 650$.

Affine distance and tidal integrals. Along the ingoing null geodesic $u = \text{const}$ the affine parameter is $\mathrm{d}\lambda = W\,\mathrm{d}v$ ($\Gamma^v_{vv} = \partial_v \ln W$), and the remaining affine distance to $v = \infty$ is $\Delta\lambda(v) = \int_v^\infty W\mathrm{d}v'$. In the simulated range $W \propto v^{-5.1}$ to $v^{-5.2}$ ($2k + 2 = 5$ expected), so $\Delta\lambda$ is finite in the slaved continuation; we complete the integral beyond $v = 380$ with a power-law tail of the local index (the difference from an exponential-tail completion is 20% in $\Delta\lambda$ at $v = 380$; the scale of λ is arbitrary, so only invariant combinations are quoted). Two exact consequences of the Raychaudhuri equation in affine form: $r_{\lambda\lambda} = -\frac{4\pi r}{h'}\phi_\lambda^2$ and therefore the transverse tidal eigenvalue (geodesic deviation of the two-sphere directions, $R_{\lambda A\lambda B} = \mathcal{T}\,g_{AB}$) is

$$\mathcal{T} = -\frac{r_{\lambda\lambda}}{r} = \frac{4\pi}{h'}\Big(\frac{\phi_v}{W}\Big)^2. \tag{31}$$

It is large ($\mathcal{T} = 8.1\times10^{20}$ at $v = 300$ and 1.5×10^{21} at 380 in the (s, v) gauge of the run, where it is not invariant) but integrable: $r_\lambda = r_v/W$ varies by 0.6% over $v \in [200, 380]$, and the change of r over the remaining affine range, $|r_\lambda|\Delta\lambda$, is 8×10^{-10}, 2.7×10^{-13}, 1.4×10^{-13} at $v = 100, 300, 350$; $\mathcal{T}\Delta\lambda^2 = 1.7\times10^{-11}$, 5.0×10^{-16}, 2.1×10^{-16}; and the double integral $\iint \mathcal{T}\,\mathrm{d}\lambda\,\mathrm{d}\lambda$ over the remaining range (invariant under a rescaling of λ; the quantity that must stay $O(1)$ for a null geodesic to be crushed, in the sense of Królak, and for volume elements to be crushed in that of Tipler) is 3.2×10^{-11}, 9.4×10^{-16}, 3.8×10^{-16} (`tidal_v600.log`). With respect to the remaining affine distance $K \propto \Delta\lambda^{e_K}$ and $\mathcal{T} \propto \Delta\lambda^{e_T}$ with $e_K = -0.59, -0.59, -0.60$ and $e_T = -0.62, -0.62, -0.62$ at $v = 320, 340, 360$ (the asymptotic structure gives $e_K = -(4k-2)/(2k+1) = -1$ and $e_T = -2k/(2k+1) = -\frac{3}{4}$ for $k = \frac{3}{2}$, approached slowly for the same reason as above). Both $e_K/2$ and e_T are above -1 (so are their asymptotic values, $-\frac{1}{2}$ and $-\frac{3}{4}$), so $\int \sqrt{K}\,\mathrm{d}\lambda$ and $\int \mathcal{T}\,\mathrm{d}\lambda$ converge: for radial *null* geodesics on this ray the divergence of K is not accompanied by a divergence of the accumulated tidal distortion, in the slaved continuation. The invariant K is dominated by $\Sigma = \sigma_{uv}/W$, a boost-weight-zero component ($R_{\ell n \ell n}$ in a null frame), which does not enter the deviation of a null geodesic; timelike observers see it through their boost. Whether a timelike geodesic reaches $v = \infty$ at finite proper time and what it feels there (the question that distinguishes weak from strong singularities in the sense of Tipler, and that Ori's analysis answers for Reissner–Nordström) is not computed here.

12.9 Curvature and slaving to $\kappa_+ v \simeq 70$: tail exponents, amplitudes and the inner mass

With code E (Sections 11 and 11.3; $h_s = 0.2$, $h_v = 0.1$, ray $U = 0.9968$, `codee/`) the quantities of Sections 12.5–12.8 can be followed to $\kappa_+ v \simeq 69$. For the case of the preceding sections ($B = 3 \times 10^{-3}$, $k = \frac{3}{2}$, $m = 5$) Table 12 gives the state behind the front up to $v = 1400$ (`kstudy2_double_v1500.log`). The invariant keeps rising, $K = 1.61 \times 10^{30}\ell^{-4}$ at $v = 1400$, and its local slope $\mathrm{d}\ln K/\mathrm{d}\ln v$ has its minimum 2.50 at $v = 340$ and then rises, 2.63 at 380, 2.97 at 500, 3.22 at 650, 3.50 at 1000 and 3.64 at 1400: the plateau of Section 12.8 (the code B value at 380 is 2.45, inside that code's error there) was a transient, as Eq. (30) said. The slope is reproduced by (30) with the measured $v\sigma_v$ and v_0 to 0.09 at $v = 300$ and 0.02 or less at $v \geq 650$ (0.13 and 0.37 at $v = 100$ and 200, where the transient of the front is not over). The terms of the formula are individually measured: $v\sigma_v = -5.416, -5.122, -5.089, -5.055, -5.038$ at $v = 300, 500, 650, 1000, 1400$ goes to -5 from below, v_0 is still drifting ($85.7 \to 96.9$ between $v = 300$ and 1400), $\mathrm{d}m/\mathrm{d}v = 4.425$ approaches the closed form 4.449 from below, and $\mathrm{d}\ln z/\mathrm{d}\ln v = -1.40 \to -1.07$. With the two $1/v$ terms of (30) added back to the measured slope, the result is 4.09, 4.04, 4.03, 4.02, 4.01 at the same v (last column of Table 12): the constant of the formula, $4k - 2 = 4$, is measured to 0.3% at $v = 1400$, while the slope itself is still 9% below it. The c_3 term of Σ is $1 + 3 \times 10^{-9}$ times Σ or less at $v \geq 300$. The product zm stays at 0.5912 ($\frac{1}{2}r_-^3 = 0.5913$).

Tail exponents, amplitudes and the inner mass. Table 13 compares the same quantities at $\kappa_+ v = 50$ for $k = \frac{1}{2}, 1, \frac{3}{2}, 2, 3$ ($k = 2$ and 3 are the exponents $2\ell + 2$ and $2\ell + 3$, $\ell = 0$, of the tail of a massless field on the event horizon of Schwarzschild for initially static and generic data [Gundlach et al., 1994a]; we use them as representative values and have not derived them for the present backgrounds), for $B = 3 \times 10^{-4}, 3 \times 10^{-3}, 3 \times 10^{-2}$ and for $m/\ell = 3, 5, 10$ ($\kappa_+ v = 40$ for $m = 10$, where $v = 1608$; `scans.log`; the discretization errors of the base case, Section 11.3, are 2×10^{-5} (h_v) and 3×10^{-4} (h_s) in K and constant in v). The generalized slope formula

$$\frac{\mathrm{d}\ln K}{\mathrm{d}\ln v} \simeq (4k-2) - \frac{6v_0}{v - v_0} + 2\big(|v\sigma_v| - (2k+2)\big) \qquad (32)$$

(Eq. (30) with $4 \to 4k - 2$ and $5 \to 2k + 2$) predicts the measured slope to ≤ 0.04 for every case with $m = 5$ and $m = 10$, including the negative slope of $k = \frac{1}{2}$ (K decreases, -0.45 against -0.46, as in Section 12.6) and the steep one of $k = 3$ (9.43 against 9.41), and to 0.04 for $m = 3$. Adding the two $1/v$ terms back to the measured slope gives the formula's constant:

Table 12: Hayward h, $B = 3\times10^{-3}$, $k = \frac{3}{2}$, $m = 5$, ray $U = 0.9968$ from code E ($h_s = 0.2$, $h_v = 0.1$, $v_1 = 1500$, double): $z = 1 - \ell^2\psi$, generalized mass m and its rate, the origin v_0 of the linear growth of m, $v\sigma_v$, K in units ℓ^{-4}, the local slope of K, the formula (30), and the slope with the two $1/v$ terms of the formula added back (the formula's constant, 4) (`kstudy2_double_v1500.log`). Changing v_1 to 1200, halving h_v, halving h_s or using long double changes K by $\le 5\times10^{-4}$ at every v (Section 11.3).

v	z	m	$\mathrm{d}m/\mathrm{d}v$	v_0	$v\sigma_v$	K	slope	Eq. (30)	slope+terms
100	5.66×10^{-3}	104	3.722	72.1	-9.409	1.12×10^{25}	-2.80	-2.67	3.87
200	1.18×10^{-3}	500	4.157	79.8	-8.767	2.20×10^{27}	7.91	7.55	4.36
300	6.38×10^{-4}	926	4.322	85.7	-5.416	1.28×10^{28}	2.52	2.43	4.09
380	4.64×10^{-4}	1273	4.354	87.5	-5.185	2.32×10^{28}	2.62	2.57	4.05
500	3.29×10^{-4}	1797	4.378	89.5	-5.122	5.01×10^{28}	2.97	2.94	4.04
650	2.41×10^{-4}	2455	4.395	91.3	-5.089	1.13×10^{29}	3.22	3.20	4.03
800	1.90×10^{-4}	3116	4.406	92.8	-5.070	2.24×10^{29}	3.37	3.35	4.02
1000	1.48×10^{-4}	3998	4.415	94.4	-5.055	4.83×10^{29}	3.50	3.48	4.02
1200	1.21×10^{-4}	4881	4.420	95.8	-5.045	9.22×10^{29}	3.58	3.57	4.01
1400	1.03×10^{-4}	5766	4.425	96.9	-5.038	1.61×10^{30}	3.64	3.63	4.01

0.010, 2.013, 4.015, 6.018, 10.023 for $k = \frac{1}{2}, \ldots, 3$ against $4k - 2 = 0$, 2, 4, 6, 10 (within 0.7% for $k \ge 1$), 4.016, 4.015 for the two other amplitudes, 4.035 ($m = 3$) and 3.994 ($m = 10$); the product $v\sigma_v$ goes to $-(2k+2)$ from below in every case (-3.032, -4.043, -5.054, -6.065, -8.088 at $\kappa_+ v = 50$), zm goes to $\frac{1}{2}r_-^3/\ell^2$ (0.59117, 0.67867, 0.54101 for $m = 5$, 3, 10, against 0.5913, 0.6790, 0.5411) independently of k and B, and $\mathrm{d}m/\mathrm{d}v$ approaches from below the closed form (27), $2r_-\beta_\infty\kappa_-/c_1^{\mathrm{II}} = 2.403$, 4.449, 9.476 for $m = 3$, 5, 10 (measured 2.367 at $\kappa_+ v = 50$, 4.415, 9.452 at $\kappa_+ v = 40$: 1.5%, 0.8%, 0.25% below), which tests that formula for two masses other than the one it was derived on. Neither zm nor $\mathrm{d}m/\mathrm{d}v$ depends on k or B (within 0.06% and 0.6%); the origin v_0 does (81, 88, 95, 102, 116 for $k = \frac{1}{2}, \ldots, 3$; 106, 95, 83 for $B = 3\times10^{-4}$, 3×10^{-3}, 3×10^{-2}) and so does the exponent of K.

Table 13: Code E at $\kappa_+ v = 50$ (for $m = 10$: $\kappa_+ v = 40$) for the cases of Section 12.9: v, zm against $\frac{1}{2}r_-^3$, $v\sigma_v$ against $-(2k+2)$, $\mathrm{d}m/\mathrm{d}v$ against the closed form (27), v_0, the local slope of K, Eq. (32), and the slope with the two $1/v$ terms added back against $4k - 2$ (`scans.log`; $h_s = 0.2$, $h_v = 0.1$).

case	v	zm ($\frac{1}{2}r_-^3$)	$v\sigma_v$	$\mathrm{d}m/\mathrm{d}v$ (closed)	v_0	slope	Eq. (32)	slope+terms ($4k-2$)
$k = \frac{1}{2}$	1021	0.59117 (0.5913)	-3.032	4.426 (4.449)	81.4	-0.45	-0.46	0.010 (0)
$k = 1$	1021	0.59117	-4.043	4.421	87.7	1.54	1.52	2.013 (2)
$k = \frac{3}{2}$	1021	0.59116	-5.054	4.415	94.6	3.51	3.49	4.015 (4)
$k = 2$	1021	0.59116	-6.065	4.410	101.6	5.48	5.47	6.018 (6)
$k = 3$	1021	0.59116	-8.088	4.399	116.0	9.43	9.41	10.023 (10)
$B = 3\times10^{-4}$	1021	0.59117	-5.054	4.415	106.2	3.43	3.41	4.016 (4)
$B = 3\times10^{-2}$	1021	0.59082	-5.054	4.426	83.1	3.59	3.58	4.015 (4)
$m = 3$	639	0.67867 (0.6790)	-5.058	2.367 (2.403)	88.3	3.19	3.15	4.035 (4)
$m = 10$	1608	0.54101 (0.5411)	-5.071	9.452 (9.476)	143.1	3.55	3.56	3.994 (4)

13 The bounded characteristic function from event-horizon-regular data

For the bounded h of (4) we use $\alpha = 1$ and $m = 1.5$ ($r_- = 1.031986$, $r_+ = 2.980943$, $\kappa_- = 0.797008$, $\kappa_+ = 0.161359$, $\psi(r_-) = 0.939$, so the static horizon is at a distance 0.061 from the wall $\alpha\psi = 1$) with the same tail and the same pipeline, $B = 3 \times 10^{-4}$, 3×10^{-3}, 3×10^{-2}, $v_1 = 110$. The static validation of Section 9(iv) applies. The results follow those of the Hayward h (Fig. 1c):

- *Exponent.* σ_v on $U \simeq 0.3$ equals $-\kappa_-(m_f)$, with m_f of (20), to 1.3×10^{-7} ($B = 3 \times 10^{-4}$), 1.8×10^{-6} (3×10^{-3}) and 5.5×10^{-5} (3×10^{-2}, short plateau); $m_f = 1.50000028, 1.50002824, 1.50283083$ (`plateau_m2_B*.log`).
- *Front.* On $U \simeq 1$ the front ($\sigma_v = -0.4$) is at $v_f = 59.5, 46.1, 31.4$ for the three amplitudes: a delay of 13.5 per decade between the two smaller amplitudes (5.85 per e-fold, against 5.7 for the Hayward h although κ_+ is 3.3 times larger) and 14.7 for the largest (`front_m2_B*.log`).
- *Behind the front* ($B = 3 \times 10^{-4}$, $U \simeq 1$): ψ approaches the wall, with $1 - \alpha\psi = 3.1 \times 10^{-3}$, 9.3×10^{-4}, 4.3×10^{-4} and $m = 6.96, 12.8, 18.8$ at $v = 70, 80, 90$ ($\mathrm{d}m/\mathrm{d}v \simeq 0.58$–$0.60$), r constant at r_- to 10^{-6}, and $\sigma_v = -0.0891, -0.0685, -0.0587$, rising as for the Hayward h; the inter-level differences of Σ are 6×10^{-9}, 1.6×10^{-6}, 3×10^{-4} (relative), and at $v = 100$ 5×10^{-2}, which sets the window $v \lesssim 95$ (`late_m2_M1.5_B*.log`). The Kretschmann invariant on the same ray is 5.0×10^{37}, 2.1×10^{36}, 3.5×10^{35} (units $\alpha = 1$) at $v = 70, 80, 90$.
- *Windows.* $v \lesssim 60$ for $B = 3 \times 10^{-3}$ (inter-level difference of Σ: 2×10^{-5} at $v = 60$, 8×10^{-3} at 70) and $v \lesssim 40$ for $B = 3 \times 10^{-2}$, where the run is non-finite beyond $v \approx 55$. The windows are shorter than for the Hayward h because κ_+ is larger and the front arrives earlier. The remedy of Section 9 applies: for $B = 3 \times 10^{-4}$ with $h_s = 0.05$ and $h_v = 0.0125, \ldots, 0.0015625$ ($v_1 = 200$) the first non-finite samples are 127, 135, 144, 153, i.e. 8.6 per halving of h_v, $\ln 4/\kappa_+ = 8.59$ with $\kappa_+ = 0.1614$, as in the Hayward case, and the window extends from $v \simeq 95$ to $v \simeq 125$ (inter-level difference of σ_v: 10^{-11} at $v = 95$, 1×10^{-4} at 125). σ_v continues to rise, $-0.0521, -0.0449, -0.0411$ at $v = 100, 115, 125$, and the relation $\sigma_v \simeq \rho_v/\rho$ of Section 12.5 holds to 4×10^{-4} for $v \geq 95$ (`bounded_aniso.log`). We have not built the kernel of the bounded model, so the prediction (26) is not tested there.

The dependence on the wall is, by scaling, a dependence on one parameter. Under $r \to \lambda r$ the bounded h with (m, α) maps to $(m/\lambda, \alpha/\lambda^2)$ (with $\psi \to \psi/\lambda^2$ and the times scaled by λ), so the equations depend on m and α only through $m/\sqrt{\alpha}$ (and on the tail through B, k and the scale); the Hayward case has $m/\ell = 5$. We sampled $m/\sqrt{\alpha} = 1.5$; the value 5 at $\alpha = 1$ is not usable (above), and a scan in $m/\sqrt{\alpha}$ is not done. The qualitative conclusion is the same for the two characteristic functions: an exponent that follows $-\kappa_-(m_f)$, a front, ψ approaching the pole or the wall without reaching it, a generalized mass growing without bound, $r \to r_-$ and σ_v rising from the static value to small negative values. In both $c_1 = 1/h' \to 0$ as ψ approaches the singular point of h, which suppresses the source of the Raychaudhuri equations (8).

14 Collapse-generated horizon tails

Up to here the tail on the event horizon was prescribed, $\phi_H = B(1+v)^{-k}$, with k and B free. This section replaces it by the horizon field of a self-gravitating collapse, in two stages, and then feeds that field to code E. What is new is the construction and its tests; the tail law itself is the known one.

What the literature fixes. For a massless $\ell = 0$ field on Schwarzschild, generic compactly supported data give a tail $\propto v^{-3}$ on $\mathcal{H}^+$ (t^{-3} at fixed r, u^{-2} for $r\phi$ at $\mathcal{I}^+$), and initially static data v^{-2} [Price, 1972, Gundlach et al., 1994a]; the coefficient of the compact-data tail was derived by Angelopoulos et al. [2018] in terms of a conserved quantity of the time integral of the field (we use only the exponent). In the nonlinear collapse the exponent is unchanged, the horizon index being measured as 2.98–2.99 by Burko and Ori [1997], while the amplitude is smaller than the linear estimate [Gundlach et al., 1994b]. Event horizons in double-null evolutions are located by integrating backwards from late time [Libson et al., 1996, Diener, 2003], and a collapse needs no regular-center treatment if the grid ends before the center and the horizon forms first [Burko and Ori, 1997]. We take these as the methods and the benchmark: the Einstein limit ($h = \psi$) of our code must reproduce the index, and the Hayward run is then compared with it.

Stage N: nonlinear collapse. The equations are those of Section 3 with $h = \psi$ (model 0) or $h = \psi/(1 - \ell^2\psi)$ (model 1), $\ell = 1$, in the gauge $\sigma = 0$ on both initial rays, evolved by the free scheme of Paper E [Ownby, 2026a] (second order; Section 9). Data: on $v = v_0 = 40$, $0 \le u \le u_{\mathrm{end}}$, flat space with $\phi = 0$ ($r = (v_0 - u)/2$, $r_u = -\frac{1}{2}$, $r_v = \frac{1}{2}$); on $u = u_0 = 0$ an ingoing pulse $\phi = A\sin^4[\pi(v - 44)/8]$, $44 \le v \le 52$, with r, r_v, r_u, σ_u, ϕ_u from the Raychaudhuri and r_{uv}, σ_{uv}, ϕ_{uv} equations along the ray. The grid ends before the center ($u_{\mathrm{end}} < v_0$), so no center is treated; the pulse is strong enough for the apparent horizon (found as the first sign change of r_v on each row) to form inside the grid, at $u_{\mathrm{AH}} = 28.04$ for $m = 5$ (the flat-space estimate $v_{\mathrm{c}} - 4m = 28$). The amplitude is tuned by bisection on the generalized mass of the ray $u = u_0$ behind the pulse (an ODE) to $m = 5.000000$ ($A = 0.030030$ in both models). Tests: (i) on the row $v = 70$ the generalized mass is 5 to 10^{-5} over $0 \le u \le 28$ (model 0); (ii) for model 1 ($h_u = h_v = 0.05$, grid to $u = 39$) the invariant $\psi = (1 + 4r_ur_v/W)/r^2$ on the row $v = 70$ equals the static Hayward function $2m/(r^3 + 2m\ell^2)$ to $\le 10^{-5}$ from $r = 30$ down to $r \approx 11$, to 4×10^{-5} at $r = 3.8$ and to 1.3×10^{-3} at $r = 1.0579$ (within 10^{-3} of $r_- = 1.05747$), beyond which, in the last ~ 5 units of u before the end of the grid, the deviation grows (1.5% to 6%, and 45% at the last point): the immediate neighborhood of the inner horizon is not resolved at this mesh (`interior.log`): *the collapse leaves, behind the pulse, the static Hayward interior of the initial row of Section 8, between the horizon and the inner horizon*, which is the physical content of the data used so far; (iii) the Hayward and Einstein tables differ by at most 2% in the tail coefficient (below), as the size of the correction $\ell^2\psi = 1/r_+^2 \le 1.02 \times 10^{-2}$ at r_+ ($m_f \ge 5$) suggests.

Stage L: the tail on the static exterior. In Eddington coordinates $u_e = t - r_*$, $v_e = t + r_*$ of the final static metric ($f = 1 - 2m/r$ or $1 - 2mr^2/(r^3 + 2m\ell^2)$, with r_* in closed form from the three roots of $r^3 - 2mr^2 + 2m\ell^2$), the field $\psi = r\phi$ obeys $\psi_{uv} = -\frac{1}{4}\,(ff'/r)\,\psi$, for either h because the scalar is minimally coupled. We integrate it with the second-order null-diamond scheme of Gundlach et al. [1994a], $\psi_N = \psi_E + \psi_W - \psi_S - \frac{1}{8}h^2V_c(\psi_E + \psi_W)$, from the data of stage N on the ray $v = v_1$ (the field $\phi = \int \phi_u \, du$ as a function of r, by a cubic spline continued through the horizon into the first points of the trapped layer) and $\psi = 0$ on $u = u_0$ ($v > 52$). The horizon field is read on the line $r_* = -9.5/(2\kappa_+)$ (error e^{-19}). Because the gauge $\sigma = 0$ on $u = u_0$ makes v non-affine in the exterior, $v_e = 2r_*(r(u_0, v))$ along that ray.

Tests of the two stages. (a) *Overlap.* Stage N run to $v = 150$ against stage L started at $v = 70$: ψ on the row $v = 150$ agrees to 5×10^{-5} ($r_* = 35$), 2.5×10^{-4} ($r_* = 20$), 1.4×10^{-3} ($r_* = 2$) and 1.6×10^{-2} at $r_* = -10$, where stage N has lost resolution ($e^{\kappa_+(v - v_c)}$ compression of the u grid) (`overlap.log`); this is the reason for the second stage, and it shows the linearization

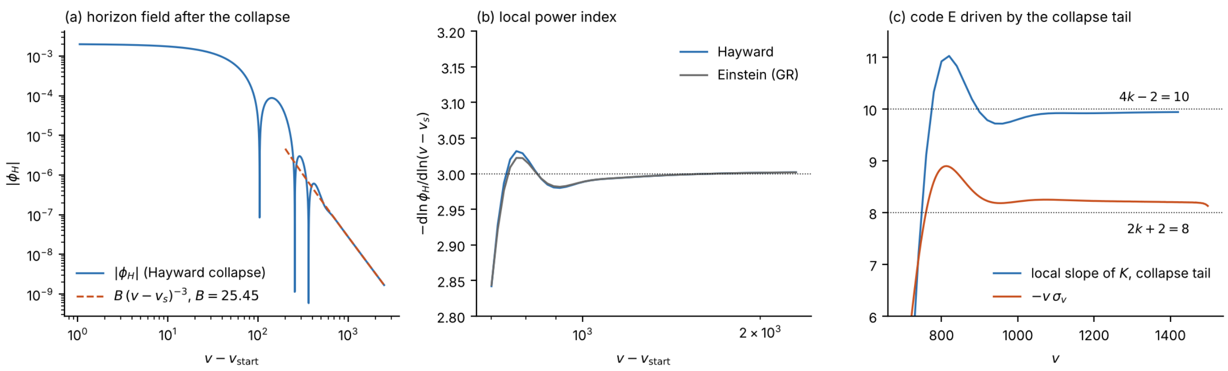

Figure 2: (a) Horizon field of the Hayward collapse ($m=5$, $\ell=1$) from the start of stage L and the fit $B(v-v_s)^{-3}$. (b) Local power index of ϕ_H for the Hayward and Einstein cases. (c) Code E driven by this table: local slope of K and $-v\sigma_v$ on $U\simeq 1$ against $4k-2=10$ and $2k+2=8$ ($k=3$).

after v_1 is accurate. (b) *Switching time and resolution.* The horizon table is independent of the switching time $v_1=60, 70, 80$ to 3×10^{-5} (relative to the maximum of $|\phi_H|$ in each window of $v-v_1$) and of the mesh ($h_N=h_L=0.1\to0.05$) to 5×10^{-5}, for both models (`tables.log`); stage L alone is converged to 10^{-6} ($h_L=0.1\to0.05$). (c) *Scale invariance.* In the Einstein limit with the geometry scaled by $m/5$ the coefficient is $B/m^3=0.2017$ for $m=3, 5, 10$ to four digits (`mscale.log`). (d) *Hayward against Einstein.* $B/m^3=0.2052, 0.2033, 0.2015$ for $m=3, 5, 10$ (window $[1000,2900]\,m/5$, hence $B=25.41$ for $m=5$ against 25.45 in the window below): the Hayward exterior changes the tail by +1.7%, +0.8%, −0.1%.

Result: exponent and amplitude. For $t=v-v_{\rm start}\in[1200,2400]$ ($v_{\rm start}$ the start of stage L, $v_1=60$) the free fit of $\phi_H=B(v-v_s)^{-k}$ gives $k=3.009$ for both models; with $k=3$, $B=25.26$ ($v_s-v_{\rm start}=23.8$) for the Einstein limit and $B=25.45$ (23.2) for the Hayward case, $m=5$; the local index $-{\rm d}\ln\phi_H/{\rm d}\ln(v-v_s)$ is within 0.01 of 3 for $t\gtrsim900$ (Fig. 2b). Before that the field oscillates through zero (three sign changes up to $t\approx400$, Fig. 2a; the fundamental $\ell=0$ quasi-normal mode of Schwarzschild has $\omega m=0.110-0.105i$, i.e. a decay rate 0.021 for $m=5$), the local index wanders by ±0.03 for $700\lesssim t\lesssim1000$, and a three-parameter fit to a window that starts at $t\le600$ is ill-conditioned ($k=3.4$, $v_s<v_{\rm start}$); we do not use it. The tail is thus the *generic* v^{-3} one; the exponent $k=2$ requires data with a nonzero Newman–Penrose constant, which a collapse from compact data does not produce, and is treated below. The amplitude of the collapse field in our family depends on the pulse: for amplitudes 0.8–1.2 of A (generalized masses 3.2–7.2) $B=16.5, 21.1, 25.4, 28.8, 30.6$, so $B\propto m^{1.0}$ at small m and saturating, not A^3 (`ascale.log`); and for the same pulse profile sent on the *static* $m=5$ Hayward exterior as a test field, $B_{\rm lin}=68.8$, so the nonlinear tail is 0.37 of the linear one (`linear.log`; the "linear" data are defined by the pulse in v_e on the same ray and are a convention). The reduction below the linear estimate is the qualitative statement of Gundlach et al. [1994b]; we do not reproduce their factor, which refers to other data.

Code E on the collapse tail. The table $\phi_H(v)$ ($v=0$ at $v_{\rm start}$, spacing 0.1, to $v=2500$) replaces $B(1+v)^{-k}$ in the horizon Raychaudhuri equation (backward shooting from $r_v=0$ at v_1+300, so the horizon column is the teleological one for this table), in ϕ_v^H and in the starting data; the static row of Section 8 is used unchanged, as the test (ii) above supports, and the field on it is the collapse profile $\mathrm{d}\phi/\mathrm{d}r$ of stage N times the static r_U (`PUOMODE=profile`) rather than $\phi_U = 0$. All else is code E (Section 11.3, $h_s = 0.2$, $h_v = 0.2$, to $v_1 = 1500$, $\kappa_+ v_1 = 73$; `genc_*.json`). Results (ray $U = 0.9968$; `collapse_E.log`):

- the front arrives at $v \simeq 53$–78 on the four rays (the first v at which $1 - \ell^2\psi$ has fallen to half its value at $v = 45$), earlier than for the prescribed $B = 3 \times 10^{-3}$, $k = 3$ (73 on $U \simeq 1$) because the quasi-normal ringing, $|\phi_H| \sim 2 \times 10^{-3}$ at the start, is a larger and more rapidly varying flux than the tail;
- behind the front, $zm = 0.59117$ against the closed form 0.59126 (and 0.59117 for the prescribed $k = 3$ run), $\mathrm{d}m/\mathrm{d}v = 4.403$ in both, $m = 5742$ at $v = 1400$; $r = r_-$ to 10^{-5};
- $-v\sigma_v = 8.246$ at $v = 1100$ and 8.20 at 1400, tending to $2k + 2 = 8$ from above in modulus (8.081 at 1100 for the prescribed run);
- the local slope of K is 9.915 at $v = 1100$ and 9.94 at 1450 on $U \simeq 1$ (9.83, 9.90, 9.94, 9.95 at $v = 1400$ on the four rays), against $4k - 2 = 10$ and 9.47 for the prescribed run at 1100; it overshoots to 11 near $v = 800$, when the oscillations of the field have died down (Fig. 2c);
- the amplitude of K is not universal: $K(1400) = 1.3 \times 10^{33}\ell^{-4}$ on $U \simeq 1$, and with $\phi_U = 0$ on the starting row 1.3×10^{31} (factors 2.6, 8.3, 99, 6×10^4 on the four rays at $v = 1100$), while ϕ_v agrees to 4×10^{-6}, σ_v to three digits and m, z to 0.6% (the field ϕ_U on a fixed ray differs by the factors 0.76, 0.53, 0.13, -0.13 on the four rays, which enters K through $g = \phi_U\phi_v/W$); the exponents are the same. K at fixed late v decreases with the tail amplitude in the prescribed runs ($K(1000) = 5 \times 10^{37}$, 5×10^{29}, 5×10^{21} for $B = 3 \times 10^{-4}$, 3×10^{-3}, 3×10^{-2}, $k = \frac{3}{2}$), so no ordering with the collapse amplitude should be read into the values.

Mesh dependence ($h_v = 0.2 \to 0.1$, $h_s = 0.2$; `conv_c.log`): the change is 4.3×10^{-6} in r (relative), 8×10^{-6} in σ_v (absolute; 1.1×10^{-5} at $v = 100$ on the first ray) and in ϕ_v (relative), $(1$–$4) \times 10^{-5}$ in $\ln m$ (at most 3.6×10^{-5}) and $\leq 7.1 \times 10^{-4}$ in $\ln K$ at $v = 100$, decreasing to $\leq 1.2 \times 10^{-4}$ at $v = 1400$, on all four rays; $-v\sigma_v = 8.19$ at $v = 1400$ for $h_v = 0.1$ (8.20 for 0.2).

The other exponent: data with a nonzero Newman–Penrose constant. *What the literature fixes.* For $\ell = 0$ on a static, asymptotically flat, spherically symmetric exterior, Angelopoulos et al. [2018] prove that if the Newman–Penrose constant $I_0 = \lim_{r\to\infty} r^2\partial_r(r\phi)$ (with ∂_r at fixed u; it is the same on every outgoing cone) does not vanish, then $\phi \sim 4I_0\,\tau^{-2}$ on and outside the horizon ($\tau = v$ on $\mathcal{H}^+$), with a coefficient that does not contain m, while for compactly supported data the leading term is $-8I_0^{(1)}\tau^{-3}$, $I_0^{(1)}$ being the constant of the time integral of the field. These are the two exponents of Price's law, $k = 2$ for initially static and $k = 3$ for generic compact data [Price, 1972, Gundlach et al., 1994a]; Gundlach et al. [1994b] found them in nonlinear collapse at fixed r ($-2.08\ldots-1.86$ for static data against -2, $-2.74\ldots-2.63$ for Gaussian data against -3), and momentarily stationary compact data fall faster, as t^{-4} [Karkowski et al., 2004, Price and Burko, 2004]. For the wave equation on a *given* dynamical black-hole exterior that settles to Schwarzschild, Gajic and Kehrberger [2025] write the leading coefficient of the compact-data tail as a time-inverted Newman–Penrose constant with a part fixed by the initial data and a part that is an integral over the horizon (through the expansion of its generators) and over null infinity (through the integral of the Bondi news, whose spherical average is the Bondi mass); this is the structure that Eqs. (33) and (34) below make explicit in spherical symmetry for strong data (initial-data charge Q_0 and horizon term Q_{abs}). Their metric is an assumption and is not required to solve the Einstein equations. For the self-gravitating scalar they refer to Gautam [2024], who proves, for the spherically symmetric Einstein–Maxwell–scalar system, for smooth compactly supported data that settle to a subextremal Reissner–Nordström hole ($0 < |e| < M$) and without smallness, that φ on the event horizon is $C_0\,\mathfrak{L}[\varphi]\,v^{-3} + O(v^{-3-\delta})$ with an explicit non-zero C_0 and a dynamical constant $\mathfrak{L}[\varphi]$ that is non-zero exactly for the generic class of data of Luk and Oh (as read, the constant is not given in terms of the data, and the uncharged case is not stated separately). We did not find a published horizon coefficient for self-gravitating collapse data with a nonzero Newman–Penrose constant; the theorem of Angelopoulos et al. [2018] is for the linear field on a fixed stationary background, and the question here is whether the strong-field collapse leaves it unchanged. *Data.* To the pulse on $u = u_0$ we add, as a function of the areal radius along the ray, $\phi_{\rm tail} = b\,S\big((r-r_a)/\Delta r\big)/r^2$, $S(x) = x^4(35-84x+70x^2-20x^3)$ (a C^3 ramp), $r_a = 25.3$, $\Delta r = 1.8$ (it ends at $r = 27.1 < r(u_0, v_1) = 27.51$): $\phi_v = \phi'_{\rm tail}(r)\,r_v$ enters the ray equations of stage N, $r\phi = b/r$ for $r \geq r_a + \Delta r$, and $I_0 = -b$ exactly, the other data being those above (A is retuned to $m = 5$: $A = 0.030015$ and $m_f = 4.99982$ at (u_0, v_1) for $b = 1$; the tail carries 1.8×10^{-4} of mass beyond v_1, which is not in the static background of stage L). In stage L the same $\psi = b\,S/r$ is the datum on $u_e = u_{e0}$, $v > v_1$, with $r(r_*)$ of the static exterior, and the row of stage N carries the constant $\phi(u_0, v_1) = b/r^2$ in $\phi = \phi(u_0, v_1) + \int \phi_u\,\mathrm{d}u$ (it is zero for the compact pulse).

Tests. (a) *Stage L alone*: $\psi = 0$ on the row, the tail on the ray, the static $m = 5$ exterior of Schwarzschild or Hayward, $h_L = 0.2$, to $t = 12800$ (`nplin.log`). $\phi_H\,t^2/(-4b) = 0.9295, 0.9649, 0.9826, 0.9914$ at $t = 800, 1600, 3200, 6400$ ($t = v - v_{\rm start}$); fitting $a_0[1 + (c_1 + c_2 \ln t)/t + (c_3 + c_4 \ln t)/t^2]$ on $t \geq 800$, 1600, 3200 gives $a_0 = 1.0000 \pm 0.0002$ (to 4×10^{-5} with the full form), the same for the two backgrounds to 10^{-4}, and a free fit $k = 2.0005$: the coefficient of the theorem, $4I_0$ with no free parameter, is reproduced, including the $\ln t/t$ correction. (b) *Nonlinear collapse* (stage N to $v_1 = 60$, $h_N = h_L = 0.1$, to $t = 12800$; `np_fit.log`): the same fits give $a_0 = 1.0000$ for $b = 1$ (Hayward and Einstein), -1, 0.5, 2 ($|a_0 - 1| \leq 3 \times 10^{-5}$ for the full form and $t \geq 1600$, $\leq 1.5 \times 10^{-4}$ with only the c_1/t correction), and the free fit gives $k = 2.0013, 2.0007, 2.0004$ for windows starting at $t = 800$, 1600, 3200 with $B/(4|b|) = 1.013, 1.007, 1.004$ (fitted shift $s \simeq -44$ in $B(t-s)^{-k}$, which is why B at a given window differs from $4|b|$: the leading coefficient is a_0). The collapse changes the transient and the subleading terms (Fig. 3a: for $t \lesssim 700$ the curves differ, with minima 0.29 and 0.52 of the limit for the collapse and the linear test, and they merge afterwards) and not the limit. The response is linear in the amplitude of the tail: $[\phi_H(b) - \phi_H(0)]/b$ for $b = 0.5, 1, -1, 2$ (the pulse is retuned to $m = 5$ each time) differs by a relative 8×10^{-3} at $t = 50$, 10^{-4} at 100, 6.5×10^{-4} at 200 and $\leq 6 \times 10^{-5}$ for $t \geq 400$ (`np_lin_b.log`). (c) *Matching*: stage N to $v = 150$ against stage L from $v = 70$ (Einstein, $b = 1$) agrees to 10^{-3} at $r_* = 35$, 1.3×10^{-4} at 20 and 1.1×10^{-4} at -9 (`overlap_np.log`; 4×10^{-5}, 2.5×10^{-4} and 1.6×10^{-2} for the compact pulse). Thus the index is 2 and the coefficient is $4I_0$ in the collapse, in both models: the strong-field stage does not change the leading coefficient, which is fixed by the data at large radius. The collapse does not enter that coefficient, in contrast with the compact case below.

Data with a nonzero scalar charge at infinity ($r\phi \to \psi_\infty \neq 0$). The datum above has $r\phi \to 0$ at infinity. The initially static case of the literature has instead a field that is the static solution of the wave equation outside a radius: on the static background of mass m the solution regular at infinity obeys $r^2 f\,\partial_r\phi = \text{const}$, $g(r) = \int_0^{1/r} \mathrm{d}s/f(1/s) = 1/r + m/r^2 + O(r^{-3})$ (for the Hayward f too, since $f = 1 - 2m/r + O(r^{-4})$), so that $r\phi \to \psi_\infty = c$ and $I_0 = -c\,m$ for $\phi = c\,g(r)$. We use $\phi_{\rm tail} = c\,S\big((r-r_a)/\Delta r\big)g(r)$ on $u = u_0$ (g for the target mass m_t and the model run; a spline of the quadrature, checked against $g\,r \to 1$ and $(g - 1/r)r^2/m \to 1$), with $c = 0.02$, the ramp of the previous datum and the same stage N/stage L pipeline; in stage L the same profile is the datum on the ray and the row carries $\phi(u_0, v_1) = c\,g(r(u_0, v_1))$ (`collapse_lib.py:np_tail`, `strong_table.py` with `TAIL_C`). The tail carries a mass of order c^2 beyond v_1 that the static background of stage L does not contain ($m_f = 4.99962$, 4.99849, 4.99397 for $c = 0.01$, 0.02, 0.04 against 5: deficits 3.8, 15.1, 60.3×10^{-4}, i.e. $\propto c^2$); the prediction below uses $m_t = 5$. Result (collapse of the pulse from flat space, $m_0 = 0 \to m_t = 5$, $h_N = h_L = 0.1$, $t \leq 12700$; `tc_fit.py`, `tc_fit.log`): the horizon field is $\phi_H = a_0 \cdot 4I_0\,t^{-2}[1 + \ldots]$ with $a_0 = 0.99993$ (Hayward, $c = 0.02$, window $t \geq 800$, full form), 0.99993 (Einstein), 0.99999 ($t \geq 1600$) and 1.0000 ($t \geq 3200$) for both; for the Hayward model at $c = 0.01$ and 0.04 $a_0 = 0.99999$ and 0.99990 ($t \geq 800$), 1.0000 and 0.99999 ($t \geq 3200$); for a pulse collapsing into an existing hole ($m_0 = 2.5 \to 5$) 0.99992, 0.99999; for a hole of twice the mass ($m_0 = 0 \to 10$, $t \leq 25400$, windows scaled by 2) 1.0000 at $t \geq 3200$ (the window $t \geq 800$ is too early on that scale: 1.0045). Free-index fits $\phi_H = B(t - s)^{-k}$ give $k = 2.0020$, 2.0010, 2.0005 for $t \geq 800$, 1600, 3200 ($c = 0.02$, Hayward and Einstein alike) with $B/(4|I_0|) = 1.020$, 1.010, 1.005 (the offset s again), and 2.0004–2.0010 for the other cases at $t \geq 3200$ (2.0021 for $m_t = 10$). Thus, to 10^{-4}, the horizon field of the strong collapse of data with $\psi_\infty \neq 0$ is the linear-theory value $4I_0t^{-2}$ with $I_0 = -c\,m$: it is linear in c (a factor 4 in c, 16 in the mass defect, leaves a_0 unchanged to 10^{-4}), the same in Einstein gravity and in the Hayward model, and independent of m_0 and of the scale of m. The constant $I_0 = -c\,m_t$ is a property of the data (g is built with m_t), and the data prefer it to $-c\,m_f$ ($a_0 = 0.99999$ against 1.0012 at $c = 0.04$, $t \geq 3200$): the mass of order c^2 that the tail carries beyond v_1 does not enter the coefficient.

Code E on the $k = 2$ tail ($b = 1$ and 0.1; ϕ_H table to $v = 2500$ with $h_N = 0.05$, $h_L = 0.1$, $v_1 = 60$; $h_s = h_v = 0.2$ to $v_1 = 1500$; `collapse_E_k2.log`). The starting row is the static one of Section 8 (the interior stays at $m = 5.0011$ and $r = r_- = 1.05746$ up to the front, as it must; a row without the constant $\phi(u_0, v_1)$ puts $m \simeq 5.4$ there and $r = 1.0526$, and was discarded). Behind the front ($v_{\text{front}} = 53$, 80, 67, 61 on the four rays for $b = 1$): $zm = 0.59114$–0.59119 ($\frac{1}{2}r_-^3 = 0.59126$), $\mathrm{d}m/\mathrm{d}v = 4.42$ at $v = 1400$ for both amplitudes (closed form 4.449; 4.41 for $k = 3$), i.e. independent of k and b as in Section 12.9; $v\sigma_v = -5.800$, -5.839, -5.867 at $v = 1000$, 1200, 1400 ($b = 1$) and -5.507, -5.612, -5.685 ($b = 0.1$), tending to $-(2k + 2) = -6$ from below; the local slope of K is 5.07, 5.23, 5.34 ($b = 1$) and 4.37, 4.68, 4.90 ($b = 0.1$) at the same v, still 11% and 18% below $4k - 2 = 6$, and with the two $1/v$ terms of Eq. (30) added back, with the measured v_0 (81 and 96) and $v\sigma_v$, it is 5.998, 5.993, 5.987 and 5.995, 5.991, 5.986 on $U \simeq 1$, and 5.99–6.00 on the other three rays (Fig. 3c): $K \propto v^{4k-2}$ with $k = 2$ holds in the sense of Section 12.9, with v_0 delaying the approach more for the weaker tail. The fit $a + c/v$ to the slope of $b = 1$ for $1000 \leq v \leq 1450$ gives $a = 6.01$ ($c = -937$). K at $v = 1400$ is 4.5×10^{21} ($b = 1$) and $1.4 \times 10^{32}\ell^{-4}$ ($b = 0.1$) on $U \simeq 1$ (and 1.3×10^{33} for the $k = 3$ collapse tail), another sign that the amplitude of K is not universal. Mesh ($h_v = 0.2 \to 0.1$, $b = 1$; `conv_c_k2.log`): $\leq 7.6 \times 10^{-4}$ in $\ln K$ at $v = 100$, $\leq 4 \times 10^{-4}$ at $200 \leq v \leq 400$ and $\leq 1.5 \times 10^{-4}$ at $v \geq 800$, 1.3×10^{-5} in σ_v, 4.3×10^{-6} in r, 5×10^{-5} in $\ln m$ and 9×10^{-6} in ϕ_v, on all four rays.

The coefficient for compact data. The field equation implies the exact conservation law $\partial_u(r^2\phi_v) + \partial_v(r^2\phi_u) = 0$ (the Noether current of $\phi \to \phi + c$), so $Q = \int r^2\,\mathrm{d}\phi$ along the ray $u = u_0$ is a property of the data, in the linear and nonlinear problems alike. In the linear problem (pulse on the ray, $\psi = 0$ on the row, static background) the fitted coefficient of the v^{-3} tail is $B_{\text{lin}} = 68.24$, 38.38, 103.9 for the three amplitudes $A/A_0 = 1$, 0.8, 1.2 ($m = 5$, 3.21, 7.18), and $8m|Q| = 68.24$, 38.38, 103.9 ($|Q| = 1.706$, 1.495, 1.810): ratio 1.0000 ± 10^{-4} for the first two, 1.001 for the third (the windows start at $t \geq 1600$; `linear_q.log`), and the same for Schwarzschild and Hayward, so that $I_0^{(1)} = +mQ$ in the notation of Angelopoulos et al. [2018] for this class of data ($Q < 0$ for the positive pulses used here, so $I_0^{(1)} < 0$ and $B = -8I_0^{(1)} = 8m|Q|$) (for the Schwarzschild exterior the integral expression of this constant is Eq. (1.5) of Gajic and Kehrberger

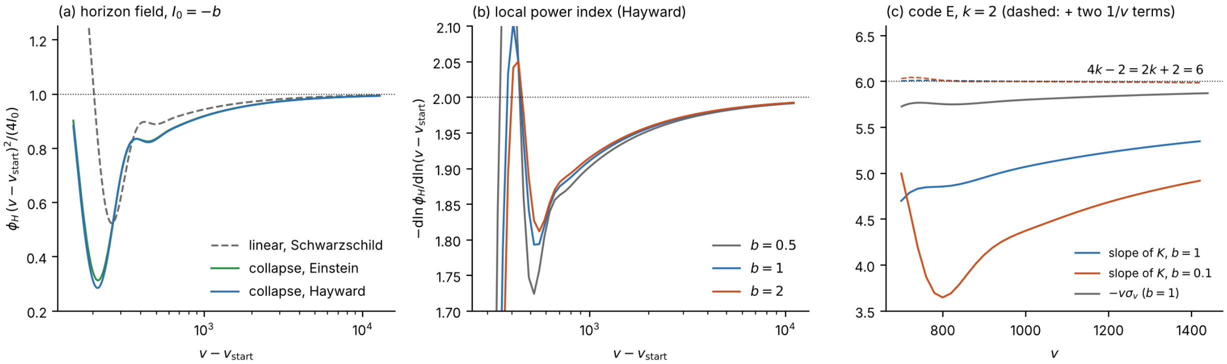

Figure 3: Collapse data with a nonzero Newman–Penrose constant ($I_0 = -b$). (a) $\phi_H t^2/(4I_0)$ for the linear problem on Schwarzschild and for the Einstein and Hayward collapses ($m = 5$, $b = 1$): the coefficient tends to 1. (b) Local index of ϕ_H for the Hayward collapse, $b = 0.5$, 1, 2. (c) Code E driven by the $k = 2$ tables: local slope of K (solid) and with the two $1/v$ terms added back (dashed) for $b = 1$, 0.1, and $-v\sigma_v$ ($b = 1$), against $4k - 2 = 2k + 2 = 6$.

[2025] with $g = g_M$, which gives $I_0 = \frac{1}{2}MQ$ in their normalization and hence $B = 8M|Q|$, see the discussion of that theorem below; for the Hayward exterior the constant is identified numerically; $Q = 0$ for momentarily stationary data, which fall as t^{-4}, is consistent). In the collapse the measured coefficient is $B/(8m|Q|) = 0.430, 0.403, 0.373, 0.336, 0.294$ for $A/A_0 = 0.8, 0.9, 1, 1.1, 1.2$ (`ascale_q.log`: 0.4302, 0.4026, 0.3727, 0.3358, 0.2944; this is the 0.37 of the linear estimate quoted above): Q is the same, m is that of the final hole, and the collapse removes a fraction of the tail that grows with the strength of the pulse; the charge on the exterior part of the row $v = 60$ ($r > r_+(m)$; `ascale_q.py`, `ascale_q.log`) is $-0.27, -0.37, -0.43, -0.39, -0.33$ and does not reproduce B either ($B/(8m|Q_{\rm row}|) = 2.4, 1.8, 1.5, 1.5, 1.6$). So the v^{-3} coefficient of a collapse is a functional of the nonlinear history, which we have not derived, while the v^{-2} one is a property of the data at infinity.

Strength of the infalling pulse and the inner radius. Shao et al. [2026] evolve a scalar field on the Hayward geometry sourced by non-linear electrodynamics in general relativity and report that a strong enough field shrinks the inner horizon to zero (we use their abstract; their threshold is a statement about a different theory). The analog in the present theory is a test of whether the inner radius depends on the strength of what falls in. *Static bound.* $r_-(m)$ is the smallest positive root of $g(r) = r^3 - 2mr^2 + 2m$ ($\ell = 1$); $g(0) = 2m$, $g(1) = 1$ and $g'(r) = r(3r - 4m) < 0$ on $(0, 4m/3)$, so $g > 0$ on $(0, 1]$ and $r_- > \ell$ for every m with horizons, with $r_- \to \ell$ only as $m \to \infty$ ($r_- = 1.0575, 1.0267$ for $m = 5, 10$). A threshold at which $r_- \to 0$ is therefore not available to a Hayward hole of this theory by the choice of its mass, and the question is whether the dynamics can do it. *Data.* To the horizon table of the $k = 3$ collapse of this section (Hayward, $m = 5$) we add a prescribed pulse $P\sin^4\bigl(\pi(v - 10)/20\bigr)$ on $10 \le v \le 30$ (ϕ_H is event-horizon data as in Section 12.9; this pulse is not generated by a collapse), and run code E on the starting row of $m = 5$ ($h_s = h_v = 0.2$, $v \le 1000$, `codec/sc/`). The horizon column is the event horizon: backward shooting of the horizon Raychaudhuri equation from $v = 1300$ ($= v_1 + 300$) with $r_v = 0$ there, for the final radius r_f for which $r(0) = r_+(5)$ (the same r_f to 10^{-4} when the shooting starts at $v = 310$), and m_f is the mass of the static hole with $r_+(m_f) = r_f$. The root exists on a monotone branch for $P \le 0.08$. For $P = 0.09$ the map $r_f \mapsto r(0)$ has a maximum of $9.19 < r_+(5) = 9.90$ near

$r_f = 30$ (a fold) and returns to $r_+(5)$ only at $r_f \simeq 145$ ($m_f \sim 70$, a branch we do not follow); at $P = 0.1$ the local maximum is 5.1 (`horizon_map.log`). This is a property of prescribing the field on the horizon: a Vaidya-like estimate $\mathrm{d}m/\mathrm{d}v = 4\pi r_+(m)^2\phi_v^2$ (not exact: it overestimates m_f by 0.3% at $P = 0.01$ and 18% at 0.08, and the Einstein version does the same) has a pole at a finite strength, between $P = 0.10$ ($m_f = 21.6$) and 0.12, because the horizon area grows with the mass (`vaidya_estimate.log`), so a field profile fixed on the horizon cannot be scaled up indefinitely on a starting row of mass 5. The scan ends at $P = 0.08$ (a horizon mass gain of 3.3 on $m = 5$).

P	r_f	m_f	r_∞	m^*	m_{ray}	$r_-(m_f)$	$\mathrm{d}m/\mathrm{d}v$	asympt.
0.01	9.946	5.024	1.05711	5.028	5.029	1.05716	4.41	4.48
0.03	10.354	5.226	1.05436	5.248	5.260	1.05462	4.61	4.70
0.05	11.357	5.723	1.04903	5.746	5.752	1.04925	5.11	5.20
0.07	13.723	6.898	1.03985	6.915	6.916	1.03996	6.33	6.38
0.08	16.597	8.329	1.03223	8.396	8.396	1.03251	7.82	7.87

Table 14: Pulse-strength scan of code E ($m = 5$ start, ray $U \simeq 1$). r_f: final radius of the event horizon; m_f: mass with $r_+(m_f) = r_f$; r_∞: radius of the ray at $v \geq 200$ (constant to 2×10^{-6} after the front); m^*: the mass with $r_-(m^*) = r_\infty$; m_{ray}: $r^3\psi/(2z)$ on the ray at the first v with $|r - r_\infty| < 3 \times 10^{-5} r_\infty$ ($v = 42, 45, 50, 64, 84$), before inflation; $\mathrm{d}m/\mathrm{d}v$: the mean over $800 \leq v \leq 900$ against the closed form (27) at r_∞.

Result (Table 14; `scan_summary.log`). The inner radius behind the front is $r_\infty = r_-(m_f)$ to 5×10^{-5} at $P = 0.01$ and to $(1\text{–}3) \times 10^{-4}$ for the others; the mass m^* implied by r_∞ differs from m_f by 0.1–0.8%, and from the ray mass just before inflation by 0.03%, 0.22%, 0.12%, 0.003% and $< 10^{-4}$ for the five strengths. The radius falls from 1.0571 to 1.0322 as the mass grows by 3.3 and stays above ℓ; $zm \to \frac{1}{2}r_-^3$ in each run (`scan_zm.log`) (0.59055, 0.58596, 0.57713, 0.56213, 0.54987 at $v = 900$ against 0.59065, 0.58605, 0.57721, 0.56219, 0.54992) and the rate $\mathrm{d}m/\mathrm{d}v$ follows the closed form (27) evaluated at the new r_-, rising from 4.4 to 7.8 (the measured value is 0.7–1.8% below it, which is of the size of the mesh dependence (below): the closed form is the limit $v \to \infty$, approached from below as in Section 12.6). The late-time quantities that depend on the tail and not on the pulse do not: $v\sigma_v = -8.30, -8.31, -8.33, -8.38, -8.45$ and $\mathrm{d}\ln K/\mathrm{d}\ln v = 10.02, 10.03, 10.02, 9.94, 9.83$ at $v = 900$ (`scan_an.log`; $-(2k+2) = -8$ and $4k-2 = 10$ for $k = 3$, approached from below as in the single-amplitude runs). So within the range reached the pulse sets the mass and, through it, r_-, $\mathrm{d}m/\mathrm{d}v$ and the amplitude of K, and not the structure behind the front; the inner radius does not vanish at any strength reached, in line with the bound above. We do not see a threshold of the kind reported for the electrodynamic model; since the Hayward form here is a property of the gravitational theory and not of a matter background, and the data are restricted to be regular on the event horizon (for which m can only grow, Proposition 1), this is a difference between the two systems and not a test of their result. *Mesh.* Halving h_v at $P = 0.08$ ($h_s = 0.2$; `conv_P0.08_hv.log`) changes, over the four rays, r by 4.4×10^{-5} ($r_\infty = 1.03227$, so the difference from $r_-(m_f)$ is not a mesh effect), σ_v by 4×10^{-5} (7×10^{-4} at $v = 100$), $\ln m$ by 9×10^{-4} and $\ln K$ by 7×10^{-3} for $v \geq 400$ (8.4×10^{-3} at $v = 200$, 2.2×10^{-2} at $v = 100$; the amplitude of K is the least converged quantity, as the pulse is the strongest); $\mathrm{d}m/\mathrm{d}v = 7.809$ against 7.816, and the slope of K at $v = 900$ is 9.825 in both. Halving h_s at $P = 0.07$ ($h_v = 0.2$; `conv_P0.07_hs.log`) moves r_∞ by $+5.3 \times 10^{-5}$ to 1.03991 (5×10^{-5} from $r_-(m_f)$), σ_v by $\leq 2.6 \times 10^{-4}$ ($\leq 1.2 \times 10^{-6}$ for $v \geq 200$), $\ln m$ by 10^{-3}–4×10^{-3}, $\ln K$ by 6×10^{-4}–1.4×10^{-2} for $v \geq 200$, $\mathrm{d}m/\mathrm{d}v$ from 6.334 to 6.279 (0.9%, so the 0.7–1.8% offset from the closed form is of the size of the mesh dependence) and the slope of K at $v = 900$ from 9.940 to 9.973. Not shown: pulses with $P > 0.08$ (no such event-horizon column exists, see above), data that are not regular on the event horizon, a pulse that changes the starting row itself (the collapse-generated pulse is the next paragraph; here the row is the static one of $m = 5$ and the pulse arrives at $v = 10$, with the horizon column carrying the anticipation).

A strong pulse generated by collapse into an existing hole. The scan above ends at a fold of the horizon column, which we attribute to prescribing ϕ on the horizon; the test is a pulse that is generated by the evolution. *Data.* In stage N the data on $v = v_0$ are now those of a static hole of mass m_0 and the data on $u = u_0$ carry the pulse as before. In the gauge $\sigma = 0$ on both rays the static solution has $r_u = -\frac{1}{2}$, $r_v = \frac{1}{2}f(r)$ and $\sigma_v = \frac{1}{2}\big(f'(r) - f'(r_c)\big)$ along $v = v_0$ ($r = r_c - u/2$, $r_c = (v_0 - u_0)/2$), and $r_v = \frac{1}{2}f(r_c)$ before the pulse and $\sigma_u = 0$ at the corner along $u = u_0$ ($f = 1 - 2m_0r^2/(r^3 + 2m_0\ell^2)$; $f' = rQ(\psi)$ makes the ray equation $\partial_v r_u = -\frac{1}{4}rQ$ consistent with the static data); $m_0 = 0$ is the flat start used so far (`collapse_lib.rays(..., m0=)`). The geometry is scaled with $s = \max(1, m_f/5)$ ($r_c = 20s$, pulse on $v_0+[4, 12]s$, $v_1 = v_0+20s$, $h_N = 0.1s$, $h_L = 0.1s$), the data on v_0 stop at $r_+(m_0) + 0.1$ (deeper cells of the old interior are amplified as $e^{\kappa_- v}$ and are not needed: the dependence is on smaller u only), and A is tuned so that the generalized mass on u_0 after the pulse is m_f (`strong.py`, `strong_table.py`; stage L with the static exterior of m_f as before). *Tests.* (i) $A = 0$, $m_0 = 5$: the row $v_1 = 60$ stays the static hole, $\max |m - m_0| = 7.1 \times 10^{-4}$, 1.8×10^{-4}, 4.5×10^{-5} for $h = 0.2$, 0.1, 0.05 (ratio 4.0; `t_static_h.py`). (ii) $m_0 = 0$ reproduces the earlier table ($A = 0.030030$, v_{1e} to 10^{-12}, ϕ_H to 3.8×10^{-5} of its maximum). (iii) After a pulse that raises $m_0 = 5$ to $m_f = 10$ or 25 the row v_1 is the static hole of m_f: $\max |\psi/\psi_s - 1| = 3 \times 10^{-6}$ (7×10^{-6} at $0.05s$) for $r > 1.5r_+$ and $\leq 4 \times 10^{-5}$ down to $0.9r_+$ for $m_f = 10$ (2.5×10^{-5} for $m_f = 25$), at $h_N = 0.1s$ (the bounds 4×10^{-5} and 2.5×10^{-5} hold at $0.05s$ as well) (`sp_interior.py`); halving h_N or h_L changes ϕ_H by $\leq 3.5 \times 10^{-5}$ and 1.5×10^{-6} of its maximum (`strong_conv.py`), and stage N to $v_1 + 80s$ against stage L from v_1 agrees to 4×10^{-5}, 3×10^{-5}, 9×10^{-5} at $r_* = 69$, 41, 12 ($m_f = 10$; `overlap_h.py`).

m_0	m_f	A	$\lvert Q\rvert$	B	$B/(8m_f\lvert Q\rvert)$	B/m_f^3
5	5.01	0.00140	0.0464	1.87	1.004	0.0148
5	5.5	0.00939	0.398	16.30	0.930	0.0980
5	6.5	0.01487	0.960	41.15	0.824	0.1498
5	8	0.01885	2.004	92.94	0.725	0.1815
0	10	0.03003	6.824	202.68	0.371	0.2027
2.5	10	0.02626	5.309	212.96	0.501	0.2130
5	10	0.02165	3.853	198.88	0.645	0.1989
7.5	15	0.02165	8.669	670.85	0.645	0.1988
10	20	0.02165	15.41	1589.9	0.645	0.1987
5	15	0.02484	10.84	711.88	0.547	0.2109
5	20	0.02626	21.23	1702.0	0.501	0.2127
2.5	12.5	0.02707	8.759	415.42	0.474	0.2127
5	25	0.02707	35.04	3321.3	0.474	0.2126

Table 15: Tail of a pulse generated by collapse into a hole of mass m_0 (Hayward, $h_N = h_L = 0.1s$, B from $\phi_H = B(v_e - v_s)^{-3}$ on $[1000, 2900]\,m_f/5$, fixed index; $|Q| = |\int r^2\phi_v\,\mathrm{d}v|$ along u_0). The free index is 3.011–3.012 in every row.

Tail. For $m_0 = 5$ the horizon field is again $B(v - v_s)^{-3}$ (free index 3.011–3.012 in all rows, the offset of the 3.009 of the flat start). For a weak pulse ($m_f = 5.01$) $B/(8m_f|Q|) = 1.004$: the linear identity $B = 8m|Q|$ of Section 14, here reproduced by the nonlinear pipeline with the hole already present (the rest of the 0.4% is the fit). For a strong pulse $F \equiv B/(8m_f|Q|)$ falls, and depends only on $\mu = m_0/m_f$: 0.645 for $\mu = \frac{1}{2}$ ($m_f = 10$, 15, 20 from $m_0 = 5$, 7.5, 10), 0.501 and 0.474 for

$\mu = \frac{1}{4}$ and $\frac{1}{5}$ (two scales each, agreement $\leq 10^{-3}$), A and $|Q|/m_f^2$ being equal for equal μ (Fig. 4a); the same table in Einstein gravity differs by $\leq 0.25\%$ ($B/m_f^3 = 0.1985, 0.2125, 0.2124, 0.2022$ for $(m_0, m_f) = (5, 10), (2.5, 10), (5, 25), (0, 10)$ against $0.1989, 0.2130, 0.2126, 0.2027$): at $m_f \geq 10$ the exterior is that of general relativity, in which the problem has no scale. $F(\mu)$ is 0.371 at $\mu = 0$ (the flat start of the previous sections, where 0.373 at $m_f = 5$), and it satisfies $F \simeq \mu + 0.23(1 - \mu)$ near $\mu = 1$ and $F - \mu = 0.29(1 - \mu)$ at $\mu = \frac{1}{2}$, i.e. B is $8|Q_0|$ times m_0 plus a fraction 0.23–0.37 of the mass that the pulse brings. B/m_f^3 has a flat maximum 0.213 at $\mu \simeq 0.2$–0.25 and then falls to 0.2027 at $\mu = 0$ and to 0 at $\mu = 1$. The next paragraph explains the reduction; the dependence on the pulse width at fixed μ is not a function of μ alone ($F = 0.363, 0.370, 0.384$ at $\mu = 0$ and $0.632, 0.644, 0.668$ at $\mu = \frac{1}{2}$ for pulses of width $4s$, $8s$, $16s$ with the same final mass, in Einstein gravity: the $8s$ values are 0.3704 and 0.6441, against 0.3713 and 0.6451 for the Hayward h), so the scale law above is for the pulse shape of Table 15.

Why $F < 1$: the charge absorbed by the growing horizon. The scalar equation $\phi_{uv} + (r_u\phi_v + r_v\phi_u)/r = 0$ is the same in all the models (the QTG corrections enter only the geometry); it is the conservation law of the Noether current of $\phi \to \phi + c$: $\omega = r^2\phi_v\,\mathrm{d}v - r^2\phi_u\,\mathrm{d}u$ is closed, and on a null ray where ϕ vanishes at both ends the charge is $\int r^2\phi_v\,\mathrm{d}v = -\int \phi\,\mathrm{d}(r^2)$. The charge Q_0 on u_0 is therefore the sum of the charge Q_S on the row v_1 outside the event horizon and the charge $Q_H = \int_{\mathcal{H}} r^2\phi_v\,\mathrm{d}v$ through the event horizon between v_0 and v_1. In the linear problem on a fixed hole r_H is constant, $Q_H = r_+^2[\phi_H] \to 0$, nothing is absorbed in the end and the identity $B = 8m|Q|$ of Section 14 holds with the charge Q_0 of the data. In the nonlinear collapse the horizon grows and takes up a net charge

$$Q_{\mathrm{abs}} = -\int_{\mathcal{H}} \phi\,\mathrm{d}(r^2) = -\int 2\,r_H\dot{r}_H\,\phi_H\,\mathrm{d}v, \qquad Q_\infty = Q_0 - Q_{\mathrm{abs}}, \tag{33}$$

(after v_1 the exterior is the static hole of mass m_f and r_H is constant, so only $v < v_1$ contributes), and what remains outside, $Q_\infty = Q_S + r_+^2\phi_H(v_1)$ (the row charge plus the charge that the static horizon still takes after v_1, $-r_+^2\phi_H(v_1)$, with the opposite sign), is the charge that sources the tail in the final static exterior. Applying the linear identity there, $B = 8m_f|Q_\infty|$, i.e.

$$F = \frac{B}{8m_f|Q_0|} = 1 - \frac{Q_{\mathrm{abs}}}{Q_0}. \tag{34}$$

The identity for data given on a slice follows from the time integral $\phi^{(1)}$ of the field ($\partial_t\phi^{(1)} = \phi$, regular at the horizon): in the static exterior $r^2 f\,\partial_r\phi^{(1)} = Q(r)$, the charge inside r, so $\phi^{(1)} = -Q/r - mQ/r^2 + \ldots$ outside the support and $r^2\partial_r(r\phi^{(1)}) \to mQ$, the Newman–Penrose constant of $\phi^{(1)}$, which is the quantity that fixes the v^{-3} tail (this is the sketch of the slice form; its numerical content is the identification $I_0^{(1)} = +mQ$ of Section 14). *Relation to the theorem of Gajic and Kehrberger.* The time-inverted Newman–Penrose constant of Gajic and Kehrberger [2025] is defined for $\Box_g\phi = F$ on a given dynamical exterior g that settles to Schwarzschild g_M; the metric is not required to solve the Einstein equations. We have read the statements of their Theorems 1.1 and 4.1 and of Assumptions 3.1, 4.1 and 4.2, with Sections 2 and 3, and, for $\ell = 0$ on Schwarzschild, the derivation of the constant (Proposition 7.1, Lemma 7.2); not the energy estimates of Section 5 or the iteration of Sections 7.4–7.8. The hypotheses of Theorem 4.1 are: g is in double null form, Eq. (3.1) near $\mathcal{H}^+$ ($r \leq \mathring{R} - 1$) and Eq. (3.2) for $r \geq \mathring{R}$, with $\mathring{R} > 100M$ the radius at which the two null foliations of their Section 2.2 are glued (not a smallness condition: the implied constants are arbitrary), and settles to g_M at a rate $\delta > 0$, namely $g_{\mu\nu} - (g_M)_{\mu\nu} \lesssim \tau^{-1-\delta}$ for $r \leq \mathring{R}$, up to the horizon (Eq. (3.6)), and, for $r \geq \mathring{R}$, $1/r$-expansions along $\mathcal{I}^+$ whose remainders are $\lesssim r^{-1}u^{-\delta}(u/r)^n$ (Eqs. (3.7)–(3.10)) in Bondi normalization (Eq. (3.12)), with finite constants that are not required to be small; the equation is written $\Box_{g_M}\phi = -(-\det g_M)^{-1/2}\partial_\mu(G^{\mu\nu}\partial_\nu\phi) + \mathfrak{F}$ with $G^{\mu\nu} = g^{\mu\nu}\sqrt{-\det g} - g_M^{\mu\nu}\sqrt{-\det g_M}$ (Eqs. (3.14), (4.1)) and $\mathfrak{F} \lesssim r^{-4}\tau^{-\beta}$, $\beta > 2$ (Eq. (4.2); $\mathfrak{F} = 0$ here); ϕ arises from smooth compactly supported data on a hyperboloidal slice Σ_{τ_0} that meets $\mathcal{H}^+$ in a sphere, with a weak a priori decay (Eq. (4.3)) which, according to their Remark 4.1, follows from the other assumptions (their Appendix B gives a sketch of the argument); the expansions are required to all orders ($N = \infty$), the metric must remain regular after commutation with τT (Eq. (3.6)), and M must be both the final Bondi mass and the limit of $r_H/2$. The conclusion for $\ell = 0$ (Theorem 4.1, Eqs. (4.6)–(4.7), with the approximate solution $\phi_{0,0}^{\mathrm{app}} = 1/[\tilde{\tau}(\tilde{\tau}+r)]$ of Eq. (4.5)) is that $r\phi_{\ell=0}$ differs from $T(r\phi_{0,0}^{\mathrm{app}})\, I_0$ by $O(\tau^{-5/2})$ in their weighted sense, with I_0 as displayed in their Eq. (1.5) for $F = 0$, $4\pi I_0 = \frac{M}{2}\int_{\mathcal{H}^+\cap\Sigma}\phi\,\mathrm{d}\mu + \frac{M}{2}\int_\Sigma n_\Sigma(\phi)\,\mathrm{d}\mu_\Sigma + \frac{M}{2}\int_{\mathcal{H}^+}\mathrm{tr}\chi\,\phi\,\mathrm{d}\mu\,\mathrm{d}v + \int_{\mathcal{I}^+}(\mathfrak{m} - M)\, r\phi\, r^{-2}\mathrm{d}\mu\,\mathrm{d}u$, with $\phi|_{r=r_0} \sim -2I_0\tau^{-3}$ for $r_0 \geq 2M$ (their Eq. (1.8); Eq. (1.6) is the equivalent form $4\pi I_0 = \int_{\mathcal{I}^+}\mathfrak{m}\, r\phi\, r^{-2}\mathrm{d}\mu\,\mathrm{d}u$, and $r\phi|_{\mathcal{I}^+} \sim -I_0\tau^{-2}$, Eq. (1.7)), and for $g = g_M$ their I_0 is the constant $I_0^{(1)}$ of Angelopoulos et al. [2018] up to a factor 2 (their footnote 2: the time functions of that paper are 2τ); the first two terms are the initial-data part and the last two the dynamical part, which vanishes for $g = g_M$ ($\mathfrak{m}$ is the mass aspect along $\mathcal{I}^+$, $\Omega\,\mathrm{tr}\underline{\chi} = -2/r + 4\mathfrak{m}/r^2 + \ldots$, whose spherical average is the Bondi mass function (their Lemma 3.2, which does not use the field equations); in vacuum $\partial_u\mathfrak{m} = -\frac{1}{4}|\hat{\chi}^{(1)}|^2$; $\mathrm{tr}\chi|_{\mathcal{H}^+}$ is the expansion of the generators of $\mathcal{H}^+$). Their conventions (Sections 1.2 and 2) are $g_M = -4D\,\mathrm{d}u\,\mathrm{d}v + r^2\mathring{\not{g}}$, $D = 1 - 2M/r$, with $v - u = r_*$ and $T = \partial_u + \partial_v = 2\partial_t$, so that $u_{\mathrm{GK}} = (t - r_*)/2$, $\tau = u$ for large r and $\tau = v$ near the horizon, i.e. $\tau \simeq t/2$ at fixed r. In spherical symmetry the first two terms are $\frac{M}{2}\,4\pi\left[\, r_H^2\phi_H + \int r^2\phi_v\,\mathrm{d}v\,\right] = 2\pi M\, Q$, the charge of the slice including the boundary term on the horizon, which is what Eq. (33) calls Q_∞ when Σ is the row v_1 (and Q_0 when Σ is the initial ray); that they appear with this relative weight is the statement that the combination is independent of the slice along a static horizon, since the flux of $\nabla\phi$ through a segment of the horizon, $4\pi r_H^2\Delta\phi_H$, cancels the change of the boundary term. The third term is, because $\mathrm{tr}\chi\,\mathrm{d}v$ is the logarithmic rate of change of the area of the generators, $\frac{M}{2}\,4\pi\int\phi\,\mathrm{d}(r_H^2) = -2\pi M\, Q_{\mathrm{abs}}$: the horizon integral of the theorem is the Noether charge taken up by the growing horizon (the area form; in spherical symmetry $\Omega\,\mathrm{tr}\chi\,\mathrm{d}v = \mathrm{d}\log\sqrt{\det\not{g}} = 2\,\mathrm{d}\log r_H$, and without symmetry, in the form (3.1) used near $\mathcal{H}^+$, $\Omega\,\mathrm{tr}\chi = \partial_v\log\sqrt{\det\not{g}} + \mathrm{div}\, b$, their Eq. (7.24); the $\mathrm{tr}\chi$ of their Eq. (1.5) is read as this expansion with respect to ∂_v; in their double-null frame $e_4 = \Omega^{-1}\partial_v$ one has $\Omega\,\mathrm{tr}\chi = \partial_v\log\sqrt{\det\not{g}}$, their Eq. (3.5), and their Remark 4.7, Eq. (4.10), writes this term with $\Omega\,\mathrm{tr}\chi\,\phi\sqrt{\det\not{g}}\,\mathrm{d}\tilde{\sigma}\,\mathrm{d}v$, i.e. with $\Omega\,\mathrm{tr}\chi\,\mathrm{d}v = \mathrm{d}\log\sqrt{\det\not{g}}$, which is the area form used here and removes the possible factor Ω^{-1} in Eq. (1.5). As printed, Eq. (4.10) has the prefactor $\frac{1}{2}$ for this term where Eq. (1.5) and the sphere term of Eq. (4.10) itself have $\frac{M}{2}$; in the proof of their Lemma 7.2 the horizon term enters through $4\pi C_0$, Eq. (7.7), and $I_0 = \cdots + MC_0$, Eq. (7.6), so that $\frac{M}{2}$ is the prefactor that follows from the proof (the dimensions require it as well) and the factor M is lost only in the printed statements (4.10) and (7.23); we use $\frac{M}{2}$, and the agreement of Eq. (34) with the measurement, 0.05% on average and 0.09% at most in 22 cases, is consistent with it). The fourth term has $\mathrm{d}u = \frac{1}{2}\mathrm{d}u_{\mathrm{phys}}$ and, with $\mathfrak{m} = M_B$ the Bondi mass and $a = r\phi|_{\mathcal{I}^+}$, equals $\frac{1}{2}\int(M_B - M)\, a\,\mathrm{d}u_{\mathrm{phys}}$. Altogether $I_0 = \frac{1}{2}\left[M(Q_0 - Q_{\mathrm{abs}}) + \int(M_B - M)\, a\,\mathrm{d}u_{\mathrm{phys}}\right]$ and

$$\phi|_{r_0} \simeq -8\Big[M\,(Q_0 - Q_{\mathrm{abs}}) + \int(M_B(u) - M)\, a(u)\,\mathrm{d}u\Big]\, t^{-3}, \tag{35}$$

so that the factor 8 and the equal weight of the two kinds of term follow from the conventions of the theorem; they agree with what we measured independently, the linear identity $B = 8m|Q|$ (10^{-4}) and, for the radiation term, the check of Bizoń et al. [2009] below. For a Schwarzschild exterior the linear identity is thus a consequence of Theorem 4.1 of GK (and of Angelopoulos et al. [2018] up to their normalization) and our 10^{-4} agreement tests the conventions used here;

for the Hayward exterior it is a numerical result. For the collapse of this section $M = m_f$ and the second term is negligible (below). *The Schwarzschild coefficient from the data (the steps we carried out, following their Proposition 7.1 and Lemma 7.2).* Take spherical data with $\mathfrak{F} = 0$ on $g = g_M$, compactly supported on a slice $\Sigma = \underline{C} \cup C$ that meets $\mathcal{H}^+$, and let w be the time integral of ϕ ($\Box_{g_M} w = 0$, $Tw = \phi$, regular on $\mathcal{H}^+$). For $g = g_M$ the correction term of their Eq. (7.6) vanishes, so $I_0 = M\,C_0$, where C_0 is the limit at $\mathcal{I}^+$ of $r^2\partial_r(rw)$ and is computed by integrating the two identities of their Eq. (2.13), $\partial_r(r^2\partial_r w)|_u = r\,\partial_r(r\phi)|_u$ and $\partial_r(r^2\partial_r w)|_v = -r\,\partial_r(r\phi)|_v$ (with ∂_r at fixed u or v), along $\underline{C}$ and C; the horizon end contributes only through $r_H^2\phi_H$ because $r^2\partial_r w|_v = 0$ on $\mathcal{H}^+$ by regularity. With $r\,\partial_r(r\phi) - \frac{1}{2}\partial_r(r^2\phi) = \frac{1}{2}r^2\partial_r\phi$ the result is $C_0 = \frac{1}{2}E$, $E = r_H^2\phi_H + Q_\Sigma$, $Q_\Sigma = \int_\Sigma r^2(\phi_v\,\mathrm{d}v - \phi_u\,\mathrm{d}u)$ oriented from the horizon to infinity, a closed one-form because $\Box\phi = 0$, so that E is the charge Q of the slice including the horizon term. Hence $4\pi I_0 = \frac{M}{2}[\int_{\mathcal{H}\cap\Sigma}\phi\,\mathrm{d}\mu + \int_\Sigma n_\Sigma(\phi)\,\mathrm{d}\mu]$ (their Eq. (1.5) with the ray terms combined), $I_0 = \frac{1}{2}MQ$ and, by Eq. (1.8) with $\tau \simeq t/2$, $\phi|_{r_0} \simeq -2I_0\tau^{-3} = -8MQ\,t^{-3}$: the factor 8 is 2 (their Eq. (1.8)) times 2^3 ($\tau = t/2$) times $\frac{1}{2}$ ($I_0 = \frac{1}{2}MQ$), and $B = 8M|Q|$ for $Q < 0$. This reproduces the linear identity measured above and is the exact-in-M statement; it holds under their hypotheses for the Schwarzschild exterior. *What this does and does not give.* Eq. (34) is Eq. (35) without the radiation term. The theorem assumes that g settles to g_M at the stated rate, with constants that are not required to be small, but its domain is metrics that settle to a Schwarzschild exterior in the coordinates of g_M ($\mathring{R} > 100M$ is only a gluing radius), and the initial ray of the collapse is not one: there the metric is close to Schwarzschild only for weak pulses, where the horizon term is the $O(\Delta m)$ integral of the weak-limit paragraph below. For a strong pulse the same conclusion follows by applying the theorem on the late slice v_1 (initial part only, the dynamical part being of the order of the back-reaction of the tail itself) together with the exact conservation law $Q_\infty = Q_0 - Q_{\mathrm{abs}}$. The back-reaction is small: after v_1 the horizon takes up $\leq 1.2 \times 10^{-5}m_f$ in the tables of the pulses and $\leq 1.2 \times 10^{-3}m_f$ in the table with $c = 0.04$, from $4\pi r_+^2 \int \phi_{H,t}^2\mathrm{d}t$ (`linbound.py`; twice as much if the factor y of the mass law is 1 rather than $\frac{1}{2}$ there), and the row sends to $\mathcal{I}^+$ an energy $E_{\mathrm{scri}} = E_{\mathrm{row}} - E_H = (0.05\text{–}8.7) \times 10^{-6}m_f$ in the 9 tables of the pulses ($E_{\mathrm{row}} = 4\pi\int r^2\phi_u^2\mathrm{d}u$ along the row, 42–43% of it; `rowflux.py`; a direct evolution of the row to $\mathcal{I}^+$ gives the same E_{scri} to 0.7%, and a charge $\int a\,\mathrm{d}u$ of the order of Q_∞, carried by a long-wavelength signal). The radiation term of Eq. (35) is bounded by this energy times the radiated charge, so it is smaller than $10^{-5}M|Q|$, and $M_\infty = m_f - E_{\mathrm{scri}}$ differs from the m_f used throughout by the same fraction. The identification of Q_{abs} with the horizon term of the time-inverted Newman–Penrose constant of Gajic and Kehrberger [2025] is conditional, for three reasons. (1) Their Theorem 1.1 is a statement about a *given* metric g that settles to Schwarzschild at the rate above (it is not required to solve the Einstein equations), with smooth compactly supported data and weak global decay of ϕ as inputs; none of these hypotheses was verified for the collapse data (the rate δ at which the collapse metric settles to its final state was not measured), and only the statements of that paper, not its proofs (Sections 5–9), were read. (2) The scalar field of the self-gravitating system obeys $\Box_g\phi = 0$ for the dynamical metric, so the theorem applies to it if that metric is in the class of Assumption 3.1. For the Einstein–scalar-field system the authors state (their Section 1.1.4) that the existence and nature of the tails was proved by Gautam [2024], whose theorems we read in the introduction of that paper and which are stated for the Einstein–Maxwell–scalar system with a subextremal Reissner–Nordström final state, and that upcoming work, cited there as [Qui] and not examined by us, produces metrics satisfying the assumptions of their Theorem 1.1. We know of no such statement for the quasi-topological models. (3) The final exterior of the Hayward model is the Hayward metric, which differs from Schwarzschild at fixed r by terms that do not decay in time (the metric function differs by $\simeq 10^{-2}$ at $r = 2m$ for $m = 5$, $\ell = 1$), so $|g - g_M| \lesssim (\tau + r)^{-1}\tau^{-\delta}$ does not hold there with g_M

Schwarzschild (Eq. (3.6) requires the decay also for $r \leq \mathring{R}$, up to the horizon); the static Hayward exterior is within the class of spherically symmetric, stationary, asymptotically flat spacetimes of Angelopoulos et al. [2018], but for the Hayward collapse the coefficient rests on the numerical result ($B = 8m|Q|$ to 10^{-4} on the static exterior) and not on the theorem. What is new here is therefore the identification of the horizon term with Q_{abs}, its evaluation on the nonlinear solution, and the test at strong field (22 cases, 0.05% on average, 0.09% at most); in the terms of Gautam [2024] it would say that his constant $\mathfrak{L}[\varphi]$ (defined for the charged system; we did not find the uncharged case stated in the text we read) is proportional to $M(Q_0 - Q_{\text{abs}})$ plus the radiation term (a conjecture: his text, as we have read it, gives no formula for $\mathfrak{L}$). The case in which the horizon takes up a large charge and a large energy is radiated at the same time is not tested: the present pipeline needs the row to carry a small energy (stage L is linear on a static exterior), and such a test needs a nonlinear evolution through the passage of the radiation. *Test.* The event horizon is the ray u_H with $r(u_H, v_1) = r_+(m_f)$; its column is followed back from v_1 to v_0 in stage N (interpolating between the two columns that bracket it; `horizon_ray.py`). The conservation law $Q_0 = Q_S + Q_H$ holds to 1.4×10^{-6}–5.3×10^{-5} of Q_0 (1.1×10^{-4}, 2.7×10^{-5}, 6.8×10^{-6} for $h_N = 0.2$, 0.1, 0.05 at $m_0 = 5 \to 10$), Q_{abs} from Eq. (33) agrees with $Q_H - r_H^2 \phi_H(v_1)$ to 10^{-4}, and Eq. (34) reproduces the measured F in all 22 cases (13 of Table 15, the Einstein tables, and the other pulse widths; the strongest is $m_0 = 0 \to 10$ with 63% of Q_0 absorbed) with $F_{\text{meas}}/F_{\text{abs}} = 1.0056 \pm 0.0003$ (range 1.0052–1.0060) for the fixed-index fit of Table 15, a constant that is the same in the linear limit ($m_f = 5.01$: $F_{\text{abs}} = 0.9983$, $F_{\text{meas}} = 1.0042$) and does not depend on μ or on the model; it is the first correction of the tail, since the fit $\phi_H = B(v_e - v_s)^{-3}(1 + C/(v_e - v_s))$ in the same window gives $F_{\text{corr}}/F_{\text{abs}} = 0.9995 \pm 0.0002$ (range 0.9991–0.9998, $C = 170$–850; `fit_corr.py`) (Table 16, Fig. 4c); changing h_N by a factor 4 does not change F_{abs} in the fourth digit. The one case that fails is a pulse wide enough that at v_1 it has not yet fallen through the horizon ($m_0 = 0$, width $16s$; the slice does not contain the event horizon: the row at v_1 ends at $r = 25 > r_+$), where Eq. (34) cannot be evaluated on that slice. So F is not an independent quantity: the reduction of the v^{-3} coefficient relative to $8m_f|Q_0|$ is the fraction of the Noether charge that the growing horizon swallows (36% of Q_0 when $m_f = 2m_0$ and 63% when the hole is formed by the pulse), and it depends on the history of the horizon, which is why it is a function of μ at fixed shape and of the width at fixed μ. For a weak pulse $1 - F \simeq 0.8\,\Delta m/m_f$ (0.83, 0.78, 0.75, 0.72 for $\Delta m/m_f = 0.09$, 0.23, 0.375, 0.5, 0.63 for the formation of the hole). What is not derived is Q_{abs} itself from (m_0, m_f, shape): Eq. (33) needs the event-horizon radius and the horizon field, which come from the nonlinear solution; the exact identity replaces the earlier statement that F is only measured, and the crude weighting $B = 8\int m\,\mathrm{d}Q$ of the local mass is not the mechanism.

An estimate of Q_{abs} from the data of the pulse alone. With $\psi = r\phi$ one has $Q = -\int 2\psi\,\mathrm{d}r$ along a null ray, and for a pulse that is short compared with the scale over which it is scattered ψ is carried unchanged along the ingoing rays, so $\psi_H(v) = \psi_{u_0}(v) = r\phi|_{u_0}$ (measured in the 22 tables: $\max|\psi_H|/\max|\psi_{u_0}| = 0.96$–$0.99$, the two lowest values, 0.96, belonging to the wide pulses, for which the approximation is the least good; `eik_test.log`). The horizon is an outgoing null ray that ends on $r_+(m_f)$, and in the Vaidya-like approximation $\dot{r}_H = \frac{1}{2}f\big(r_H, m(v)\big)$ with $m(v)$ the generalized mass on u_0 (a quantity of the data, from the constraint along the ray) and f the static function, integrated backwards from $r_H(v_1) = r_+(m_f)$ (the integration is stable backwards). Then $Q_{\text{abs}} \simeq -\int 2\psi_{u_0}\dot{r}_H\,\mathrm{d}v$ and $F \simeq 1 - Q_{\text{abs}}/Q_0$ require no run of stage N (`eik_test.py`). Against Eq. (34) evaluated on the nonlinear solution, the estimate with the true horizon radius is low by 1.5% on average (at most 3.6%) and the complete estimate by 2.3% on average (at most 4.1%, rms 2.6%) in the same 22 cases, and the radius $r_H(v_0)$ of the horizon at the start (for example 16.37 against 16.44 for $m_0 = 5 \to 10$, i.e. $1.66\,r_+(m_0)$) to 0.7%; the weak-pulse value is reproduced to 2×10^{-4} (0.9981 against 0.9983). In the weak-field limit $m_f - m_0 = O(A^2)$ and

m_0	m_f	Q_0	$Q_{\rm abs}$	$1 - Q_{\rm abs}/Q_0$	$F_{\rm meas}$	ratio
5	5.01	−0.0464	−0.0001	0.9983	1.0042	1.0059
5	5.5	−0.398	−0.030	0.9249	0.9303	1.0059
5	8	−2.004	−0.561	0.7203	0.7245	1.0058
5	10	−3.853	−1.382	0.6414	0.6451	1.0059
5	15	−10.84	−4.945	0.5439	0.5472	1.0059
5	25	−35.04	−18.53	0.4712	0.4740	1.0060
2.5	10	−5.309	−2.662	0.4985	0.5014	1.0058
0	10	−6.824	−4.305	0.3691	0.3713	1.0058
0	10 (Einstein)	−6.824	−4.309	0.3685	0.3704	1.0052
0	10 (Einstein, width $4s$)	−2.385	−1.524	0.3611	0.3630	1.0052
5	10 (Einstein, width $16s$)	−10.93	−3.669	0.6644	0.6679	1.0052

Table 16: Charge absorbed by the event horizon (Eq. (33), from the horizon column of stage N) and the measured coefficient: $F_{\rm meas} = B/(8m_f|Q_0|)$ of Table 15 against $1 - Q_{\rm abs}/Q_0$. Hayward unless stated ("Einstein": model 0), pulse width $8s$ unless stated; the full list of 22 cases is in `sp/horizon_ray.log`.

$Q_0 = O(A)$, so that $8m|Q|$ is of the third order in the amplitude, as the tail of a self-gravitating massless scalar field of small compact data computed by Bizoń et al. [2009] at third order; their data are smooth and have no hole at the start, so Eq. (34) cannot be tested against their formula with our characteristic setup, which has no regular center; their formula was therefore tested separately (below). The two statements are, however, the same structure in opposite regimes. Writing their first-order field as $\phi_1 = (a(u) - a(v))/r$ (their Eq. (23)) and using their second-order mass (Eq. (25), whose limit $r \to \infty$ is) $m_2(u, \infty) = 8\pi \int_u^\infty a'^2 \mathrm{d}s$ ($m = 2M_{\rm MS}$ in their convention, so the Bondi mass is $M_B(u) = 4\pi \int_u^\infty a'^2 \mathrm{d}s$), and for the coefficient (their Eq. (35), with I_1^0 from Eq. (28)) $\Gamma_0 = -2^5\pi \int I_1^0(s)\, a(s)\, \mathrm{d}s$ with $I_1^0(u) = \int_u^\infty a'^2 \mathrm{d}s$, one finds $\Gamma_0 = -8\int M_B(u)\,(r\phi)|_{\mathcal{I}^+}(u)\,\mathrm{d}u$ with $(r\phi)|_{\mathcal{I}^+} = a(u)$: the null-infinity term of the Gajic and Kehrberger [2025] coefficient, with no initial-data term because the background is flat. It reduces to $-8M_{\rm ADM}\, Q$ ($Q = \int a\, \mathrm{d}s$ the shift charge of the data, $M_{\rm ADM} = 4\pi \int a'^2 \mathrm{d}s$) only when the Bondi mass is constant over the support of a; for dispersing data it is not, because the radiation carries mass and charge to $\mathcal{I}^+$ together (as read from the equations of that paper, whose tail is $\phi_3 \simeq \Gamma_0 t^{-3}$ at fixed r, their Eq. (34); we have not rederived it). We checked this with an independent code for the regular-center problem (`codec/bcr/`: polar–areal gauge, conservative second-order discretization, RK4, Dirichlet wall at $R = 400$, ϵ-differencing $[\phi_\epsilon/\epsilon - \phi_{\epsilon/2}/(\epsilon/2)]/(\frac{3}{4}\epsilon^2)$ to isolate the third-order field; units $G_{ab} = 8\pi T_{ab}$, data $\phi = \epsilon(a(-r) - a(r))/r$, $a \propto [x(1-x)]^6$ on a support of width 6). The code is second-order convergent ($h = 0.1, 0.05, 0.025$: ratio of differences 3.9), its mass is the ADM mass of the data ($14.17\epsilon^2$), and the fixed-r field $t^3\phi_3$ ($r = 5, 10, 20$) tends, after the $h \to 0$ and $\epsilon \to 0$ extrapolations and a fit $c + a/t + b/t^2$ over $100 \leq t \leq 300$, at $r = 5$, 10, 20 to $c = -118.2$, -118.3, -119.1 (three-parameter fit) and -116.5, -115.6, -112.1 (two-parameter fit $c + a/t$) for the symmetric pulse, against $\Gamma_0 = -8\int M_B a\, \mathrm{d}u = -116.0$ (and $-8M_{\rm ADM}Q = -232.0$), and to -131.6, -131.7, -132.6 and -129.2, -128.3, -124.4 for a pulse with $a \to a\,[1 + 1.5(x - \frac{1}{2})]$, against -128.2 (-247.7 for $-8M_{\rm ADM}Q$, and -119.5 if the mass passed rather than the mass radiated weights the integral). The formula holds to 3.5% in all twelve fits (1.9–3.4% above for the three-parameter ones, 0.1–3.4% off for the two-parameter ones), which is the size of the fitting systematics: the two forms differ by 1.5% at $r = 5$ and by 6% at $r = 20$, and $t^3\phi_3$ is still drifting at the end of the window ($-133.4 \to -122.2$ between $t = 100$ and 300 at $r = 5$), so that c is an extrapolation in t (`codec/bcr/tailfit_results.log`). The weighting by the Bondi mass of the outgoing radiation is confirmed against the naive one at that accuracy (-232.0 and -247.7 are 100% off). This is a check of the cited formula and of our reading of it, not a new result; it covers one family of data and one observer range. The all-order form of the same law (Eq. (35) with $M = 0$, i.e. $\phi|_{r_0} \to -8\int M_B(u)\,a(u)\,\mathrm{d}u\;t^{-3}$ with the true Bondi mass and the true a) was then tested at strong field with the same code (`codec/bcr/s2a_an.py`): at $\epsilon = 0.04, 0.06, 0.08, 0.10$ ($M_{\mathrm{ADM}} = 0.023, 0.051, 0.092, 0.146$; the data collapse to a hole for $\epsilon \gtrsim 0.13$) the late-time field $t^3\phi_{\mathrm{NL}}$ at $r = 5, 10, 20$, with $\phi_{\mathrm{NL}} = [\phi_\epsilon - (\epsilon/\epsilon_0)\phi_{\epsilon_0}]/(1-\epsilon_0^2/\epsilon^2)$, $\epsilon_0 = 0.01$, extrapolated in h (Richardson, $h = 0.1, 0.05$) and fitted by $c + a/t + b/t^2$ on $100 \le t \le 300$, is $-0.0080, -0.0296, -0.0801, -0.1892$ against $-8\int M_B\,a\,\mathrm{d}u = -0.0080, -0.0296, -0.0799, -0.1883$ computed from $m(t, R_{\mathrm{obs}})$ and $r\phi(t, R_{\mathrm{obs}})$ of the same run ($R_{\mathrm{obs}} = 100, 200$ extrapolated linearly in $1/R_{\mathrm{obs}}$), a ratio 1.0003–1.0051 with no free parameter; the same data give $-0.015, -0.051, -0.123, -0.246$ for $-8M_{\mathrm{ADM}}\int a\,\mathrm{d}u$ and $-0.0074, -0.025, -0.059, -0.116$ for the third-order formula $-116\epsilon^3$. So the Bondi-mass weighting of the radiation term holds in the nonlinear regime, not only at order ϵ^3; it is a flat-space check (no horizon). In our collapse the pulse falls into the hole, the outgoing flux through $\mathcal{I}^+$ is of second order, and the term that matters is the horizon one; Eq. (34) is the corresponding statement for strong data. Including the scattering of the pulse by the hole as a linear transfer (`eik2.py`: the field $\phi(u_0, v)$ of the pulse is a test field on the static exterior of mass m_f, propagated to the horizon column with stage L and mapped to the label v by $v_e = 2r_*(r(u_0, v))$, the horizon radius being the same Vaidya-like one) changes the result in a way that depends on the history. For a pulse thrown into an existing hole with $m_f \le 2m_0$ it is accurate to 0.8% (for $m_0 = 5 \to 5.5, 6.5, 8, 10$: -0.4, -0.7, -0.3, $+0.7$%; $7.5 \to 15$ and $10 \to 20$: $+0.7$%) against 0.8–3.1% for the eikonal field; for larger ratios it is worse than the eikonal one, with $+3.4$, $+6.9$% at $m_f = 15, 25$ ($m_0 = 5$), $+5.4$% for $2.5 \to 10$ and $+16$% for the collapse from flat space ($0 \to 10$), where the eikonal estimate is -0.5%; the linear transfer assumes that the field propagates on the final geometry, which a pulse that makes the hole does not see. So the two first approximations bracket the exact F in the formation regime and the linear one is better in the perturbative regime; neither is more than a few percent from Eq. (34) where it is valid. *The weak limit.* Linearize the Vaidya-like horizon about the final hole: with $r_H = 2m_f + \delta$ and $\kappa = 1/(4m_f)$, $\delta' = \kappa[\delta + 2(m_f - m(v))]$, $\delta(v_1) = 0$, so that $\dot r_H = 2\kappa(m_f - m) + \kappa\delta$. Eq. (33) with $\psi_H \simeq \psi_{u_0}$ gives $Q_{\mathrm{abs}} = -2\int \psi\,\dot r_H\,\mathrm{d}v$, while on u_0 one has exactly $Q_0 = -\int \psi f\,\mathrm{d}v$ ($\mathrm{d}r = \frac{1}{2}f\,\mathrm{d}v$, $f = 2r_v$). For a pulse that is short against $4m_f$ the δ term is small, and since m rises monotonically across the pulse its ψ-weighted average $\langle m_f - m\rangle_\psi$ is about $\Delta m/2$, so that

$$1 - F \simeq \frac{\Delta m}{2\,m_f\,\langle f\rangle_\psi}, \qquad \langle f\rangle_\psi = \frac{\int \psi f\,\mathrm{d}v}{\int \psi\,\mathrm{d}v}, \tag{36}$$

with $\langle f\rangle_\psi$ the value of f on u_0 where the pulse sits ($r \simeq 20$–27, $f = 0.50$–0.61 for $m = 5$). Against Eq. (34) (`weak_limit.py`, `weak_limit.log`): for $m_0 = 5 \to m_f = 5.01, 5.5, 6.5, 8$ the measured $1 - F$ is 0.0017, 0.0751, 0.1808, 0.2797, the linearized-horizon value 0.0019, 0.0823, 0.1940, 0.2935 ($+12$, $+10$, $+7$, $+5$%), and Eq. (36) 0.0020, 0.0862, 0.2031, 0.3072 ($\langle f\rangle_\psi = 0.501, 0.527, 0.568, 0.610$). The coefficient 0.8 (measured 0.85, 0.83, 0.75 at $\Delta m/m_f = 0.002, 0.09, 0.375$) is therefore $1/(2\langle f\rangle_\psi) \simeq 1$ reduced by the horizon history and by $\psi_H \ne \psi_{u_0}$; it is not a universal number: it depends on where the pulse sits on u_0 and on its shape (compare the dependence on the pulse width at fixed μ above). Since Q_0 is conserved but ψ is carried along ingoing rays only to the eikonal accuracy, Eq. (36) depends on the slice u_0 at that level (5–10%). This accounts for the form and the size of the weak limit and not for the percent. That accuracy is that of a first approximation (it ignores the difference between the null coordinate v and the advanced time of a Vaidya solution, and the scattering in the eikonal variant), and those are the only a priori statements made here; the exact results are Eqs. (33) and (34).

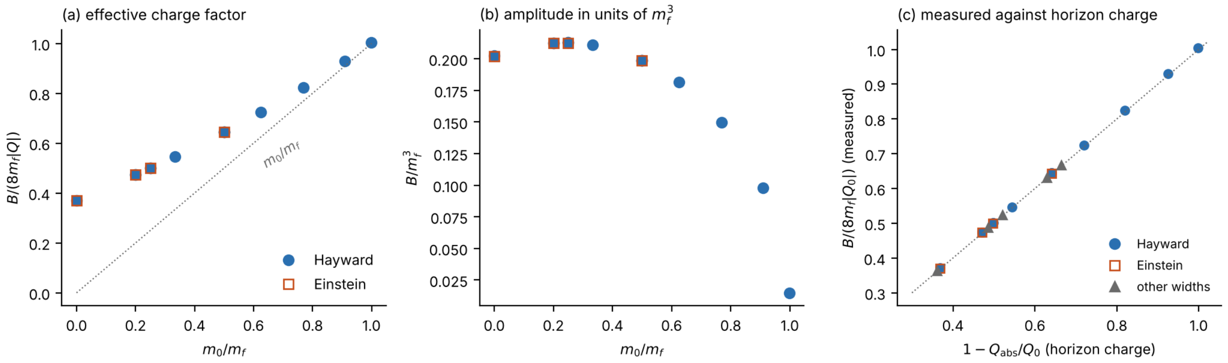

Figure 4: Tails of collapse pulses into an existing hole: (a) $F = B/(8m_f|Q|)$ and (b) B/m_f^3 against $\mu = m_0/m_f$; filled circles Hayward, open squares Einstein ($m_f = 5.01$ to 25; points with equal μ and different m_f coincide); (c) F measured against $1 - Q_{\rm abs}/Q_0$ from the charge absorbed by the event horizon, Eq. (33), including pulses of other widths (the line is the identity; B from the fixed-index fit, the points lie 0.56% above it on average (0.52–0.60%), and 0.05% below on average (0.09% at most) with the first correction).

Why the fold does not occur, and code E. The starting row of code E is the static hole of m_f, which test (iii) shows is what the strong collapse leaves behind at v_1; the field enters code E only through $\phi_H(v)$ after the pulse. The fold of the previous paragraph belongs to a pulse placed inside the code-E interval, where the column is shot backwards from a final state, so no event-horizon finder is needed here. Code E on the tables for $m_0 = 5 \to m_f = 10$ and $m_0 = 0 \to 10$ ($h_s = h_v = 0.2$, $v \le 2000$, `MASS=10`, `PUOMODE=zero`) has the same final state: $r_\infty = 1.02653$ (against $r_-(10) = 1.02670$; the mass $m^* = 10.059$ implied by r_∞ agrees with the ray mass 10.055 at the freezing time $v = 94$), $zm \to 0.54085$ ($\frac{1}{2}r_\infty^3 = 0.54086$), $\mathrm{d}m/\mathrm{d}v = 9.466$ over $1600 \le v \le 1800$ against 9.535 from Eq. (27) at r_∞ (0.7% below), $v\sigma_v = -8.44$ and -8.43 and $\mathrm{d}\ln K/\mathrm{d}\ln v = 10.13$ and 10.10 at $v = 1800$ (-8 and 10 for $k = 3$), the same for the two histories; the amplitude of K differs as B^{-4} (K at $v = 1500$: 8.98×10^{25} against 8.37×10^{25}, ratio 1.072, $(B_0/B_5)^4 = 1.079$ for the tables' $B = 198.9$ and 202.7; `pscan_an.py`, `pscan_zm.py`, `pscan_freeze.py`). For $m_f = 15$ ($h_s = h_v = 0.2$, $v \le 2800$) $r_\infty = 1.01665$ against $r_-(15) = 1.01740$ (implied front mass 15.65, 4.3% above m_f), $zm = \frac{1}{2}r_\infty^3$ to 10^{-5}, and $\mathrm{d}m/\mathrm{d}v$ is 0.992 of Eq. (27) at r_∞. *The case* $m_f = 25$*.* Here $r_-(25) = 1.01026$ is only 1% above ℓ, and at $h_s = h_v = 0.2$ the run is not converged: the implied front mass is 0.6%, 4.3% and 92% above m_f for $m_f = 10, 15, 25$ (it grows as $r_- \to \ell$, the front being under-resolved). Refining the mesh moves r_∞ toward $r_-(25)$: 1.00527, 1.00633 ($h_s = 0.1$), 1.00888 ($h_v = 0.1$) and 1.00993 ($h_s = h_v = 0.1$), i.e. distances from $r_-(25)$ of 5.0×10^{-3}, 3.9×10^{-3}, 1.4×10^{-3} and 3.3×10^{-4}, implied front masses 48.1, 40.2, 28.8, 25.8 (92%, 61%, 15%, 3.2% above m_f), $zm = \frac{1}{2}r_\infty^3$ to 10^{-5} in each, and with both meshes at 0.1 the rate $\mathrm{d}m/\mathrm{d}v = 25.21$ is 0.997 of Eq. (27) at r_∞ (the ratios were 1.20, 0.97, 1.09 on the other meshes). The bump at the front was therefore a resolution effect; at $h_s = h_v = 0.1$ the case is as converged as $m_f = 15$ is at 0.2 (4.3%), which is the accuracy claimed for it. What remains: no further refinement was run, $v\sigma_v = -19.8$ and $\mathrm{d}\ln K/\mathrm{d}\ln v = 32.4$ at $v = 2666$ ($\kappa_+ v \simeq 26$) are far from -8 and 10 and follow the $m_f = 10$ curve at the same $\kappa_+ v$ (-16.6 and 26.6 at $v = 1200$), so the exponents are not tested at $m_f = 25$ (`strong_codeE.log`; `pscan_freeze.py`, `pscan_zm.py` with `VS=4.4444`).

What this establishes and what it does not. The conclusions of Section 12.9 (the front, the linear growth of m, $zm \to \frac{1}{2}r_-^3/\ell^2$, $v\sigma_v \to -(2k+2)$ and $K \propto v^{4k-2}$ with $k=3$) hold when the horizon field is the one generated by a spherically symmetric collapse into the Hayward hole of a compact scalar pulse ($k=3$, the generic Price exponent, with the amplitude as measured, the quasi-normal transient included) or of a pulse with a b/r^2 field or a static field $c\,g(r)$ ($r\phi \to c$, $I_0 = -cm$) on the initial cone ($k=2$, with $I_0 = -b$ resp. $-cm$ and the coefficient $4I_0$ of the linear theorem; $v\sigma_v \to -6$ and $K \propto v^{4k-2}$ with $k=2$ in the sense of Section 12.9). Not established: (i) the tail for other collapse data (a thicker shell; data that are static from the center outwards rather than outside a radius, a case that requires a regular static field through the hole formed; the nonlinear value of the $k=3$ coefficient, for which only the linear identity $B = 8m|Q|$ and the measured reduction are given); (ii) the full nonlinear evolution to the late times of the tail in one code (stage L is linear on the static exterior, justified by test (a) and by the independence of the switching time, not by a nonlinear run to $v \sim 10^3$); (iii) the interior beyond the thin layer $U \leq 1.5$ of the starting row and the region between the pulse and the core; (iv) the amplitude of K (it depends on the profile on the starting row, and the profile used is from a nonlinear row at $v=60$ continued to the interior of that layer only); (v) the Hawking flux and the back-reaction of the Hayward collapse on the shell itself, which is not followed to the core ($u_{\mathrm{end}} < v_0$).

15 What this establishes, what it does not, and what is next

Established, by the kind of support. *Derived.* The generalized-mass law (16) for any h; the sign theorem (Proposition 1) for a real scalar field and data regular on the event horizon whenever $h' > 0$; the classification of Section 7; the branch $y < 0$, to which the finite-advanced-time singularity of interior characteristic data belongs, is excluded for event-horizon-regular data (existence of the solution for all v is not proved). *Checked against independent computations.* The double-null equations in the Einstein limit (10^{-16}), on static interiors (10^{-7}–10^{-6}), through the integrability and compatibility identities (2×10^{-11}, 3×10^{-12}; 7×10^{-50} in 80-digit arithmetic near the pole); as the Euler–Lagrange equations of the reduced action, including the $\phi_u^2\phi_v^2$ term and the Raychaudhuri coefficient $4\pi/h'$ (exact rational arithmetic); the covariant scalars P, H, T, K that lift that action reduce to $\psi, \eta, \tau, \mathcal{R}$ on random spherical metrics ($\leq 8 \times 10^{-11}$, computed from the four-dimensional Riemann tensor, Section 4(g)); the evolution against exact static solutions (Hayward, bounded h), against the r_{uv} equation it does not use, against the linear-response kernel of Paper G [Ownby, 2026b], against a second discretization (mode 2 of code A) and against an independent fourth-order code (code B: 10^{-7} in r and 4×10^{-6} absolute in σ_v after the h_s correction for $120 \leq v \leq 240$). *Measured.* The metric exponent $-\kappa_-(m_f)$ (2×10^{-6}); the front and its delay (13.1 per decade of B for the Hayward h, 13.5 for the bounded h); the difference from the additive closure ($\propto B^2$ in the flux, $O(1)$ in σ_v behind the front); the state behind the front for $B = 3 \times 10^{-3}$ to $v = 300$ (code A), $v = 380$ (code B) and $v = 1400$ (code E): $\psi \to 1/\ell^2$ with $1 - \ell^2\psi \simeq 0.59/m$, m growing linearly with $\mathrm{d}m/\mathrm{d}v = 3.7 \to 4.4$, $r \to r_-$, σ_v rising monotonically from -0.094 to -0.018 without a zero crossing, K from 10^{25} to $1.6 \times 10^{30}\ell^{-4}$ at $U \simeq 1$ with a local slope of 2.5 at $v = 340$ that rises to 3.64 at 1400, as Eq. (30) predicts (and its generalization (32) for $k = \frac{1}{2}, \ldots, 3$, three amplitudes and $m/\ell = 3, 5, 10$, Section 12.9); the slaving structure of Section 12.5; the error law (21) (28.3 per halving of h_v at second order and 56.6 at fourth order, as $p \ln 2/\kappa_+$ predicts), which moved the limit of the run from $v \approx 170$ to 300 (code A), 380 (code B) and beyond 1400 (code E, Sections 11 and 11.3); the transverse null tidal integrals over the remaining affine range ($\iint \mathcal{T}\,\mathrm{d}\lambda^2 \leq 10^{-15}$ for $v \geq 300$). The tail of a collapse of a compact scalar pulse into the Hayward hole: exponent 3.009, coefficient $B = 25.45$ ($m = 5$; $B/m^3 = 0.2052$, 0.2033, 0.2015 for $m = 3, 5, 10$

against 0.2017 in the Einstein limit), the static Hayward interior left behind the pulse (to 10^{-5} for $11 \lesssim r \lesssim 30$, 4×10^{-5} at $r = 3.8$, 1.3×10^{-3} at $r = 1.058$), and code E on that tail to $v = 1400$ with the exponents of Section 14. *Derived and tested against runs not used to fit it.* The mechanism of the error law (Section 11): the forward integration of the horizon Raychaudhuri equation, whose one-dimensional model predicts the advanced time of failure of code B at two other amplitudes to 0.4 units (one fitted constant, the threshold $\delta^* = 0.062$); the closed form (27) for $m/\ell = 3$ and 10 (1.5% and 0.25%); the seed of the late-time departure of code E (Section 11.3): an offset ϵ in the initial row advances the departure by -19.8 per e-fold of ϵ over 10^{-15}–10^{-12}, against $-1/\kappa_+ = -20.4$. *Derived from the measured structure.* The closed forms (26), (27) and (30), the sign $\sigma_v < 0$ for a decreasing flux, and the asymptotics (28), which hold in the slaved description and are tested up to the simulated v.

Not established. (i) *The endpoint and the nature of* $v = \infty$. $\sigma_v \to 0^-$ as $-(2k+2)/v$, a finite affine range, and $K \propto v^{4k-2}$ (the local slope of K is 3.64 at $v = 1400$, as Eq. (30) predicts; with the two $1/v$ terms of that equation removed it is 4.01 for $k = \frac{3}{2}$ and 0.01, 2.01, 6.02, 10.02 for $k = \frac{1}{2}$, 1, 2, 3 against $4k-2$ at $v = 1021$, but the slope itself comes within 10% of $4k-2$ only at the end of the simulated range, $\kappa_+ v \gtrsim 70$) are conjectures about $v \to \infty$ based on a verified structure; extendibility of the metric, the strength of the singularity for timelike observers, and linear perturbations of the solution at $v = \infty$ are not studied. (ii) *Parameters.* Tail exponents $k = \frac{1}{2}, 1, \frac{3}{2}, 2, 3$ (prescribed on the horizon in Section 12.9; Section 14 generates $k = 3$ and its amplitude from a collapse of a compact pulse, and $k = 2$ from collapse data with $I_0 \neq 0$ ($r\phi \to 0$ and $r\phi \to c$ at infinity); code E is driven by the $k = 3$ and the b/r^2 $k = 2$ horizon fields, not by the $r\phi \to c$ one, whose horizon field is the same to 10^{-4} in the coefficient), amplitudes 3×10^{-4}–3×10^{-2} and inner masses $m/\ell = 3, 5, 10$ with code E to $\kappa_+ v \simeq 50$–69 (Section 12.9; the equations depend on m/ℓ only); with code A $k = \frac{1}{2}, \frac{3}{2}, 3$ (windows $v \simeq 160$–380) and $m = 3$ ($zm \to 0.677$); for the bounded h one value of $m/\sqrt{\alpha}$, a window $v \lesssim 125$, no kernel and no run of code E. Codes B and E were run for the Hayward h only. (iii) *The theory.* The four-dimensional theory, its Hayward resummation and the analytic continuation of the time-dependent equations outside $0 < \psi < 1/\ell^2$ are taken from Refs. Bueno et al. [2026a], Borissova and Carballo-Rubio [2026]; the lifts are non-polynomial and not unique (Section 4(g)), and the denominators of I_{C^2} vanish on conformally flat configurations, which the solution behind the front avoids ($\Omega \geq 8\times10^{11}$); hyperbolicity and well-posedness of the full (non-spherical) theory are not studied. For the spherical sector the system (8)–(11) has the form $X_{uv} = F(X, X_u, X_v)$ for $X = (r, \sigma, \phi)$ plus two constraints, with F smooth for $0 < \psi < 1/\ell^2$, $r > 0$, $W > 0$, i.e. a Goursat problem whose characteristics are $u, v = \text{const}$ whatever the size of $c_3\phi_u^2\phi_v^2$; the change from hyperbolic to elliptic type that Ripley and Pretorius [2019] find in shift-symmetric Einstein-dilaton-Gauss–Bonnet collapse requires a derivative coupling of the scalar and has no counterpart for this minimally coupled scalar as long as $\psi < 1/\ell^2$ (an observation about the form of the equations, not a theorem about their solutions; the coefficients degenerate at the pole of h, which the solution approaches as $1/v$). The equations, not only the scheme, are untested by a code in a different gauge or variables. The solution is very sensitive to the c_3 term (10% moves σ_v by 0.024 at $v = 70$; without it there is no front), which is fixed by the action. (iv) *Validity.* A curvature of 10^{25}–$10^{28}\ell^{-4}$ is far outside the regime of a classical four-dimensional effective theory; the result is the classical evolution of the theory with a minimally coupled scalar. In particular the dynamical curvature bound that holds for static solutions (the limiting-curvature property) does not hold for this matter coupling. (v) *Physics not included.* The Hawking flux, matter other than a massless scalar, charge and rotation, and any thin-shell (Ori) argument: Di Filippo, Kubižnák and Srinivasan Di Filippo et al. [2026] find that null thin shells are not available in the standard distributional theory for vacuum regular black holes of this kind, whereas Ref. Frolov and Zelnikov [2026] uses them with junction conditions; we take no side. (vi) *The numerics.* The error law is derived (Section 11), but the threshold $\delta^* = 0.062$ is a fitted constant (from the five failure times of code B at $B = 3 \times 10^{-3}$) and is not derived; the former limit of code E, a step in the initial row amplified at the rate κ_+ from $\kappa_+ v \simeq 33$, is located and removed (Section 11.3), and no other limit appears to $\kappa_+ v = 69$, but no run beyond that has been made; the free (unconstrained) evolution fails in both codes by $v \simeq 100$ and that mechanism is not analyzed. (vii) *The tail coefficient of the strong collapse.* The identification of $Q_{\rm abs}$ with the horizon term of the time-inverted Newman–Penrose constant of Gajic and Kehrberger [2025] is conditional: the hypotheses of their theorem (a given metric settling to Schwarzschild at a polynomial rate, smooth compactly supported data) were not verified for the collapse data, the final exterior of the Hayward model is not Schwarzschild, and of that paper the energy estimates of Section 5 and the iteration of Sections 7.4–7.8 were not read (see Section 14). The agreement of F with $1 - Q_{\rm abs}/Q_0$ (0.05% on average) is a numerical fact about our runs; the a priori estimates of $Q_{\rm abs}$ (Vaidya-like horizon, eikonal field; low by 1.5% and 2.3% on average, 3.6% and 4.1% at most, in the same 22 cases) are empirical figures and not bounds.

Consequence for the series. The question left open by Paper E [Ownby, 2026a], whether the mass-tracking metric can be tested nonlinearly, has a partial answer. In the theory that realizes its static sector, with data regular on the event horizon, the metric exponent follows the generalized mass accumulated on the horizon, as in the additive closure; the flux through the interior is the linear response of the static interior (Paper G) up to a correction quadratic in the tail amplitude; then a front arrives, behind which the generalized mass inflates linearly in v, ψ approaches the pole of h without reaching it, the Misner–Sharp mass stays at $\frac{1}{2}r_-^3/\ell^2$, r stays at r_-, the exponent σ_v is slaved to the flux and rises toward zero from below, and the curvature, carried by the term of the theory quartic in the scalar gradient, is far above ℓ^{-4} and grows with a slowly varying power of v. This differs from the additive closure of Papers E and G [Ownby, 2026a,b], where the Misner–Sharp mass inflates and σ_v keeps the exponent $-\kappa_-(\mu_f)$. The inner horizon of the Hayward interior is thus unstable in this theory in the sense that the generalized mass grows linearly and the curvature as a power of v while the Misner–Sharp mass and r stay bounded; the geometry behind the front is neither the static interior nor the singular branch of interior data; in the slaved continuation (power-law tail completion) the accumulated transverse tidal distortion along radial null geodesics on the ray stays finite and small.

Next steps. (1) The strong collapse pulse at $m_f > 25$ (where $r_- \to \ell$ and the front needs $h \leq 0.1$ and a still finer mesh, $h = 0.05$, for the exponents), an estimate of the absorbed charge $Q_{\rm abs}$ (Eq. (33)) better than the 2–4% observed in the 22 cases (an empirical figure, not a bound) of the Vaidya-eikonal one (scattering of the pulse by the hole), and an event-horizon finder for pulses that do not end in a static hole; a thicker shell, initially static fields through the center, a collapse followed to the core, and an a priori value of the nonlinear v^{-3} coefficient, the full nonlinear evolution to $v \sim 10^3$ in one code (a grid adapted to the horizon), the dependence of the amplitude of K on the interior profile, the kernel of the bounded model, and the bounded h with code E. (2) Timelike geodesics and the singularity strength at $v = \infty$, and the question whether a null thin shell regularizes or strengthens the front. (3) A continuation criterion for the spherical Goursat system (the solution exists as long as ψ stays below $1/\ell^2$ and r, W stay positive) and the hyperbolicity of the non-spherical theory, which is the open one. (4) A code in the gauge $\varphi = r$ of Ref. Bueno et al. [2026a]. (5) Non-minimally coupled matter that restores a curvature bound (Ref. Bueno et al.

[2026b]) and its effect on the front. (6) The Hawking flux (a separate study). (7) Code E beyond $\kappa_+ v \simeq 70$ ($v \gtrsim 1.5 \times 10^3$ for $m/\ell = 5$), to see whether a further rounding-level seed exists and to reach the regime where the slope of K itself is within a few percent of $4k - 2$.

Level of reading of the sources. We use the time-dependent equations of Ref. Bueno et al. [2026a] (Eqs. (36)–(38)) and the characteristic functions (Eq. (51) for the bounded model there); the static identification of the Hayward metric with $h = \psi/(1 - \ell^2\psi)$ and the covariant densities (Eqs. (53)–(66)) are those of Ref. Borissova and Carballo-Rubio [2026], checked here directly (Section 2 and 4(g)). Of Ref. Borissova and Carballo-Rubio [2026] the text was read through its Eq. (156); the rest of its Section IV and its Discussion were not, and the statement about Ref. Di Filippo et al. [2026] rests on its abstract. Refs. Bueno et al. [2025a,b], Frolov and Zelnikov [2026], Di Filippo et al. [2026], Carballo-Rubio et al. [2022], Bueno et al. [2026b], Bonanno et al. [2026], Shao et al. [2026], Iofa [2022] are used at the level of the abstract and the statements attributed to them; we have not checked their numerical results. Refs. Poisson and Israel [1989, 1990], Dafermos [2005], Ori [1991], Brady and Smith [1995], Burko [1997], Hod and Piran [1998], Marolf and Ori [2012] are cited for the mass-inflation context. Ref. [Gajic and Kehrberger, 2025], which Section 14 uses, was read in part: the abstract, Section 1 (Theorem 1.1, Eqs. (1.5)–(1.8), Remarks 1.1–1.3, footnotes 1 and 2, Section 1.1.4 and the sketch of Section 1.2), Sections 2 and 3 and the statements of Section 4 (Assumptions 4.1 and 4.2, Theorems 4.1–4.5, Remarks 4.1–4.15); the derivation of the $\ell = 0$ constant (Proposition 7.1, Lemma 7.2), Sections 6, 8 and 9 and the appendices were also read; the energy estimates of Section 5 and the iteration of Sections 7.4–7.8 were not. Ref. [Gautam, 2024] was read through the statements of its Theorems 1.1 and 1.15 in the introduction, Ref. [Bizoń et al., 2009] for its Eqs. (23), (25), (28), (34) and (35), and Ref. [Gundlach et al., 1994b] for the exponents quoted. The identification of $Q_{\rm abs}$ with the horizon term of the time-inverted Newman–Penrose constant of Gajic and Kehrberger is conditional: the hypotheses of their theorem (a given metric settling to Schwarzschild at a polynomial rate, smooth compactly supported data, weak global decay of ϕ) were not verified for the collapse data, the final exterior of the Hayward model is not Schwarzschild (Assumption 3.1 requires $g - g_M \lesssim \tau^{-1-\delta}$ up to the horizon), and the energy estimates of Section 5 and the iteration of Sections 7.4–7.8 of that paper were not read. For $m = 5$, $\ell = 1$ one has $r_+ = 9.898$ against $2m = 10$: with the null coordinates of g and g_M identified and r_* matched at infinity, $\not{g}_{AB} - (\not{g}_M)_{AB} \to (r_+^2 - 4m^2)\mathring{g}_{AB} = -2.03\,\mathring{g}_{AB}$ on $\mathcal{H}^+$ for all τ, so that Eq. (3.6) fails for every $\delta > 0$, and so does the weaker form $g - g_M \lesssim r^{-1}\tau^{-\delta}$ of their Remark 4.6; choosing $M = r_+/2$ instead violates the Bondi normalization (an r^{-1} term $-2(m - M)/r$ in $\Omega^2 - \Omega_M^2$). The expansions along $\mathcal{I}^+$ are not the obstruction. The Hayward coefficient therefore rests on the numerical result $B/(8m|Q|) = 1.000$ to 10^{-3} and on the stationary theory of Angelopoulos et al. [2018], whose hypotheses on f we have not checked.

A Interior characteristic data: the ray tests

Before event-horizon-regular data are available (Section 8), the equations can be tested on characteristic data given on interior rays, as in the nonlinear test of Paper E [Ownby, 2026a]. This section reports those runs. Their role changes after Proposition 1: they are tests of the ray equations and of the scheme, and the singular behavior they show belongs to the data.

Scheme and data. We solve (9)–(11) for (r, σ, ϕ) with the second-order scheme of Paper E (predictor–corrector on the null diamond, three corrector iterations; `qtg_dn.c`), in the streaming form that stores only selected columns $u = \text{const}$. The data are characteristic data inside the hole as in the nonlinear test of Paper E: the static Hayward interior ($m = 5$, $\ell = 1$; σ static on both rays), $\phi = B + A\,e^{-u^2}$ on $v = 0$ ($A = 0.02$) and the power-law tail $\phi = B(1+v)^{-k}$ on $u = u_0 = -6$ ($k = 3/2$ unless noted); r, r_u, r_v and the σ-derivatives on the data rays follow from (8)–(9) along the rays by an ODE integration to 10^{-12}. These are not data regular on the event horizon (Section 8); they are the data of the (u, v) runs of Paper E, for which the additive closure gave mass inflation at finite radius.

The data ray. On the ray $u = u_0$ the ray ODE is closed (the data are the tail $\phi_v(v)$ and the static σ_v of the data surface). We integrate it to the event $|\psi| > 10^6$ (`ray_blowup.py`, `ray_blowup.log`) and find

B	2×10^{-2}	2×10^{-3}	2×10^{-4}	2×10^{-5}	2×10^{-6}	2×10^{-7}
v_*	46.5	57.3	70.1	82.8	95.2	107.6
$r - r_-$ at v_*	-0.56	-5.6×10^{-3}	-5.6×10^{-5}	-5.6×10^{-7}	-5.6×10^{-9}	-5.6×10^{-11}

with $\psi(v_*) = -10^6$ (the event) in every case: v_* increases by 12.3 per decade of B in a linear fit over the six values (local slopes 12.8, 12.6, 12.5, 12.3 per decade, i.e. 5.58 → 5.36 per e-fold of $1/B$), i.e. logarithmically, as expected for a quantity that grows like $e^{\lambda v}$ from an initial value $\propto B$ or B^2; we have not derived λ (a rate $\lambda \simeq 1/5.4$ would fit an initial departure $\propto B$). The negative sign of $r - r_-$ is the Hayward curvature singularity of Section 7(ii): the ray has crossed below r_- by an amount $\propto B^2$ at v_* (the tabulated differences fall by 10^2 per decade of B), and in the limit $B \to 0$ the singular radius tends to r_-. The dependence on the tail exponent at $B = 2 \times 10^{-3}$ is weak: $v_* = 51.5, 53.6, 57.3, 61.8, 72.1$ for $k = 0.5, 1, 3/2, 2, 3$. For comparison the Einstein-additive ray ODE of Paper E has no finite-v singularity at this stage ($M \sim e^{\kappa_- v}$ at finite radius to $v = 3000$).

Classification of rays. `run_scan_u.py` scans $u \in [-5, 6]$ at $B = 2 \times 10^{-4}$ (table below; $h = 0.0025$; the values at $v = 64$ are converged to about 10^{-3} in ψ on $y > 0$ rays and are quoted mainly for the sign of y; see Appendix C). The sign of $y = -r_u/W$ is negative for $u \le -2$ and positive for $u \ge -1$, as in Paper E (there $u_\star \in [-2.04, -1.0]$ for the additive closure); on the $y < 0$ rays ψ stays at the static value $1/r_-^2 = 0.8943$ until $v \sim 60$ and then decreases, on the $y > 0$ rays ψ rises toward $1/\ell^2 = 1$ (and σ_v rises from $-\kappa_-$ toward 0), as (16) predicts for the two signs.

u	sign y	$\psi(40)$	$\psi(56)$	$\psi(60)$	$\psi(64)$	$\sigma_v(60)$, $\sigma_v(64)$
-5	–	0.8943	0.8943	0.8942	0.8930	-0.796, -0.796
-4	–	0.8943	0.8943	0.8942	0.8929	-0.796, -0.796
-3	–	0.8943	0.8942	0.8942	0.8928	-0.796, -0.796
-2	–	0.8943	0.8943	0.8942	0.8936	-0.795, -0.782
-1	+	0.8950	0.8966	0.9136	0.9532	-0.564, -0.197
0	+	0.8977	0.9069	0.9435	0.9694	-0.294, -0.111
1	+	0.8984	0.9107	0.9488	0.9714	-0.258, -0.102
2	+	0.8988	0.9131	0.9515	0.9724	-0.240, -0.098
4	+	0.8989	0.9135	0.9517	0.9722	-0.237, -0.097
6	+	0.8989	0.9147	0.9524	0.9716	-0.227, -0.094

The $y < 0$ side in the evolution. The data ray is the first to reach $v_*(u_0)$, and the evolved solution cannot be continued past it on any rectangle $[u_0, u_1] \times [0, v]$. Below that time the interior $y < 0$ rays behave as (16) requires: at $B = 2 \times 10^{-3}$ ($v_*(u_0) = 57.3$) the run to $v = 56.5$ on $h = 0.02$ gives ψ passing a maximum of 0.894–0.896 at $v \approx 32$ (its static value is $1/r_-^2 =$

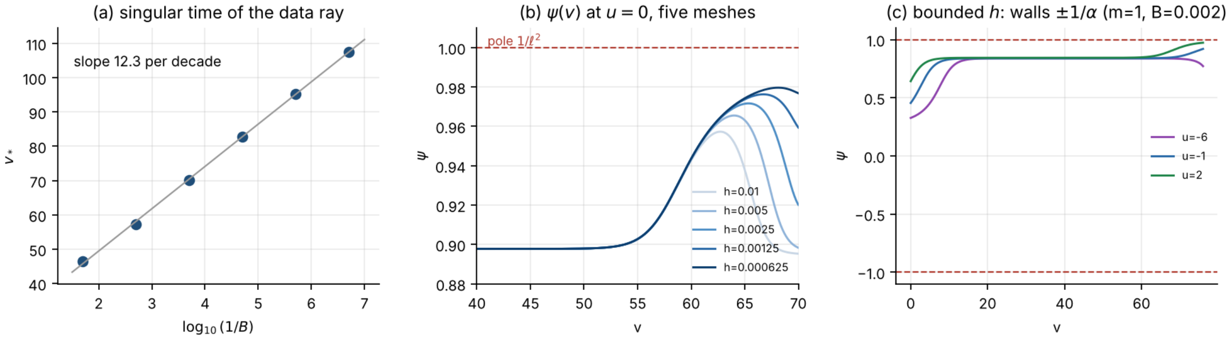

Figure 5: (a) Advanced time v_* at which $|\psi|$ reaches 10^6 on the data ray $u = u_0$ against $\log_{10}(1/B)$ (Hayward h, $k = 3/2$); the line is the linear fit of Appendix A. (b) $\psi(v)$ at $u = 0$ for $B = 2 \times 10^{-4}$ on five meshes; the pole of the Hayward h is $1/\ell^2 = 1$. (c) Bounded h ($m = 1$, $\alpha = 1$, $B = 2\times 10^{-3}$, $h = 0.02$): ψ on rays with $y < 0$ ($u = -6$) and $y > 0$ ($u = -1$, 2) stays within the walls $\pm 1/\alpha$.

0.8943) and then decreasing monotonically to 0.855, 0.853, 0.852, 0.850, 0.846 at $v = 55.5$ on $u = -6, -5, -4, -3, -2$, with no $|\psi| > 10^3$ anywhere, while the $y > 0$ rays rise to 0.905–0.945 (`run_pde_blowup2.py`, `pde_blowup_h0.02.log`; $h = 0.02$ is not a converged mesh at these times, and the statement is qualitative). We have not evolved the interior rays through their own singular times; that would need data on a surface that is regular beyond $v_*(u_0)$.

Reading after Proposition 1. The rays with $y < 0$ above arise because the characteristic data on $u = u_0$ and $v = 0$ do not come from a surface regular on the event horizon: their y is whatever the static interior and the pulse at $u \sim 0$ give, and it is negative for $u \leq -2$. For event-horizon-regular data $y \geq \frac{1}{2}$ and the negative-mass branch cannot occur. The finite-advanced-time singularity of this section is therefore an artifact of the data, and it should not be read as the fate of the interior of the Hayward metric in this theory. What these runs do establish is that the ray equations behave as (16) requires for both signs of y and that the scheme is second-order convergent (Section 4).

Bounded h on interior data. For the bounded h of (4) with $\alpha = 1$ and $m = 1$ ($r_- = 1.0927$, $\kappa_- = 0.5054$, $\psi(r_-) = 0.8376$) the same data ($A = 0.02$, $B = 2 \times 10^{-3}$, $k = 3/2$) run to $v = 76$ at $h = 0.02$ without any failure (`run_model2.py`, `model2_h0.02.log`), well beyond the v_* of the Hayward h for the same data (57.3), and ψ stays inside the wall: its range over the whole grid is [0.326, 0.9727]. On the rays $u = -6, -3$ ($y < 0$, $y \simeq -9 \times 10^{11}$ at $v = 76$) ψ falls from 0.8376 to 0.771, 0.770 (the generalized mass decreasing, as (16) says, so that ψ moves toward the lower wall $-1/\alpha$, which is far), while on $u = 0, 2, 4, 6$ ($y > 0$, $y \simeq 1.7$–1.9×10^{12}) ψ rises to 0.959, 0.9726, 0.9726, 0.9724 and σ_v rises from $-\kappa_- = -0.505$ to +0.003, +0.026, +0.026, +0.026. The picture is that of the Hayward $y > 0$ side: ψ goes to the positive wall (here $1/\alpha = 1$) as the generalized mass grows, σ_v rises through zero. The convergence is second order for both models (Section 4). The Hayward and the bounded h therefore agree qualitatively on the positive-mass side and differ on the negative-mass side, exactly where the range of h differs.

B Curvature invariants in double-null form

For $ds^2 = g_{ab}dx^a dx^b + r^2 d\Omega^2$, with $g = -W\,du\,dv$, the Kretschmann invariant is $K = R_2^2 + 8\,H_{ab}H^{ab}/r^2 + 4(1 - (\nabla r)^2)^2/r^4$, with R_2 the curvature scalar of the two-metric and $H_{ab} = \nabla_a \nabla_b r$.

Here $(1-(\nabla r)^2)/r^2 = \psi$ and $R_2 = 4\sigma_{uv}/W \equiv 4\Sigma$. With $\Gamma^u_{uu} = \sigma_u$, $\Gamma^v_{vv} = \sigma_v$ one has $H_{uu} = r_{uu} - \sigma_u r_u$, $H_{vv} = r_{vv} - \sigma_v r_v$, $H_{uv} = r_{uv}$, and $g^{uv} = -2/W$, so $H_{ab}H^{ab} = (8/W^2)(H_{uu}H_{vv} + H_{uv}^2)$. Using (8), (9): $H_{uu}H_{vv} = 16\pi^2 r^2 c_1^2 \phi_u^2 \phi_v^2$ and $H_{uv}^2 = W^2 r^2 Q^2/16$, which gives (29), and Σ follows from (10). The formula reproduces $48M^2/r^6$ for the Schwarzschild metric and $24/\ell^4$ at the Hayward pole with no flux, and agrees with $f''^2 + 4f'^2/r^2 + 4(1-f)^2/r^4$ on the static interiors (`check_curv_static.log`). It is a geometric statement about the metric (r, σ) that was evolved; it depends on h through Q, c_1, c_2, c_3.

C Resolution study of the interior-data run

Resolution study at $u = 0$. The $y > 0$ side is where ψ approaches the pole $1/\ell^2$, and where the scheme is most demanding. Table 17 gives ψ and σ_v at $u = 0$ for $B = 2 \times 10^{-4}$ on five meshes. At $v \leq 60$ the solution is converged to 5×10^{-5} in ψ and 6×10^{-4} in σ_v already at $h = 0.0025$. At $v = 64$ the successive differences in ψ are 1.37×10^{-2}, 3.9×10^{-3}, 1.0×10^{-3}, 2.6×10^{-4}, with ratios 3.5, 3.9, 3.9, i.e. second order, and the two finest meshes differ by 2.6×10^{-4} in ψ and 2.6×10^{-3} in σ_v. The coarse meshes ($h \geq 0.005$) show a spurious maximum of ψ near $v =$ 64–66 followed by a decrease and a spurious sign change of σ_v; this is a resolution artifact (the converged ψ increases monotonically through $v = 68$). At $v = 68$ the two finest meshes still differ by 5.5×10^{-3} in ψ ($\sigma_v = +0.169$ against $+0.120$): the solution is not converged there. As an independent diagnostic we evaluate the exact identity (15) on the stored columns (`check_psi_identity.py`): the defect of $\partial_v \psi$ relative to the largest term is 2.1×10^{-3}, 0.39, 2.8 for $h = 0.01$ and 5.9×10^{-5}, 1.6×10^{-3}, 6.1×10^{-2} for $h = 0.000625$ in $v \in [40, 56]$, $[56, 62]$, $[62, 66]$ (falling by about 4 per halving in the middle window), so the ψ-equation is satisfied to 2×10^{-3} for $v \leq 62$ at the finest mesh and to 6% in $[62, 66]$.

Table 17: ψ and σ_v at $u = 0$ ($B = 2 \times 10^{-4}$, $A = 0.02$, Hayward h) on five meshes (`run_conv.py`, `conv_B2e-4.log`).

h	$\psi(56)$	$\psi(60)$	$\psi(64)$	$\psi(66)$	$\psi(68)$	$\sigma_v(64)$, $\sigma_v(68)$
0.01	0.90684	0.94268	0.95176	0.91842	0.89799	+0.0939, +1.1879
0.005	0.90691	0.94337	0.96550	0.95392	0.91531	-0.0705, +0.9805
0.0025	0.90693	0.94355	0.96942	0.97093	0.95445	-0.1114, +0.3855
0.00125	0.90693	0.94359	0.97042	0.97575	0.97413	-0.1215, +0.1687
0.000625	0.90693	0.94360	0.97068	0.97698	0.97962	-0.1241, +0.1197

We do not claim a converged late-time endpoint for $y > 0$: the converged range is $v \lesssim 64$ ($\psi = 0.971$, $\sigma_v = -0.12$ at $v = 64$), the finest mesh gives $\psi = 0.980$ and $\sigma_v = +0.12$ at $v = 68$ with the 5×10^{-3} uncertainty stated above, and the singular behavior on the $y < 0$ side (which sets $v_* \simeq 70$ for this amplitude) is the limit on useful v for any run in this setup. Whether $\sigma_v \to 0^+$ and the $y > 0$ side tends to a regular configuration with $\psi \to 1/\ell^2$, as (16) suggests for $m \to \infty$, or something else, is open.

Code and data availability

All code, logs and data tables behind the numbers and figures of this paper are archived at Zenodo, doi:10.5281/zenodo.23455595, (code: MIT license; text, figures and data: CC-BY-4.0). The archive is self-contained: the modules of Paper E that it needs are copied in `code/vendor_E/`

(copied from the Paper E archive, doi:10.5281/zenodo.23455587 [Ownby, 2026a]), and no other directory is required. It needs Python 3 with numpy, scipy and matplotlib and a C compiler (`requirements.txt`). `code/README.md` gives the dependency graph of the scripts, one measured run time per script and the logs that have no generator in the archive.

`code/eh_qtg/run_analysis.sh` re-runs the analysis scripts and compares each new log with the shipped one (`ONLY=<substring>` selects scripts, `RUN_LONG=1` adds the long evolutions). `bash verify.sh` in the archive runs 162 quick-tier jobs (about 55 CPU-minutes) and, with `--full`, 40 further long jobs; it compares 2149 quoted numbers with the computed ones and writes `verification/REPORT.md`; `B_ASYM=1.5` is the default for the `asym1` fits. Interior-data runs (Sections 4 and A): `code/run_all.sh` regenerates every log quoted there. `qtg_dn.c` is the C evolution routine (`evolve_qtg`, `evolve_stream_qtg` with models $0, 1, 2$ and, new here, `evolve_stream_qtg_cy`, the constrained evolution of Section 9 with modes 1–3); `qtg_lib.py` the Python side (data, ray ODE, `coef`, `hfun`, `run_stream_cy`); `qtg_bg2.py` the static interior of the bounded model; `check_*.py` the checks of Section 4; `ray_blowup.py`, `run_scan_u.py`, `run_conv.py`, `run_model2.py`, `run_pde_blowup2.py` Appendix A; `make_figs.py` its figure.

Code E, the mechanism, the seed and the scans (Sections 11, 11.3 and 12.9): `code/eh_qtg/codee/` (`qtg_e.c`, `qtg_el.c` in long double, `gen_e.py`, `t_static_e.py`, `mech_vb.py` and `vb_prediction.py`, `conv_e.py`, `conv_e2.py`, `seed_amplitude.py`, `noise_tests.py`, `scan_an.py`, `scans.sh`, `kstudy2.py`; thinned data in `data/`, logs in `logs/`; `setup.sh` builds the libraries). Event-horizon-regular runs (Sections 8–13): `code/eh_qtg/` (README there).

`setup.sh` builds the three evolution libraries and the data-construction library from the modules in `code/vendor_E/` (no sibling directory is needed); `run_eh.sh` regenerates the logs in `logs/` (the runs are independent; the longest are the runs on four meshes to $h = 0.00625$ for three amplitudes and two models). The modules: `static_gen.py` (exact static solution for any f), `dnU_qtg.py` (horizon data and rows), `gen_q.py` (four-mesh runs, writes the JSON in `data/`), `analyze_q.py` (Romberg extrapolation); the analyses `late_q.py`, `endpoint_q.py`, `front_q.py`, `width_q.py`, `plateau_q.py`, `kap_check.py`, `kap_check_b.py`, `kernel_ratio_q.py`, `cmp_add.py`, `masslaw_q.py`, `curv_q.py`, `curv_terms_q.py`, `check_curv_static.py`, `check_ruv_q.py`, `t_static_*.py`, `free_vs_con.py`, `fig_eh.py`. The JSON files shipped are the extrapolation inputs.

Collapse-generated tails (Section 14). `code/eh_qtg/codec/`: `qtg_col.c` (stage N: free evolution of `qtg_dn.c` with row snapshots and an apparent-horizon tracker), `qtg_L.c` and `tail_L.py` (stage L and the static Hayward/Schwarzschild background in closed form), `collapse_lib.py` (data on the two rays), `pipeline.py`, `collapse_tail.py` (the table of $\phi_H(v)$), `gen_c.py` (code E driven by a table; needs `eh_qtg/setup.sh`), the tests `interior.py`, `overlap.py`, `tables.py`, `mscale.py`, `ascale.py`, `linear.py`, `fit_tables.py`, the analyses `front.py`, `collapse_E.py`, `conv_c.py`, `fig_collapse.py`; `run_all.sh` regenerates the logs of stages N and L (the data of the runs quoted are in `data/`). The $k = 2$ data of Section 14: `nplin.py` (stage L alone), `np_nl.py` (stage N+L with the tail, option `tail=(b,r_a,`Δr`)` of `collapse_lib.rays`, `pipeline` and `collapse_tail`), `np_fit.py`, `np_lin_b.py`, `overlap_np.py`, `linear_q.py` (the $8m|Q|$ test), `an_k.py`, `fig_collapse_k2.py`, and `np_all.sh`, with the tables and the code-E runs in `npdata/`, `ce/`. The pulse-strength scan of Section 14: `pulse_table.py` (adds $P \sin^4$ to a horizon table), `sc/runq.sh` (the code-E runs, P as arguments), `pscan_summary.py`, `pscan_an.py`, `pscan_zm.py`, `pscan_conv.py` (with `sc/runh.sh`, `sc/runs.sh` for the two mesh checks), `horizon_map.py` and `vaidya_estimate.py` (run from `codec/sc/`; the tables and the JSON of the runs are in the second archive, `archive/paperF_pulse_scan_data/`, and the logs in `sc/`); `gen_c.py` has event stops on r in the horizon shooting.

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